



```
In [5]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.style as style
import seaborn as sns
import itertools
%matplotlib inline
```

```
In [6]: import warnings
warnings.filterwarnings('ignore')
```

```
In [7]: pd.set_option('display.max_rows', 500)
pd.set_option('display.max_columns', 500)
pd.set_option('display.width', 1000)
pd.set_option('display.expand_frame_repr', False)
```

```
In [8]: import os
for dirname, _, filenames in os.walk(r'C:\Users\HP\Downloads/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
In [9]: applicationDF = pd.read_csv(r'C:\Users\HP\Downloads/application_data.csv')
previousDF = pd.read_csv(r'C:\Users\HP\Downloads/previous_application.csv')
applicationDF.head()
```

```
Out[9]:
```

|   | SK_ID_CURR | TARGET | NAME_CONTRACT_TYPE | CODE_GENDER | FLAG_OWN_CAR |
|---|------------|--------|--------------------|-------------|--------------|
| 0 | 100002     | 1      | Cash loans         | M           | 1            |
| 1 | 100003     | 0      | Cash loans         | F           | 1            |
| 2 | 100004     | 0      | Revolving loans    | M           | 1            |
| 3 | 100006     | 0      | Cash loans         | F           | 1            |
| 4 | 100007     | 0      | Cash loans         | M           | 1            |

```
In [10]: previousDF.head()
```

```
Out[10]:
```

|   | SK_ID_PREV | SK_ID_CURR | NAME_CONTRACT_TYPE | AMT_ANNUITY | AMT_APPLI |
|---|------------|------------|--------------------|-------------|-----------|
| 0 | 2030495    | 271877     | Consumer loans     | 1730.430    | ...       |
| 1 | 2802425    | 108129     | Cash loans         | 25188.615   | 60        |
| 2 | 2523466    | 122040     | Cash loans         | 15060.735   | 10        |
| 3 | 2819243    | 176158     | Cash loans         | 47041.335   | 40        |
| 4 | 1784265    | 202054     | Cash loans         | 31924.395   | 30        |

```
In [11]: # Database dimension
print("Database dimension - applicationDF      :",applicationDF.shape)
print("Database dimension - previousDF        :",previousDF.shape)

#Database size
print("Database size - applicationDF          :",applicationDF.size)
print("Database size - previousDF            :",previousDF.size)

Database dimension - applicationDF      : (307511, 122)
Database dimension - previousDF        : (1670214, 37)
Database size - applicationDF          : 37516342
Database size - previousDF            : 61797918
```

```
In [12]: # Database column types
applicationDF.info(verbose=True)
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 307511 entries, 0 to 307510
Data columns (total 122 columns):
#   Column                                          Dtype
---  -
0   SK_ID_CURR                                   int64
1   TARGET                                        int64
2   NAME_CONTRACT_TYPE                           object
3   CODE_GENDER                                  object
4   FLAG_OWN_CAR                                 object
5   FLAG_OWN_REALTY                             object
6   CNT_CHILDREN                                 int64
7   AMT_INCOME_TOTAL                            float64
8   AMT_CREDIT                                  float64
9   AMT_ANNUITY                                 float64
10  AMT_GOODS_PRICE                             float64
11  NAME_TYPE_SUITE                             object
12  NAME_INCOME_TYPE                            object
13  NAME_EDUCATION_TYPE                         object
14  NAME_FAMILY_STATUS                          object
15  NAME_HOUSING_TYPE                           object
16  REGION_POPULATION_RELATIVE                  float64
17  DAYS_BIRTH                                  int64
18  DAYS_EMPLOYED                               int64
19  DAYS_REGISTRATION                           float64
20  DAYS_ID_PUBLISH                             int64
21  OWN_CAR_AGE                                 float64
22  FLAG_MOBIL                                   int64
23  FLAG_EMP_PHONE                              int64
24  FLAG_WORK_PHONE                             int64
25  FLAG_CONT_MOBILE                           int64
26  FLAG_PHONE                                  int64
27  FLAG_EMAIL                                  int64
28  OCCUPATION_TYPE                             object
29  CNT_FAM_MEMBERS                             float64
30  REGION_RATING_CLIENT                        int64
31  REGION_RATING_CLIENT_W_CITY                 int64
32  WEEKDAY_APPR_PROCESS_START                  object
33  HOUR_APPR_PROCESS_START                     int64
34  REG_REGION_NOT_LIVE_REGION                  int64
35  REG_REGION_NOT_WORK_REGION                  int64
36  LIVE_REGION_NOT_WORK_REGION                 int64
37  REG_CITY_NOT_LIVE_CITY                      int64
38  REG_CITY_NOT_WORK_CITY                      int64
39  LIVE_CITY_NOT_WORK_CITY                     int64
40  ORGANIZATION_TYPE                           object
41  EXT_SOURCE_1                                float64
42  EXT_SOURCE_2                                float64
43  EXT_SOURCE_3                                float64
44  APARTMENTS_AVG                             float64
45  BASEMENTAREA_AVG                           float64
46  YEARS_BEGINEXPLUATATION_AVG                float64
47  YEARS_BUILD_AVG                             float64
48  COMMONAREA_AVG                             float64

```

|     |                              |         |
|-----|------------------------------|---------|
| 49  | ELEVATORS_AVG                | float64 |
| 50  | ENTRANCES_AVG                | float64 |
| 51  | FLOORSMAX_AVG                | float64 |
| 52  | FLOORSMIN_AVG                | float64 |
| 53  | LANDAREA_AVG                 | float64 |
| 54  | LIVINGAPARTMENTS_AVG         | float64 |
| 55  | LIVINGAREA_AVG               | float64 |
| 56  | NONLIVINGAPARTMENTS_AVG      | float64 |
| 57  | NONLIVINGAREA_AVG            | float64 |
| 58  | APARTMENTS_MODE              | float64 |
| 59  | BASEMENTAREA_MODE            | float64 |
| 60  | YEARS_BEGINEXPLUATATION_MODE | float64 |
| 61  | YEARS_BUILD_MODE             | float64 |
| 62  | COMMONAREA_MODE              | float64 |
| 63  | ELEVATORS_MODE               | float64 |
| 64  | ENTRANCES_MODE               | float64 |
| 65  | FLOORSMAX_MODE               | float64 |
| 66  | FLOORSMIN_MODE               | float64 |
| 67  | LANDAREA_MODE                | float64 |
| 68  | LIVINGAPARTMENTS_MODE        | float64 |
| 69  | LIVINGAREA_MODE              | float64 |
| 70  | NONLIVINGAPARTMENTS_MODE     | float64 |
| 71  | NONLIVINGAREA_MODE           | float64 |
| 72  | APARTMENTS_MEDI              | float64 |
| 73  | BASEMENTAREA_MEDI            | float64 |
| 74  | YEARS_BEGINEXPLUATATION_MEDI | float64 |
| 75  | YEARS_BUILD_MEDI             | float64 |
| 76  | COMMONAREA_MEDI              | float64 |
| 77  | ELEVATORS_MEDI               | float64 |
| 78  | ENTRANCES_MEDI               | float64 |
| 79  | FLOORSMAX_MEDI               | float64 |
| 80  | FLOORSMIN_MEDI               | float64 |
| 81  | LANDAREA_MEDI                | float64 |
| 82  | LIVINGAPARTMENTS_MEDI        | float64 |
| 83  | LIVINGAREA_MEDI              | float64 |
| 84  | NONLIVINGAPARTMENTS_MEDI     | float64 |
| 85  | NONLIVINGAREA_MEDI           | float64 |
| 86  | FONDKAPREMONT_MODE           | object  |
| 87  | HOUSETYPE_MODE               | object  |
| 88  | TOTALAREA_MODE               | float64 |
| 89  | WALLSMATERIAL_MODE           | object  |
| 90  | EMERGENCYSTATE_MODE          | object  |
| 91  | OBS_30_CNT_SOCIAL_CIRCLE     | float64 |
| 92  | DEF_30_CNT_SOCIAL_CIRCLE     | float64 |
| 93  | OBS_60_CNT_SOCIAL_CIRCLE     | float64 |
| 94  | DEF_60_CNT_SOCIAL_CIRCLE     | float64 |
| 95  | DAYS_LAST_PHONE_CHANGE       | float64 |
| 96  | FLAG_DOCUMENT_2              | int64   |
| 97  | FLAG_DOCUMENT_3              | int64   |
| 98  | FLAG_DOCUMENT_4              | int64   |
| 99  | FLAG_DOCUMENT_5              | int64   |
| 100 | FLAG_DOCUMENT_6              | int64   |
| 101 | FLAG_DOCUMENT_7              | int64   |
| 102 | FLAG_DOCUMENT_8              | int64   |

```
103 FLAG_DOCUMENT_9          int64
104 FLAG_DOCUMENT_10         int64
105 FLAG_DOCUMENT_11         int64
106 FLAG_DOCUMENT_12         int64
107 FLAG_DOCUMENT_13         int64
108 FLAG_DOCUMENT_14         int64
109 FLAG_DOCUMENT_15         int64
110 FLAG_DOCUMENT_16         int64
111 FLAG_DOCUMENT_17         int64
112 FLAG_DOCUMENT_18         int64
113 FLAG_DOCUMENT_19         int64
114 FLAG_DOCUMENT_20         int64
115 FLAG_DOCUMENT_21         int64
116 AMT_REQ_CREDIT_BUREAU_HOUR float64
117 AMT_REQ_CREDIT_BUREAU_DAY float64
118 AMT_REQ_CREDIT_BUREAU_WEEK float64
119 AMT_REQ_CREDIT_BUREAU_MON float64
120 AMT_REQ_CREDIT_BUREAU_QRT float64
121 AMT_REQ_CREDIT_BUREAU_YEAR float64
dtypes: float64(65), int64(41), object(16)
memory usage: 286.2+ MB
```

```
In [13]: previousDF.info(verbose=True)
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1670214 entries, 0 to 1670213
Data columns (total 37 columns):
#   Column                                          Non-Null Count  Dtype
---  -
0   SK_ID_PREV                                     1670214 non-null  int64
1   SK_ID_CURR                                     1670214 non-null  int64
2   NAME_CONTRACT_TYPE                           1670214 non-null  object
3   AMT_ANNUITY                                   1297979 non-null  float64
4   AMT_APPLICATION                             1670214 non-null  float64
5   AMT_CREDIT                                   1670213 non-null  float64
6   AMT_DOWN_PAYMENT                             774370 non-null   float64
7   AMT_GOODS_PRICE                             1284699 non-null  float64
8   WEEKDAY_APPR_PROCESS_START                  1670214 non-null  object
9   HOUR_APPR_PROCESS_START                     1670214 non-null  int64
10  FLAG_LAST_APPL_PER_CONTRACT                 1670214 non-null  object
11  NFLAG_LAST_APPL_IN_DAY                     1670214 non-null  int64
12  RATE_DOWN_PAYMENT                           774370 non-null   float64
13  RATE_INTEREST_PRIMARY                       5951 non-null     float64
14  RATE_INTEREST_PRIVILEGED                   5951 non-null     float64
15  NAME_CASH_LOAN_PURPOSE                      1670214 non-null  object
16  NAME_CONTRACT_STATUS                       1670214 non-null  object
17  DAYS_DECISION                               1670214 non-null  int64
18  NAME_PAYMENT_TYPE                           1670214 non-null  object
19  CODE_REJECT_REASON                          1670214 non-null  object
20  NAME_TYPE_SUITE                             849809 non-null   object
21  NAME_CLIENT_TYPE                           1670214 non-null  object
22  NAME_GOODS_CATEGORY                        1670214 non-null  object
23  NAME_PORTFOLIO                             1670214 non-null  object
24  NAME_PRODUCT_TYPE                          1670214 non-null  object
25  CHANNEL_TYPE                               1670214 non-null  object
26  SELLERPLACE_AREA                           1670214 non-null  int64
27  NAME_SELLER_INDUSTRY                       1670214 non-null  object
28  CNT_PAYMENT                                1297984 non-null   float64
29  NAME_YIELD_GROUP                           1670214 non-null  object
30  PRODUCT_COMBINATION                        1669868 non-null  object
31  DAYS_FIRST_DRAWING                         997149 non-null   float64
32  DAYS_FIRST_DUE                             997149 non-null   float64
33  DAYS_LAST_DUE_1ST_VERSION                  997149 non-null   float64
34  DAYS_LAST_DUE                              997149 non-null   float64
35  DAYS_TERMINATION                           997149 non-null   float64
36  NFLAG_INSURED_ON_APPROVAL                  997149 non-null   float64
dtypes: float64(15), int64(6), object(16)
memory usage: 471.5+ MB

```

```

In [14]: # Checking the numeric variables of the dataframes
         applicationDF.describe()

```

| Out[14]: |              | SK_ID_CURR    | TARGET        | CNT_CHILDREN  | AMT_INCOME_TOTAL | AMT   |
|----------|--------------|---------------|---------------|---------------|------------------|-------|
|          | <b>count</b> | 307511.000000 | 307511.000000 | 307511.000000 | 3.075110e+05     | 3.075 |
|          | <b>mean</b>  | 278180.518577 | 0.080729      | 0.417052      | 1.687979e+05     | 5.990 |
|          | <b>std</b>   | 102790.175348 | 0.272419      | 0.722121      | 2.371231e+05     | 4.024 |
|          | <b>min</b>   | 100002.000000 | 0.000000      | 0.000000      | 2.565000e+04     | 4.500 |
|          | <b>25%</b>   | 189145.500000 | 0.000000      | 0.000000      | 1.125000e+05     | 2.700 |
|          | <b>50%</b>   | 278202.000000 | 0.000000      | 0.000000      | 1.471500e+05     | 5.135 |
|          | <b>75%</b>   | 367142.500000 | 0.000000      | 1.000000      | 2.025000e+05     | 8.086 |
|          | <b>max</b>   | 456255.000000 | 1.000000      | 19.000000     | 1.170000e+08     | 4.050 |

In [15]: `previousDF.describe()`

| Out[15]: |              | SK_ID_PREV   | SK_ID_CURR   | AMT_ANNUITY  | AMT_APPLICATION | AMT_CRI   |
|----------|--------------|--------------|--------------|--------------|-----------------|-----------|
|          | <b>count</b> | 1.670214e+06 | 1.670214e+06 | 1.297979e+06 | 1.670214e+06    | 1.670213e |
|          | <b>mean</b>  | 1.923089e+06 | 2.783572e+05 | 1.595512e+04 | 1.752339e+05    | 1.961140e |
|          | <b>std</b>   | 5.325980e+05 | 1.028148e+05 | 1.478214e+04 | 2.927798e+05    | 3.185746e |
|          | <b>min</b>   | 1.000001e+06 | 1.000010e+05 | 0.000000e+00 | 0.000000e+00    | 0.000000e |
|          | <b>25%</b>   | 1.461857e+06 | 1.893290e+05 | 6.321780e+03 | 1.872000e+04    | 2.416050e |
|          | <b>50%</b>   | 1.923110e+06 | 2.787145e+05 | 1.125000e+04 | 7.104600e+04    | 8.054100e |
|          | <b>75%</b>   | 2.384280e+06 | 3.675140e+05 | 2.065842e+04 | 1.803600e+05    | 2.164185e |
|          | <b>max</b>   | 2.845382e+06 | 4.562550e+05 | 4.180581e+05 | 6.905160e+06    | 6.905160e |

In [16]: `%pip install missingno`

Requirement already satisfied: missingno in c:\users\hp\anaconda3\lib\site-packages (0.5.2)  
Requirement already satisfied: numpy in c:\users\hp\anaconda3\lib\site-packages (from missingno) (2.5.0)  
Requirement already satisfied: matplotlib in c:\users\hp\anaconda3\lib\site-packages (from missingno) (3.10.6)  
Requirement already satisfied: scipy in c:\users\hp\anaconda3\lib\site-packages (from missingno) (1.16.3)  
Requirement already satisfied: seaborn in c:\users\hp\anaconda3\lib\site-packages (from missingno) (0.13.2)  
Requirement already satisfied: contourpy>=1.0.1 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (1.3.3)  
Requirement already satisfied: cycler>=0.10 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (0.11.0)  
Requirement already satisfied: fonttools>=4.22.0 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (4.60.1)  
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (1.4.9)  
Requirement already satisfied: packaging>=20.0 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (25.0)  
Requirement already satisfied: pillow>=8 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (12.0.0)  
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (3.2.5)  
Requirement already satisfied: python-dateutil>=2.7 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib->missingno) (2.9.0.post0)  
Requirement already satisfied: six>=1.5 in c:\users\hp\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib->missingno) (1.17.0)  
Requirement already satisfied: pandas>=1.2 in c:\users\hp\anaconda3\lib\site-packages (from seaborn->missingno) (2.3.3)  
Requirement already satisfied: pytz>=2020.1 in c:\users\hp\anaconda3\lib\site-packages (from pandas>=1.2->seaborn->missingno) (2025.2)  
Requirement already satisfied: tzdata>=2022.7 in c:\users\hp\anaconda3\lib\site-packages (from pandas>=1.2->seaborn->missingno) (2025.2)  
Note: you may need to restart the kernel to use updated packages.

```
In [17]: import pandas as pd  
applicationDF = pd.read_csv(r'C:\Users\HP\Downloads\application_data.csv')
```

```
In [18]: applicationDF.head()
```

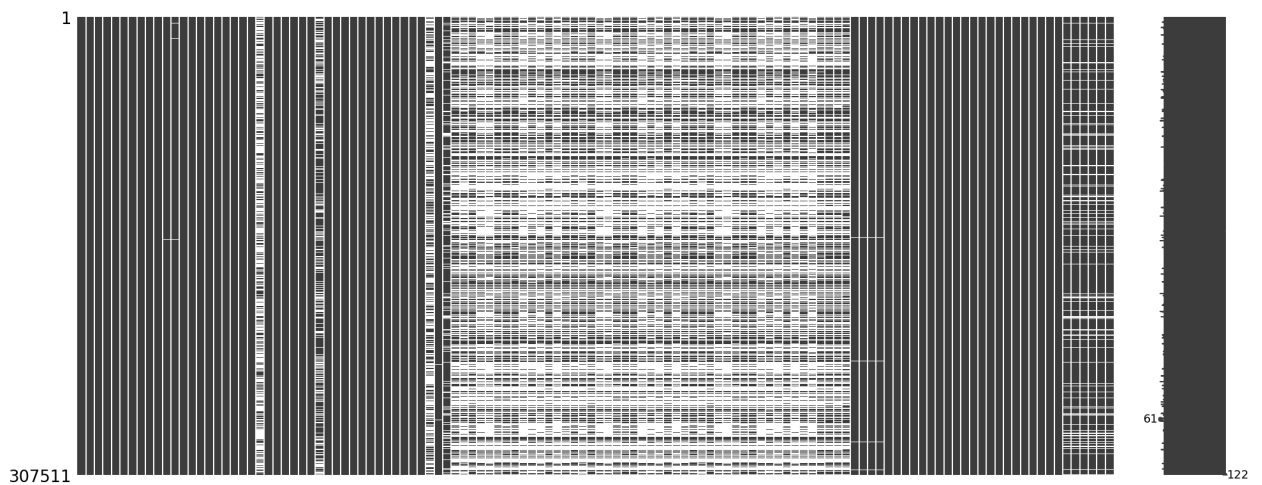


```
Out[18]:
```

|   | SK_ID_CURR | TARGET | NAME_CONTRACT_TYPE | CODE_GENDER | FLAG_OWN_CAR |
|---|------------|--------|--------------------|-------------|--------------|
| 0 | 100002     | 1      | Cash loans         | M           | 1            |
| 1 | 100003     | 0      | Cash loans         | F           | 1            |
| 2 | 100004     | 0      | Revolving loans    | M           | 1            |
| 3 | 100006     | 0      | Cash loans         | F           | 1            |
| 4 | 100007     | 0      | Cash loans         | M           | 1            |

```
In [19]: import missingno as mn
mn.matrix(applicationDF)
```

```
Out[19]: <Axes: >
```



```
In [20]: # % null value in each column
round(applicationDF.isnull().sum() / applicationDF.shape[0] * 100.00,2)
```

```

Out[20]: SK_ID_CURR      0.00
          TARGET         0.00
          NAME_CONTRACT_TYPE 0.00
          CODE_GENDER     0.00
          FLAG_OWN_CAR     0.00
          FLAG_OWN_REALTY  0.00
          CNT_CHILDREN     0.00
          AMT_INCOME_TOTAL 0.00
          AMT_CREDIT       0.00
          AMT_ANNUITY      0.00
          AMT_GOODS_PRICE  0.09
          NAME_TYPE_SUITE   0.42
          NAME_INCOME_TYPE  0.00
          NAME_EDUCATION_TYPE 0.00
          NAME_FAMILY_STATUS 0.00
          NAME_HOUSING_TYPE  0.00
          REGION_POPULATION_RELATIVE 0.00
          DAYS_BIRTH       0.00
          DAYS_EMPLOYED    0.00
          DAYS_REGISTRATION 0.00
          DAYS_ID_PUBLISH  0.00
          OWN_CAR_AGE      65.99
          FLAG_MOBIL       0.00
          FLAG_EMP_PHONE   0.00
          FLAG_WORK_PHONE  0.00
          FLAG_CONT_MOBILE  0.00
          FLAG_PHONE       0.00
          FLAG_EMAIL       0.00
          OCCUPATION_TYPE  31.35
          CNT_FAM_MEMBERS  0.00
          REGION_RATING_CLIENT 0.00
          REGION_RATING_CLIENT_W_CITY 0.00
          WEEKDAY_APPR_PROCESS_START 0.00
          HOUR_APPR_PROCESS_START 0.00
          REG_REGION_NOT_LIVE_REGION 0.00
          REG_REGION_NOT_WORK_REGION 0.00
          LIVE_REGION_NOT_WORK_REGION 0.00
          REG_CITY_NOT_LIVE_CITY 0.00
          REG_CITY_NOT_WORK_CITY 0.00
          LIVE_CITY_NOT_WORK_CITY 0.00
          ORGANIZATION_TYPE 0.00
          EXT_SOURCE_1     56.38
          EXT_SOURCE_2     0.21
          EXT_SOURCE_3     19.83
          APARTMENTS_AVG   50.75
          BASEMENTAREA_AVG  58.52
          YEARS_BEGINEXPLUATATION_AVG 48.78
          YEARS_BUILD_AVG  66.50
          COMMONAREA_AVG   69.87
          ELEVATORS_AVG    53.30
          ENTRANCES_AVG    50.35
          FLOORSMAX_AVG    49.76
          FLOORSMIN_AVG    67.85
          LANDAREA_AVG     59.38

```

|                              |       |
|------------------------------|-------|
| LIVINGAPARTMENTS_AVG         | 68.35 |
| LIVINGAREA_AVG               | 50.19 |
| NONLIVINGAPARTMENTS_AVG      | 69.43 |
| NONLIVINGAREA_AVG            | 55.18 |
| APARTMENTS_MODE              | 50.75 |
| BASEMENTAREA_MODE            | 58.52 |
| YEARS_BEGINEXPLUATATION_MODE | 48.78 |
| YEARS_BUILD_MODE             | 66.50 |
| COMMONAREA_MODE              | 69.87 |
| ELEVATORS_MODE               | 53.30 |
| ENTRANCES_MODE               | 50.35 |
| FLOORSMAX_MODE               | 49.76 |
| FLOORSMIN_MODE               | 67.85 |
| LANDAREA_MODE                | 59.38 |
| LIVINGAPARTMENTS_MODE        | 68.35 |
| LIVINGAREA_MODE              | 50.19 |
| NONLIVINGAPARTMENTS_MODE     | 69.43 |
| NONLIVINGAREA_MODE           | 55.18 |
| APARTMENTS_MEDI              | 50.75 |
| BASEMENTAREA_MEDI            | 58.52 |
| YEARS_BEGINEXPLUATATION_MEDI | 48.78 |
| YEARS_BUILD_MEDI             | 66.50 |
| COMMONAREA_MEDI              | 69.87 |
| ELEVATORS_MEDI               | 53.30 |
| ENTRANCES_MEDI               | 50.35 |
| FLOORSMAX_MEDI               | 49.76 |
| FLOORSMIN_MEDI               | 67.85 |
| LANDAREA_MEDI                | 59.38 |
| LIVINGAPARTMENTS_MEDI        | 68.35 |
| LIVINGAREA_MEDI              | 50.19 |
| NONLIVINGAPARTMENTS_MEDI     | 69.43 |
| NONLIVINGAREA_MEDI           | 55.18 |
| FONDKAPREMONT_MODE           | 68.39 |
| HOUSETYPE_MODE               | 50.18 |
| TOTALAREA_MODE               | 48.27 |
| WALLSMATERIAL_MODE           | 50.84 |
| EMERGENCYSTATE_MODE          | 47.40 |
| OBS_30_CNT_SOCIAL_CIRCLE     | 0.33  |
| DEF_30_CNT_SOCIAL_CIRCLE     | 0.33  |
| OBS_60_CNT_SOCIAL_CIRCLE     | 0.33  |
| DEF_60_CNT_SOCIAL_CIRCLE     | 0.33  |
| DAYS_LAST_PHONE_CHANGE       | 0.00  |
| FLAG_DOCUMENT_2              | 0.00  |
| FLAG_DOCUMENT_3              | 0.00  |
| FLAG_DOCUMENT_4              | 0.00  |
| FLAG_DOCUMENT_5              | 0.00  |
| FLAG_DOCUMENT_6              | 0.00  |
| FLAG_DOCUMENT_7              | 0.00  |
| FLAG_DOCUMENT_8              | 0.00  |
| FLAG_DOCUMENT_9              | 0.00  |
| FLAG_DOCUMENT_10             | 0.00  |
| FLAG_DOCUMENT_11             | 0.00  |
| FLAG_DOCUMENT_12             | 0.00  |
| FLAG_DOCUMENT_13             | 0.00  |

|                            |       |
|----------------------------|-------|
| FLAG_DOCUMENT_14           | 0.00  |
| FLAG_DOCUMENT_15           | 0.00  |
| FLAG_DOCUMENT_16           | 0.00  |
| FLAG_DOCUMENT_17           | 0.00  |
| FLAG_DOCUMENT_18           | 0.00  |
| FLAG_DOCUMENT_19           | 0.00  |
| FLAG_DOCUMENT_20           | 0.00  |
| FLAG_DOCUMENT_21           | 0.00  |
| AMT_REQ_CREDIT_BUREAU_HOUR | 13.50 |
| AMT_REQ_CREDIT_BUREAU_DAY  | 13.50 |
| AMT_REQ_CREDIT_BUREAU_WEEK | 13.50 |
| AMT_REQ_CREDIT_BUREAU_MON  | 13.50 |
| AMT_REQ_CREDIT_BUREAU_QRT  | 13.50 |
| AMT_REQ_CREDIT_BUREAU_YEAR | 13.50 |

dtype: float64

```
In [21]: import matplotlib.pyplot as plt
import seaborn as sns
```

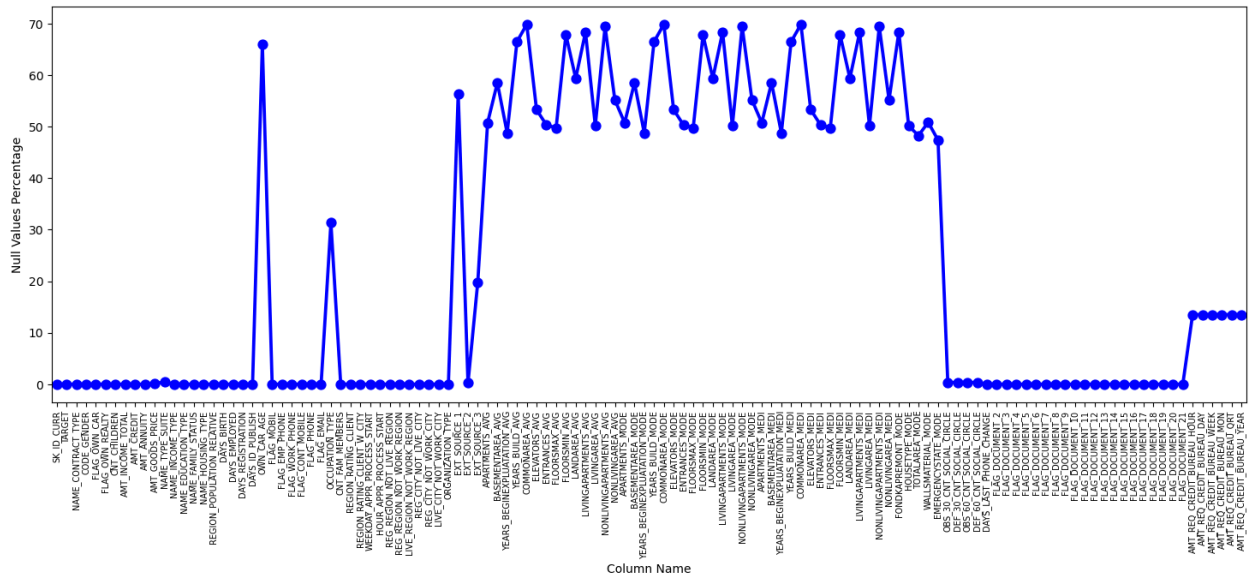
```
In [22]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

null_applicationDF = pd.DataFrame(
    (applicationDF.isnull().sum()) * 100 / applicationDF.shape[0]
).reset_index()

null_applicationDF.columns = ['Column Name', 'Null Values Percentage']

fig = plt.figure(figsize=(18,6))
ax = sns.pointplot(
    x="Column Name",
    y="Null Values Percentage",
    data=null_applicationDF,
    color='blue'
)

plt.xticks(rotation=90, fontsize=7)
plt.show()
```



```
In [23]: # more than or equal to 40% empty rows columns
nullcol_40_application = null_applicationDF[null_applicationDF["Null Values Percentage"] >= 40]
nullcol_40_application
```

Out[23]:

|    | Column Name                  | Null Values Percentage |
|----|------------------------------|------------------------|
| 21 | OWN_CAR_AGE                  | 65.990810              |
| 41 | EXT_SOURCE_1                 | 56.381073              |
| 44 | APARTMENTS_AVG               | 50.749729              |
| 45 | BASEMENTAREA_AVG             | 58.515956              |
| 46 | YEARS_BEGINEXPLUATATION_AVG  | 48.781019              |
| 47 | YEARS_BUILD_AVG              | 66.497784              |
| 48 | COMMONAREA_AVG               | 69.872297              |
| 49 | ELEVATORS_AVG                | 53.295980              |
| 50 | ENTRANCES_AVG                | 50.348768              |
| 51 | FLOORSMAX_AVG                | 49.760822              |
| 52 | FLOORSMIN_AVG                | 67.848630              |
| 53 | LANDAREA_AVG                 | 59.376738              |
| 54 | LIVINGAPARTMENTS_AVG         | 68.354953              |
| 55 | LIVINGAREA_AVG               | 50.193326              |
| 56 | NONLIVINGAPARTMENTS_AVG      | 69.432963              |
| 57 | NONLIVINGAREA_AVG            | 55.179164              |
| 58 | APARTMENTS_MODE              | 50.749729              |
| 59 | BASEMENTAREA_MODE            | 58.515956              |
| 60 | YEARS_BEGINEXPLUATATION_MODE | 48.781019              |
| 61 | YEARS_BUILD_MODE             | 66.497784              |
| 62 | COMMONAREA_MODE              | 69.872297              |
| 63 | ELEVATORS_MODE               | 53.295980              |
| 64 | ENTRANCES_MODE               | 50.348768              |
| 65 | FLOORSMAX_MODE               | 49.760822              |
| 66 | FLOORSMIN_MODE               | 67.848630              |
| 67 | LANDAREA_MODE                | 59.376738              |
| 68 | LIVINGAPARTMENTS_MODE        | 68.354953              |
| 69 | LIVINGAREA_MODE              | 50.193326              |
| 70 | NONLIVINGAPARTMENTS_MODE     | 69.432963              |
| 71 | NONLIVINGAREA_MODE           | 55.179164              |
| 72 | APARTMENTS_MEDI              | 50.749729              |

|    | Column Name                  | Null Values Percentage |
|----|------------------------------|------------------------|
| 73 | BASEMENTAREA_MEDI            | 58.515956              |
| 74 | YEARS_BEGINEXPLUATATION_MEDI | 48.781019              |
| 75 | YEARS_BUILD_MEDI             | 66.497784              |
| 76 | COMMONAREA_MEDI              | 69.872297              |
| 77 | ELEVATORS_MEDI               | 53.295980              |
| 78 | ENTRANCES_MEDI               | 50.348768              |
| 79 | FLOORSMAX_MEDI               | 49.760822              |
| 80 | FLOORSMIN_MEDI               | 67.848630              |
| 81 | LANDAREA_MEDI                | 59.376738              |
| 82 | LIVINGAPARTMENTS_MEDI        | 68.354953              |
| 83 | LIVINGAREA_MEDI              | 50.193326              |
| 84 | NONLIVINGAPARTMENTS_MEDI     | 69.432963              |
| 85 | NONLIVINGAREA_MEDI           | 55.179164              |
| 86 | FONDKAPREMONT_MODE           | 68.386172              |
| 87 | HOUSETYPE_MODE               | 50.176091              |
| 88 | TOTALAREA_MODE               | 48.268517              |
| 89 | WALLSMATERIAL_MODE           | 50.840783              |
| 90 | EMERGENCYSTATE_MODE          | 47.398304              |

```
In [24]: # How many columns have more than or euqal to 40% null values ?
len(nullcol_40_application)
```

Out[24]: 49

```
In [25]: import pandas as pd
previousDF = pd.read_csv(r'C:\Users\HP\Downloads/previous_application.csv')
```

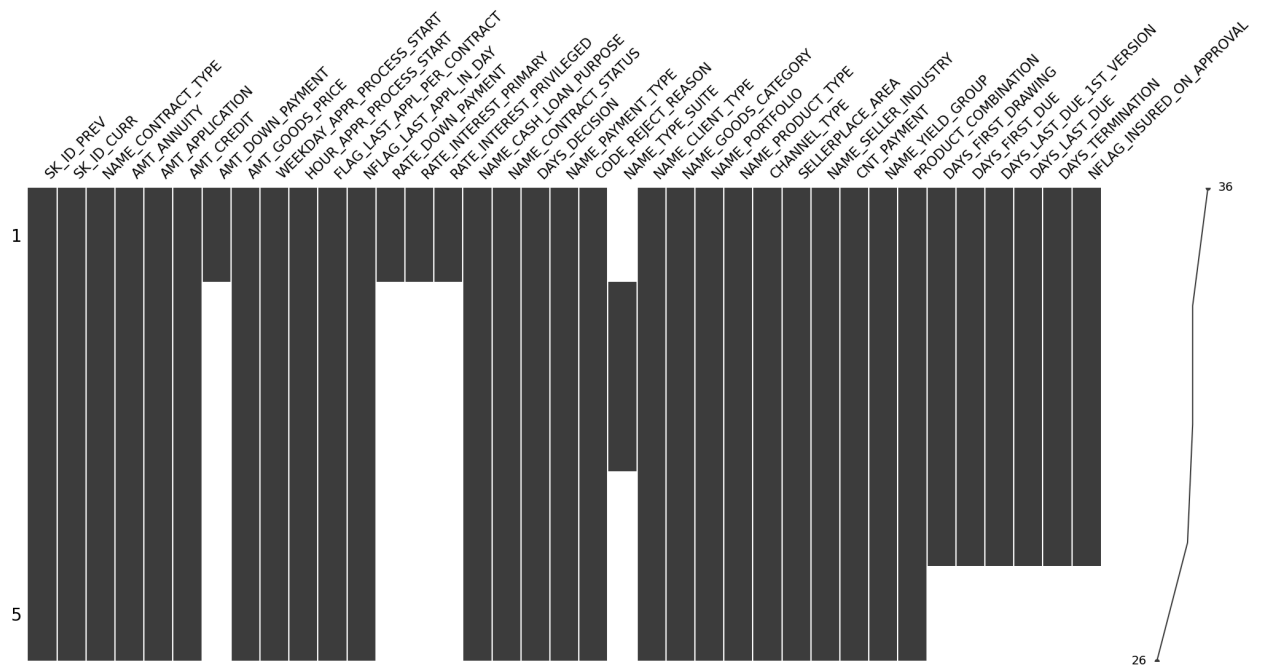
```
In [26]: previousDF.head()
```

Out[26]:

|   | SK_ID_PREV | SK_ID_CURR | NAME_CONTRACT_TYPE | AMT_ANNUITY | AMT_APPLI |
|---|------------|------------|--------------------|-------------|-----------|
| 0 | 2030495    | 271877     | Consumer loans     | 1730.430    | :         |
| 1 | 2802425    | 108129     | Cash loans         | 25188.615   | 60        |
| 2 | 2523466    | 122040     | Cash loans         | 15060.735   | 1:        |
| 3 | 2819243    | 176158     | Cash loans         | 47041.335   | 4:        |
| 4 | 1784265    | 202054     | Cash loans         | 31924.395   | 3:        |

```
In [27]: import missingno as mn
mn.matrix(previousDF.head())
```

Out[27]: <Axes: >



```
In [28]: # checking the null value % of each column in previousDF dataframe
round(previousDF.isnull().sum() / previousDF.shape[0] * 100.00,2)
```



```

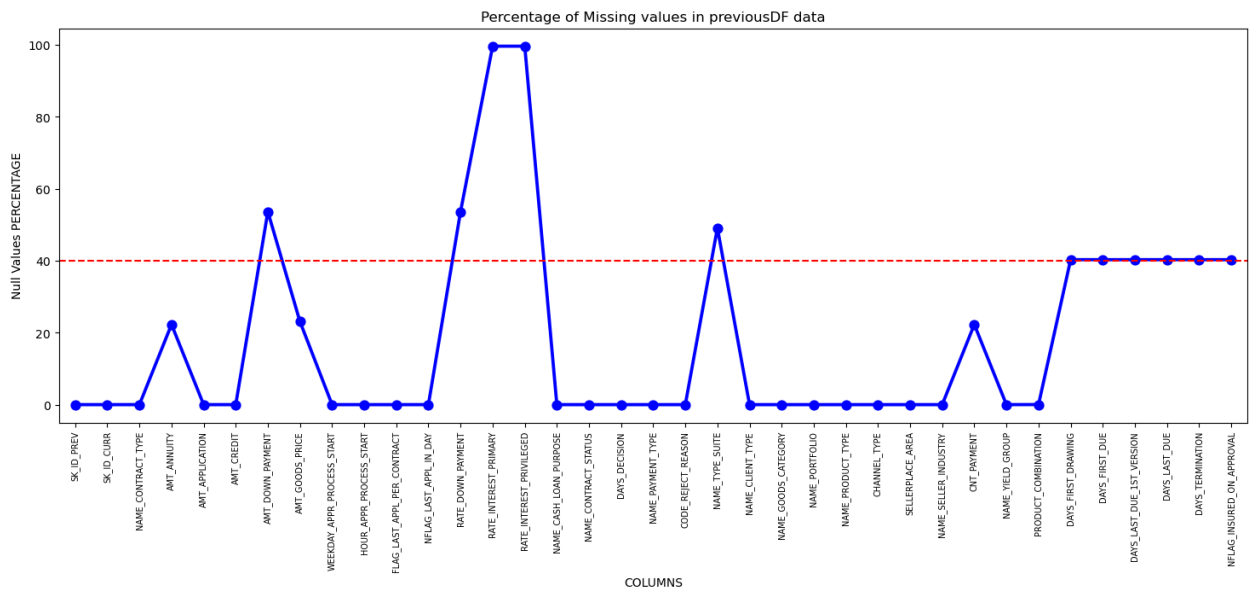
Out[28]: SK_ID_PREV                0.00
          SK_ID_CURR                0.00
          NAME_CONTRACT_TYPE        0.00
          AMT_ANNUITY               22.29
          AMT_APPLICATION            0.00
          AMT_CREDIT                 0.00
          AMT_DOWN_PAYMENT           53.64
          AMT_GOODS_PRICE            23.08
          WEEKDAY_APPR_PROCESS_START 0.00
          HOUR_APPR_PROCESS_START    0.00
          FLAG_LAST_APPL_PER_CONTRACT 0.00
          NFLAG_LAST_APPL_IN_DAY     0.00
          RATE_DOWN_PAYMENT           53.64
          RATE_INTEREST_PRIMARY       99.64
          RATE_INTEREST_PRIVILEGED    99.64
          NAME_CASH_LOAN_PURPOSE      0.00
          NAME_CONTRACT_STATUS        0.00
          DAYS_DECISION               0.00
          NAME_PAYMENT_TYPE           0.00
          CODE_REJECT_REASON          0.00
          NAME_TYPE_SUITE             49.12
          NAME_CLIENT_TYPE            0.00
          NAME_GOODS_CATEGORY         0.00
          NAME_PORTFOLIO              0.00
          NAME_PRODUCT_TYPE           0.00
          CHANNEL_TYPE                0.00
          SELLERPLACE_AREA            0.00
          NAME_SELLER_INDUSTRY        0.00
          CNT_PAYMENT                22.29
          NAME_YIELD_GROUP            0.00
          PRODUCT_COMBINATION         0.02
          DAYS_FIRST_DRAWING           40.30
          DAYS_FIRST_DUE              40.30
          DAYS_LAST_DUE_1ST_VERSION   40.30
          DAYS_LAST_DUE              40.30
          DAYS_TERMINATION            40.30
          NFLAG_INSURED_ON_APPROVAL   40.30
          dtype: float64

```

```

In [29]: null_previousDF = pd.DataFrame((previousDF.isnull().sum()*100/previousDF.shape[0]),
          null_previousDF.columns = ['Column Name', 'Null Values Percentage']
          fig = plt.figure(figsize=(18,6))
          ax = sns.pointplot(x="Column Name",y="Null Values Percentage",data=null_previousDF)
          plt.xticks(rotation =90,fontsize =7)
          ax.axhline(40, ls='--',color='red')
          plt.title("Percentage of Missing values in previousDF data")
          plt.ylabel("Null Values PERCENTAGE")
          plt.xlabel("COLUMNS")
          plt.show()

```



```
In [30]: # more than or equal to 40% empty rows columns
nullcol_40_previous = null_previousDF[null_previousDF["Null Values Percentage"]
nullcol_40_previous
```

Out[30]:

|    | Column Name               | Null Values Percentage |
|----|---------------------------|------------------------|
| 6  | AMT_DOWN_PAYMENT          | 53.636480              |
| 12 | RATE_DOWN_PAYMENT         | 53.636480              |
| 13 | RATE_INTEREST_PRIMARY     | 99.643698              |
| 14 | RATE_INTEREST_PRIVILEGED  | 99.643698              |
| 20 | NAME_TYPE_SUITE           | 49.119754              |
| 31 | DAYS_FIRST_DRAWING        | 40.298129              |
| 32 | DAYS_FIRST_DUE            | 40.298129              |
| 33 | DAYS_LAST_DUE_1ST_VERSION | 40.298129              |
| 34 | DAYS_LAST_DUE             | 40.298129              |
| 35 | DAYS_TERMINATION          | 40.298129              |
| 36 | NFLAG_INSURED_ON_APPROVAL | 40.298129              |

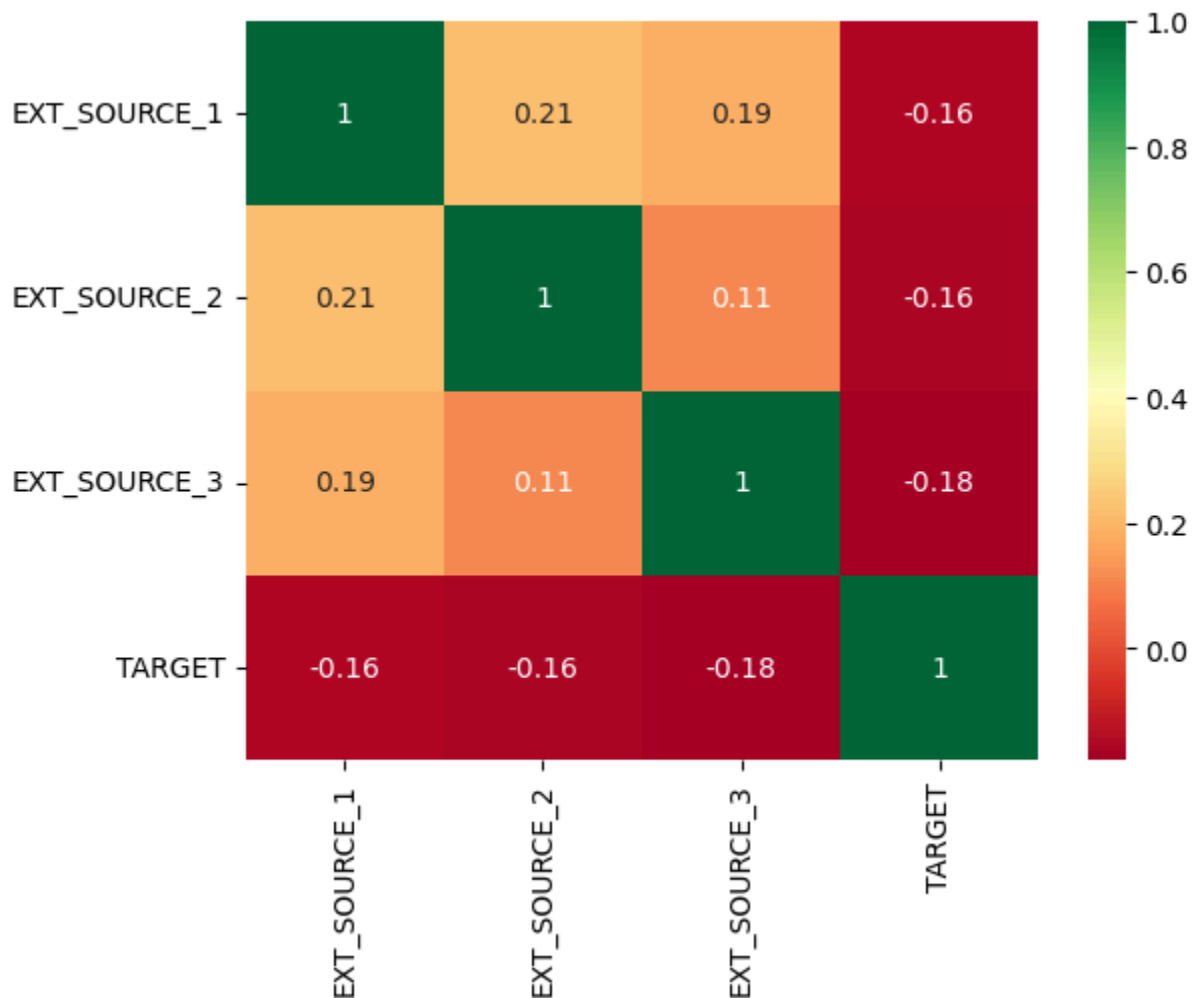
|    | Column Name               | Null Values Percentage |
|----|---------------------------|------------------------|
| 6  | AMT_DOWN_PAYMENT          | 53.636480              |
| 12 | RATE_DOWN_PAYMENT         | 53.636480              |
| 13 | RATE_INTEREST_PRIMARY     | 99.643698              |
| 14 | RATE_INTEREST_PRIVILEGED  | 99.643698              |
| 20 | NAME_TYPE_SUITE           | 49.119754              |
| 31 | DAYS_FIRST_DRAWING        | 40.298129              |
| 32 | DAYS_FIRST_DUE            | 40.298129              |
| 33 | DAYS_LAST_DUE_1ST_VERSION | 40.298129              |
| 34 | DAYS_LAST_DUE             | 40.298129              |
| 35 | DAYS_TERMINATION          | 40.298129              |
| 36 | NFLAG_INSURED_ON_APPROVAL | 40.298129              |

```
In [31]: # How many columns have more than or euqal to 40% null values ?
len(nullcol_40_previous)
```

Out[31]: 11

```
In [32]: # Checking correlation of EXT_SOURCE_X columns vs TARGET column
Source = applicationDF[["EXT_SOURCE_1","EXT_SOURCE_2","EXT_SOURCE_3","TARGET"]]
source_corr = Source.corr()
ax = sns.heatmap(source_corr,
```

```
xticklabels=source_corr.columns,
yticklabels=source_corr.columns,
annot = True,
cmap = "RdYlGn")
```



```
In [33]: # create a list of columns that needs to be dropped including the columns with
Unwanted_application = nullcol_40_application["Column Name"].tolist() + ['EXT_S
# as EXT_SOURCE_1 column is already included in nullcol_40_application
len(Unwanted_application)
```

Out[33]: 51

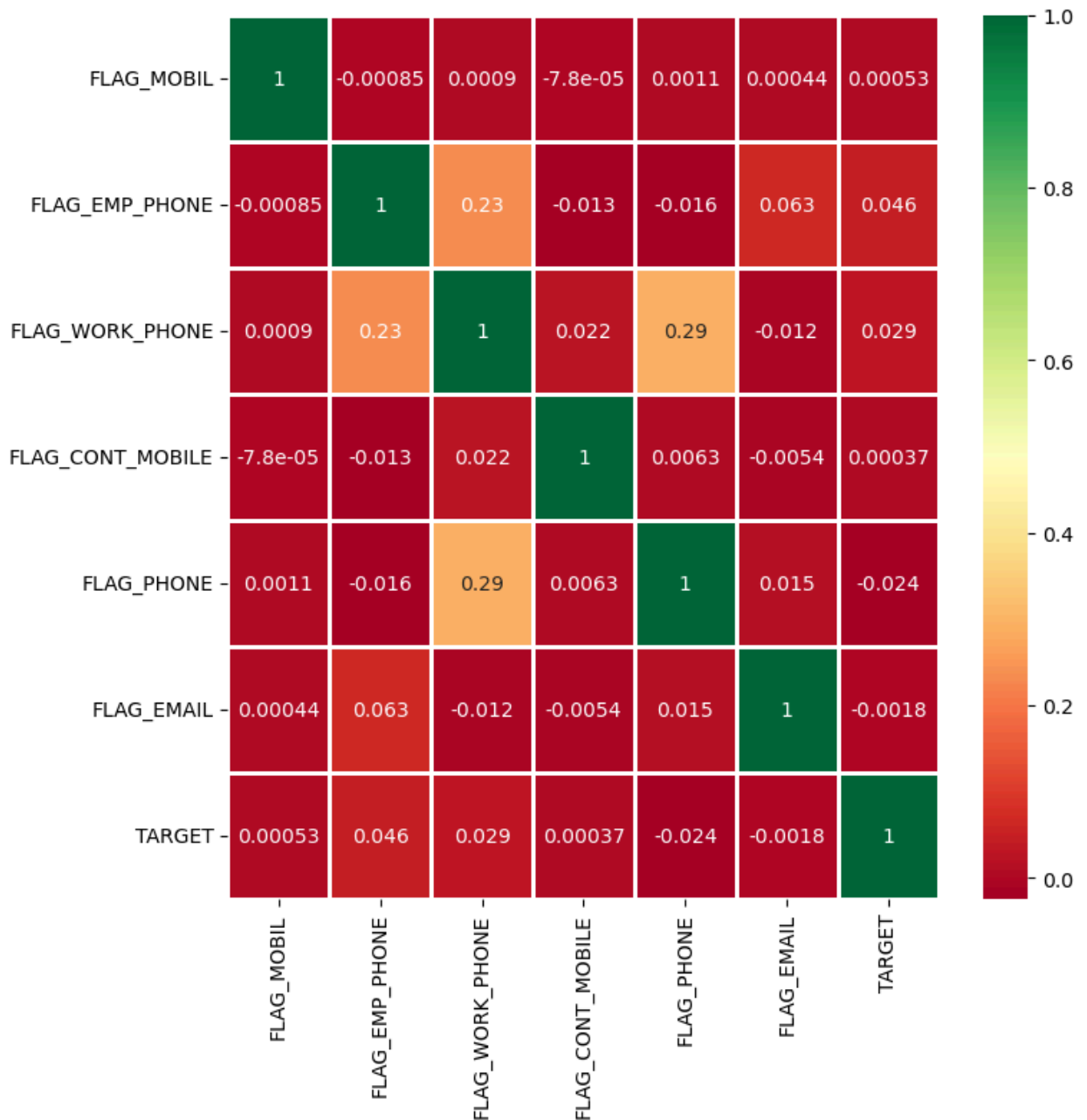
```
In [34]: import pandas as pd
applicationDF = pd.read_csv(r'C:\Users\HP\Downloads\application_data.csv')
```

```
In [35]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import itertools
import seaborn as sns
```

```
In [36]: fig = plt.figure(figsize=(21,24))
```

<Figure size 2100x2400 with 0 Axes>

```
In [37]: # checking is there is any correlation between mobile phone, work phone etc, e
contact_col = ['FLAG_MOBIL', 'FLAG_EMP_PHONE', 'FLAG_WORK_PHONE', 'FLAG_CONT_M
            'FLAG_PHONE', 'FLAG_EMAIL', 'TARGET']
Contact_corr = applicationDF[contact_col].corr()
fig = plt.figure(figsize=(8,8))
ax = sns.heatmap(Contact_corr,
                  xticklabels=Contact_corr.columns,
                  yticklabels=Contact_corr.columns,
                  annot = True,
                  cmap = "RdYlGn",
                  linewidth=1)
```



```
In [38]: # Inspecting the dataframe after removal of unnecessary columns
```

```
applicationDF.shape
```

```
Out[38]: (307511, 122)
```

```
In [39]: # inspecting the column types after removal of unnecessary columns
applicationDF.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 307511 entries, 0 to 307510
Columns: 122 entries, SK_ID_CURR to AMT_REQ_CREDIT_BUREAU_YEAR
dtypes: float64(65), int64(41), object(16)
memory usage: 286.2+ MB
```

```
In [40]: import pandas as pd
previousDF = pd.read_csv(r'C:\Users\HP\Downloads/previous_application.csv')
```

```
In [41]: previousDF.head()
```

```
Out[41]:
```

|   | SK_ID_PREV | SK_ID_CURR | NAME_CONTRACT_TYPE | AMT_ANNUITY | AMT_APPLI |
|---|------------|------------|--------------------|-------------|-----------|
| 0 | 2030495    | 271877     | Consumer loans     | 1730.430    | :         |
| 1 | 2802425    | 108129     | Cash loans         | 25188.615   | 60        |
| 2 | 2523466    | 122040     | Cash loans         | 15060.735   | 1:        |
| 3 | 2819243    | 176158     | Cash loans         | 47041.335   | 4:        |
| 4 | 1784265    | 202054     | Cash loans         | 31924.395   | 3:        |

```
In [42]: Unnecessary_previous = ['WEEKDAY_APPR_PROCESS_START', 'HOUR_APPR_PROCESS_START',
                                'FLAG_LAST_APPL_PER_CONTRACT', 'NFLAG_LAST_APPL_IN_DAY']
```

```
In [43]: date_col = ['DAYS_BIRTH', 'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH']

for col in date_col:
    applicationDF[col] = abs(applicationDF[col])
```

```
In [44]: # Binning Numerical Columns to create a categorical column

# Creating bins for income amount
applicationDF['AMT_INCOME_TOTAL'] = applicationDF['AMT_INCOME_TOTAL'] / 100000

bins = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
slot = ['0-100k', '100K-200K', '200k-300k', '300k-400k', '400k-500k', '500k-600k',
        '600k-700k', '700k-800k', '800k-900k', '900k-1000k', '1000k-1100k', '1100k-1200k']

applicationDF['AMT_INCOME_RANGE'] = pd.cut(applicationDF['AMT_INCOME_TOTAL'], bins, labels=slot)
```

```
In [45]: applicationDF['AMT_INCOME_RANGE'].value_counts(normalize=True) * 100
```

```
Out[45]: AMT_INCOME_RANGE
100K-200K    50.735000
200k-300k    21.210691
0-100K       20.729695
300k-400k     4.776116
400k-500k     1.744669
500k-600k     0.356354
600k-700k     0.282805
800k-900k     0.096980
700k-800k     0.052721
900k-1M       0.009112
1M Above      0.005858
Name: proportion, dtype: float64
```

```
In [46]: # Creating bins for Credit amount
applicationDF['AMT_CREDIT'] = applicationDF['AMT_CREDIT'] / 100000

bins = [0,1,2,3,4,5,6,7,8,9,10,100]
slots = ['0-100K', '100K-200K', '200k-300k', '300k-400k', '400k-500k', '500k-600k',
         '800k-900k', '900k-1M', '1M Above']

applicationDF['AMT_CREDIT_RANGE'] = pd.cut(applicationDF['AMT_CREDIT'], bins=bins
```

```
In [47]: #checking the binning of data and % of data in each category
applicationDF['AMT_CREDIT_RANGE'].value_counts(normalize=True)*100
```

```
Out[47]: AMT_CREDIT_RANGE
200k-300k    17.824728
1M Above     16.254703
500k-600k    11.131960
400k-500k    10.418489
100K-200K     9.801275
300k-400k     8.564897
600k-700k     7.820533
800k-900k     7.086576
700k-800k     6.241403
900k-1M       2.902986
0-100K        1.952450
Name: proportion, dtype: float64
```

```
In [48]: applicationDF['AGE'] = applicationDF['DAYS_BIRTH'] // 365
bins = [0,20,30,40,50,100]
slots = ['0-20', '20-30', '30-40', '40-50', '50 above']

applicationDF['AGE_GROUP'] = pd.cut(applicationDF['AGE'], bins=bins, labels=slots)
```

```
In [49]: #checking the binning of data and % of data in each category
applicationDF['AGE_GROUP'].value_counts(normalize=True)*100
```

```
Out[49]: AGE_GROUP
50 above    31.604398
30-40       27.028952
40-50       24.194582
20-30       17.171743
0-20        0.000325
Name: proportion, dtype: float64
```

```
In [50]: # Creating bins for Employment Time
applicationDF['YEARS_EMPLOYED'] = applicationDF['DAYS_EMPLOYED'] // 365
bins = [0,5,10,20,30,40,50,60,150]
slots = ['0-5','5-10','10-20','20-30','30-40','40-50','50-60','60 above']

applicationDF['EMPLOYMENT_YEAR']=pd.cut(applicationDF['YEARS_EMPLOYED'],bins=b
```

```
In [51]: #checking the binning of data and % of data in each category
applicationDF['EMPLOYMENT_YEAR'].value_counts(normalize=True)*100
```

```
Out[51]: EMPLOYMENT_YEAR
0-5      55.582363
5-10     24.966441
10-20    14.564315
20-30     3.750117
30-40     1.058720
40-50     0.078044
50-60     0.000000
60 above  0.000000
Name: proportion, dtype: float64
```

```
In [52]: #Checking the number of unique values each column possess to identify categori
applicationDF.nunique().sort_values()
```

|          |                             |   |
|----------|-----------------------------|---|
| Out[52]: | TARGET                      | 2 |
|          | NAME_CONTRACT_TYPE          | 2 |
|          | FLAG_OWN_REALTY             | 2 |
|          | FLAG_OWN_CAR                | 2 |
|          | FLAG_WORK_PHONE             | 2 |
|          | FLAG_CONT_MOBILE            | 2 |
|          | FLAG_EMP_PHONE              | 2 |
|          | FLAG_MOBIL                  | 2 |
|          | FLAG_PHONE                  | 2 |
|          | FLAG_EMAIL                  | 2 |
|          | LIVE_CITY_NOT_WORK_CITY     | 2 |
|          | REG_CITY_NOT_WORK_CITY      | 2 |
|          | LIVE_REGION_NOT_WORK_REGION | 2 |
|          | REG_CITY_NOT_LIVE_CITY      | 2 |
|          | REG_REGION_NOT_LIVE_REGION  | 2 |
|          | REG_REGION_NOT_WORK_REGION  | 2 |
|          | FLAG_DOCUMENT_18            | 2 |
|          | FLAG_DOCUMENT_19            | 2 |
|          | FLAG_DOCUMENT_20            | 2 |
|          | FLAG_DOCUMENT_21            | 2 |
|          | FLAG_DOCUMENT_6             | 2 |
|          | FLAG_DOCUMENT_7             | 2 |
|          | FLAG_DOCUMENT_8             | 2 |
|          | FLAG_DOCUMENT_9             | 2 |
|          | FLAG_DOCUMENT_10            | 2 |
|          | FLAG_DOCUMENT_11            | 2 |
|          | FLAG_DOCUMENT_12            | 2 |
|          | FLAG_DOCUMENT_13            | 2 |
|          | FLAG_DOCUMENT_14            | 2 |
|          | FLAG_DOCUMENT_15            | 2 |
|          | FLAG_DOCUMENT_16            | 2 |
|          | EMERGENCYSTATE_MODE         | 2 |
|          | FLAG_DOCUMENT_2             | 2 |
|          | FLAG_DOCUMENT_3             | 2 |
|          | FLAG_DOCUMENT_4             | 2 |
|          | FLAG_DOCUMENT_5             | 2 |
|          | FLAG_DOCUMENT_17            | 2 |
|          | CODE_GENDER                 | 3 |
|          | REGION_RATING_CLIENT_W_CITY | 3 |
|          | REGION_RATING_CLIENT        | 3 |
|          | HOUSETYPE_MODE              | 3 |
|          | FONDKAPREMONT_MODE          | 4 |
|          | AMT_REQ_CREDIT_BUREAU_HOUR  | 5 |
|          | NAME_EDUCATION_TYPE         | 5 |
|          | AGE_GROUP                   | 5 |
|          | NAME_FAMILY_STATUS          | 6 |
|          | EMPLOYMENT_YEAR             | 6 |
|          | NAME_HOUSING_TYPE           | 6 |
|          | NAME_TYPE_SUITE             | 7 |
|          | WALLSMATERIAL_MODE          | 7 |
|          | WEEKDAY_APPR_PROCESS_START  | 7 |
|          | NAME_INCOME_TYPE            | 8 |
|          | AMT_REQ_CREDIT_BUREAU_DAY   | 9 |
|          | DEF_60_CNT_SOCIAL_CIRCLE    | 9 |



|                              |      |
|------------------------------|------|
| AMT_REQ_CREDIT_BUREAU_WEEK   | 9    |
| DEF_30_CNT_SOCIAL_CIRCLE     | 10   |
| AMT_CREDIT_RANGE             | 11   |
| AMT_REQ_CREDIT_BUREAU_QRT    | 11   |
| AMT_INCOME_RANGE             | 11   |
| CNT_CHILDREN                 | 15   |
| CNT_FAM_MEMBERS              | 17   |
| OCCUPATION_TYPE              | 18   |
| HOURLY_APPR_PROCESS_START    | 24   |
| AMT_REQ_CREDIT_BUREAU_MON    | 24   |
| FLOORSMIN_MODE               | 25   |
| AMT_REQ_CREDIT_BUREAU_YEAR   | 25   |
| FLOORSMAX_MODE               | 25   |
| ELEVATORS_MODE               | 26   |
| ENTRANCES_MODE               | 30   |
| OBS_30_CNT_SOCIAL_CIRCLE     | 33   |
| OBS_60_CNT_SOCIAL_CIRCLE     | 33   |
| ENTRANCES_MEDI               | 46   |
| ELEVATORS_MEDI               | 46   |
| FLOORSMIN_MEDI               | 47   |
| FLOORSMAX_MEDI               | 49   |
| AGE                          | 50   |
| YEARS_EMPLOYED               | 51   |
| ORGANIZATION_TYPE            | 58   |
| OWN_CAR_AGE                  | 62   |
| REGION_POPULATION_RELATIVE   | 81   |
| YEARS_BUILD_AVG              | 149  |
| YEARS_BUILD_MEDI             | 151  |
| YEARS_BUILD_MODE             | 154  |
| NONLIVINGAPARTMENTS_MODE     | 167  |
| NONLIVINGAPARTMENTS_MEDI     | 214  |
| YEARS_BEGINEXPLUATATION_MODE | 221  |
| YEARS_BEGINEXPLUATATION_MEDI | 245  |
| ELEVATORS_AVG                | 257  |
| YEARS_BEGINEXPLUATATION_AVG  | 285  |
| ENTRANCES_AVG                | 285  |
| FLOORSMIN_AVG                | 305  |
| NONLIVINGAPARTMENTS_AVG      | 386  |
| FLOORSMAX_AVG                | 403  |
| LIVINGAPARTMENTS_MODE        | 736  |
| APARTMENTS_MODE              | 760  |
| EXT_SOURCE_3                 | 814  |
| AMT_GOODS_PRICE              | 1002 |
| LIVINGAPARTMENTS_MEDI        | 1097 |
| APARTMENTS_MEDI              | 1148 |
| LIVINGAPARTMENTS_AVG         | 1868 |
| APARTMENTS_AVG               | 2339 |
| AMT_INCOME_TOTAL             | 2548 |
| COMMONAREA_MODE              | 3128 |
| COMMONAREA_AVG               | 3181 |
| COMMONAREA_MEDI              | 3202 |
| NONLIVINGAREA_AVG            | 3290 |
| NONLIVINGAREA_MEDI           | 3323 |
| NONLIVINGAREA_MODE           | 3327 |

|                        |        |
|------------------------|--------|
| LANDAREA_AVG           | 3527   |
| LANDAREA_MEDI          | 3560   |
| LANDAREA_MODE          | 3563   |
| BASEMENTAREA_MEDI      | 3772   |
| DAYS_LAST_PHONE_CHANGE | 3773   |
| BASEMENTAREA_AVG       | 3780   |
| BASEMENTAREA_MODE      | 3841   |
| TOTALAREA_MODE         | 5116   |
| LIVINGAREA_AVG         | 5199   |
| LIVINGAREA_MEDI        | 5281   |
| LIVINGAREA_MODE        | 5301   |
| AMT_CREDIT             | 5603   |
| DAYS_ID_PUBLISH        | 6168   |
| DAYS_EMPLOYED          | 12574  |
| AMT_ANNUITY            | 13672  |
| DAYS_REGISTRATION      | 15688  |
| DAYS_BIRTH             | 17460  |
| EXT_SOURCE_1           | 114584 |
| EXT_SOURCE_2           | 119831 |
| SK_ID_CURR             | 307511 |
| dtype:                 | int64  |

In [53]: *# inspecting the column types if they are in correct data type using the above*  
`applicationDF.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 307511 entries, 0 to 307510
Columns: 128 entries, SK_ID_CURR to EMPLOYMENT_YEAR
dtypes: category(4), float64(65), int64(43), object(16)
memory usage: 292.1+ MB
```

In [54]: *#Conversion of Object and Numerical columns to Categorical Columns*

```
categorical_columns = ['NAME_CONTRACT_TYPE', 'CODE_GENDER', 'NAME_TYPE_SUITE', 'NAME_FAMILY_STATUS', 'NAME_HOUSING_TYPE', 'OCCUPATION_TYPE', 'ORGANIZATION_TYPE', 'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'LIVE_CITY_NOT_WORK_CITY', 'REG_CITY_NOT_LIVE_CITY', 'REG_CITY_NOT_WORK_CITY', 'REG_REGION_NOT_WORK_REGION', 'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY']

for col in categorical_columns:
    applicationDF[col] = pd.Categorical(applicationDF[col])
```

In [55]: *# inspecting the column types if the above conversion is reflected*  
`applicationDF.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 307511 entries, 0 to 307510
Columns: 128 entries, SK_ID_CURR to EMPLOYMENT_YEAR
dtypes: category(23), float64(65), int64(36), object(4)
memory usage: 253.1+ MB
```

In [56]: *#Checking the number of unique values each column possess to identify categorical*  
`previousDF.nunique().sort_values()`

```

Out[56]: FLAG_LAST_APPL_PER_CONTRACT      2
         NFLAG_LAST_APPL_IN_DAY           2
         NFLAG_INSURED_ON_APPROVAL        2
         NAME_PRODUCT_TYPE                3
         NAME_PAYMENT_TYPE                4
         NAME_CLIENT_TYPE                 4
         NAME_CONTRACT_STATUS             4
         NAME_CONTRACT_TYPE               4
         NAME_PORTFOLIO                   5
         NAME_YIELD_GROUP                  5
         WEEKDAY_APPR_PROCESS_START       7
         NAME_TYPE_SUITE                   7
         CHANNEL_TYPE                      8
         CODE_REJECT_REASON                9
         NAME_SELLER_INDUSTRY             11
         PRODUCT_COMBINATION               17
         HOUR_APPR_PROCESS_START           24
         NAME_CASH_LOAN_PURPOSE            25
         RATE_INTEREST_PRIVILEGED          25
         NAME_GOODS_CATEGORY              28
         CNT_PAYMENT                       49
         RATE_INTEREST_PRIMARY             148
         SELLERPLACE_AREA                  2097
         DAYS_TERMINATION                  2830
         DAYS_FIRST_DRAWING                2838
         DAYS_LAST_DUE                     2873
         DAYS_FIRST_DUE                    2892
         DAYS_DECISION                     2922
         DAYS_LAST_DUE_1ST_VERSION         4605
         AMT_DOWN_PAYMENT                  29278
         AMT_CREDIT                        86803
         AMT_GOODS_PRICE                   93885
         AMT_APPLICATION                   93885
         RATE_DOWN_PAYMENT                 207033
         SK_ID_CURR                        338857
         AMT_ANNUITY                       357959
         SK_ID_PREV                        1670214
         dtype: int64

```

```

In [57]: # inspecting the column types if the above conversion is reflected
         previousDF.info()

```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1670214 entries, 0 to 1670213
Data columns (total 37 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   SK_ID_PREV                            1670214 non-null int64
1   SK_ID_CURR                            1670214 non-null int64
2   NAME_CONTRACT_TYPE                    1670214 non-null object
3   AMT_ANNUITY                           1297979 non-null float64
4   AMT_APPLICATION                       1670214 non-null float64
5   AMT_CREDIT                            1670213 non-null float64
6   AMT_DOWN_PAYMENT                      774370 non-null float64
7   AMT_GOODS_PRICE                       1284699 non-null float64
8   WEEKDAY_APPR_PROCESS_START            1670214 non-null object
9   HOUR_APPR_PROCESS_START               1670214 non-null int64
10  FLAG_LAST_APPL_PER_CONTRACT            1670214 non-null object
11  NFLAG_LAST_APPL_IN_DAY                 1670214 non-null int64
12  RATE_DOWN_PAYMENT                      774370 non-null float64
13  RATE_INTEREST_PRIMARY                  5951 non-null float64
14  RATE_INTEREST_PRIVILEGED               5951 non-null float64
15  NAME_CASH_LOAN_PURPOSE                 1670214 non-null object
16  NAME_CONTRACT_STATUS                  1670214 non-null object
17  DAYS_DECISION                          1670214 non-null int64
18  NAME_PAYMENT_TYPE                     1670214 non-null object
19  CODE_REJECT_REASON                    1670214 non-null object
20  NAME_TYPE_SUITE                        849809 non-null object
21  NAME_CLIENT_TYPE                       1670214 non-null object
22  NAME_GOODS_CATEGORY                   1670214 non-null object
23  NAME_PORTFOLIO                        1670214 non-null object
24  NAME_PRODUCT_TYPE                     1670214 non-null object
25  CHANNEL_TYPE                           1670214 non-null object
26  SELLERPLACE_AREA                       1670214 non-null int64
27  NAME_SELLER_INDUSTRY                  1670214 non-null object
28  CNT_PAYMENT                           1297984 non-null float64
29  NAME_YIELD_GROUP                       1670214 non-null object
30  PRODUCT_COMBINATION                   1669868 non-null object
31  DAYS_FIRST_DRAWING                     997149 non-null float64
32  DAYS_FIRST_DUE                         997149 non-null float64
33  DAYS_LAST_DUE_1ST_VERSION              997149 non-null float64
34  DAYS_LAST_DUE                         997149 non-null float64
35  DAYS_TERMINATION                       997149 non-null float64
36  NFLAG_INSURED_ON_APPROVAL              997149 non-null float64
dtypes: float64(15), int64(6), object(16)
memory usage: 471.5+ MB
```

```
In [58]: #Converting negative days to positive days
previousDF['DAYS_DECISION'] = abs(previousDF['DAYS_DECISION'])
```

```
In [59]: #age group calculation e.g. 388 will be grouped as 300-400
previousDF['DAYS_DECISION_GROUP'] = (previousDF['DAYS_DECISION']-(previousDF['
```

```
In [60]: previousDF['DAYS_DECISION_GROUP'].value_counts(normalize=True)*100
```

```
Out[60]: DAYS_DECISION_GROUP
0-400      37.490525
400-800    22.944724
800-1200   12.444753
1200-1600   7.904556
2400-2800   6.297456
1600-2000   5.795784
2000-2400   5.684960
2800-3200   1.437241
Name: proportion, dtype: float64
```

```
In [61]: #Converting Categorical columns from Object to categorical
Categorical_col_p = ['NAME_CASH_LOAN_PURPOSE', 'NAME_CONTRACT_STATUS', 'NAME_PAYM
                  'CODE_REJECT_REASON', 'NAME_CLIENT_TYPE', 'NAME_GOODS_CATEGOR
                  'NAME_PRODUCT_TYPE', 'CHANNEL_TYPE', 'NAME_SELLER_INDUSTRY', '
                  'NAME_CONTRACT_TYPE', 'DAYS_DECISION_GROUP']

for col in Categorical_col_p:
    previousDF[col] = pd.Categorical(previousDF[col])
```

```
In [62]: # inspecting the column types after conversion
previousDF.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1670214 entries, 0 to 1670213
Data columns (total 38 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   SK_ID_PREV                            1670214 non-null  int64
1   SK_ID_CURR                            1670214 non-null  int64
2   NAME_CONTRACT_TYPE                    1670214 non-null  category
3   AMT_ANNUITY                           1297979 non-null  float64
4   AMT_APPLICATION                       1670214 non-null  float64
5   AMT_CREDIT                            1670213 non-null  float64
6   AMT_DOWN_PAYMENT                      774370 non-null   float64
7   AMT_GOODS_PRICE                       1284699 non-null  float64
8   WEEKDAY_APPR_PROCESS_START            1670214 non-null  object
9   HOUR_APPR_PROCESS_START                1670214 non-null  int64
10  FLAG_LAST_APPL_PER_CONTRACT            1670214 non-null  object
11  NFLAG_LAST_APPL_IN_DAY                 1670214 non-null  int64
12  RATE_DOWN_PAYMENT                      774370 non-null   float64
13  RATE_INTEREST_PRIMARY                   5951 non-null     float64
14  RATE_INTEREST_PRIVILEGED               5951 non-null     float64
15  NAME_CASH_LOAN_PURPOSE                  1670214 non-null  category
16  NAME_CONTRACT_STATUS                   1670214 non-null  category
17  DAYS_DECISION                           1670214 non-null  int64
18  NAME_PAYMENT_TYPE                       1670214 non-null  category
19  CODE_REJECT_REASON                     1670214 non-null  category
20  NAME_TYPE_SUITE                         849809 non-null   object
21  NAME_CLIENT_TYPE                       1670214 non-null  category
22  NAME_GOODS_CATEGORY                    1670214 non-null  category
23  NAME_PORTFOLIO                         1670214 non-null  category
24  NAME_PRODUCT_TYPE                      1670214 non-null  category
25  CHANNEL_TYPE                           1670214 non-null  category
26  SELLERPLACE_AREA                       1670214 non-null  int64
27  NAME_SELLER_INDUSTRY                   1670214 non-null  category
28  CNT_PAYMENT                            1297984 non-null   float64
29  NAME_YIELD_GROUP                       1670214 non-null  category
30  PRODUCT_COMBINATION                    1669868 non-null  category
31  DAYS_FIRST_DRAWING                     997149 non-null   float64
32  DAYS_FIRST_DUE                         997149 non-null   float64
33  DAYS_LAST_DUE_1ST_VERSION              997149 non-null   float64
34  DAYS_LAST_DUE                          997149 non-null   float64
35  DAYS_TERMINATION                       997149 non-null   float64
36  NFLAG_INSURED_ON_APPROVAL              997149 non-null   float64
37  DAYS_DECISION_GROUP                    1670214 non-null  category
dtypes: category(14), float64(15), int64(6), object(3)
memory usage: 328.1+ MB

```

```

In [63]: # checking the null value % of each column in applicationDF dataframe
round(applicationDF.isnull().sum() / applicationDF.shape[0] * 100.00,2)

```

```

Out[63]: SK_ID_CURR      0.00
          TARGET         0.00
          NAME_CONTRACT_TYPE 0.00
          CODE_GENDER     0.00
          FLAG_OWN_CAR     0.00
          FLAG_OWN_REALTY  0.00
          CNT_CHILDREN     0.00
          AMT_INCOME_TOTAL 0.00
          AMT_CREDIT       0.00
          AMT_ANNUITY      0.00
          AMT_GOODS_PRICE  0.09
          NAME_TYPE_SUITE   0.42
          NAME_INCOME_TYPE  0.00
          NAME_EDUCATION_TYPE 0.00
          NAME_FAMILY_STATUS 0.00
          NAME_HOUSING_TYPE 0.00
          REGION_POPULATION_RELATIVE 0.00
          DAYS_BIRTH       0.00
          DAYS_EMPLOYED    0.00
          DAYS_REGISTRATION 0.00
          DAYS_ID_PUBLISH  0.00
          OWN_CAR_AGE      65.99
          FLAG_MOBIL       0.00
          FLAG_EMP_PHONE   0.00
          FLAG_WORK_PHONE  0.00
          FLAG_CONT_MOBILE 0.00
          FLAG_PHONE       0.00
          FLAG_EMAIL       0.00
          OCCUPATION_TYPE  31.35
          CNT_FAM_MEMBERS  0.00
          REGION_RATING_CLIENT 0.00
          REGION_RATING_CLIENT_W_CITY 0.00
          WEEKDAY_APPR_PROCESS_START 0.00
          HOUR_APPR_PROCESS_START 0.00
          REG_REGION_NOT_LIVE_REGION 0.00
          REG_REGION_NOT_WORK_REGION 0.00
          LIVE_REGION_NOT_WORK_REGION 0.00
          REG_CITY_NOT_LIVE_CITY 0.00
          REG_CITY_NOT_WORK_CITY 0.00
          LIVE_CITY_NOT_WORK_CITY 0.00
          ORGANIZATION_TYPE 0.00
          EXT_SOURCE_1     56.38
          EXT_SOURCE_2     0.21
          EXT_SOURCE_3     19.83
          APARTMENTS_AVG   50.75
          BASEMENTAREA_AVG 58.52
          YEARS_BEGINEXPLUATATION_AVG 48.78
          YEARS_BUILD_AVG  66.50
          COMMONAREA_AVG   69.87
          ELEVATORS_AVG    53.30
          ENTRANCES_AVG    50.35
          FLOORSMAX_AVG     49.76
          FLOORSMIN_AVG     67.85
          LANDAREA_AVG     59.38

```

|                              |       |
|------------------------------|-------|
| LIVINGAPARTMENTS_AVG         | 68.35 |
| LIVINGAREA_AVG               | 50.19 |
| NONLIVINGAPARTMENTS_AVG      | 69.43 |
| NONLIVINGAREA_AVG            | 55.18 |
| APARTMENTS_MODE              | 50.75 |
| BASEMENTAREA_MODE            | 58.52 |
| YEARS_BEGINEXPLUATATION_MODE | 48.78 |
| YEARS_BUILD_MODE             | 66.50 |
| COMMONAREA_MODE              | 69.87 |
| ELEVATORS_MODE               | 53.30 |
| ENTRANCES_MODE               | 50.35 |
| FLOORSMAX_MODE               | 49.76 |
| FLOORSMIN_MODE               | 67.85 |
| LANDAREA_MODE                | 59.38 |
| LIVINGAPARTMENTS_MODE        | 68.35 |
| LIVINGAREA_MODE              | 50.19 |
| NONLIVINGAPARTMENTS_MODE     | 69.43 |
| NONLIVINGAREA_MODE           | 55.18 |
| APARTMENTS_MEDI              | 50.75 |
| BASEMENTAREA_MEDI            | 58.52 |
| YEARS_BEGINEXPLUATATION_MEDI | 48.78 |
| YEARS_BUILD_MEDI             | 66.50 |
| COMMONAREA_MEDI              | 69.87 |
| ELEVATORS_MEDI               | 53.30 |
| ENTRANCES_MEDI               | 50.35 |
| FLOORSMAX_MEDI               | 49.76 |
| FLOORSMIN_MEDI               | 67.85 |
| LANDAREA_MEDI                | 59.38 |
| LIVINGAPARTMENTS_MEDI        | 68.35 |
| LIVINGAREA_MEDI              | 50.19 |
| NONLIVINGAPARTMENTS_MEDI     | 69.43 |
| NONLIVINGAREA_MEDI           | 55.18 |
| FONDKAPREMONT_MODE           | 68.39 |
| HOUSETYPE_MODE               | 50.18 |
| TOTALAREA_MODE               | 48.27 |
| WALLSMATERIAL_MODE           | 50.84 |
| EMERGENCYSTATE_MODE          | 47.40 |
| OBS_30_CNT_SOCIAL_CIRCLE     | 0.33  |
| DEF_30_CNT_SOCIAL_CIRCLE     | 0.33  |
| OBS_60_CNT_SOCIAL_CIRCLE     | 0.33  |
| DEF_60_CNT_SOCIAL_CIRCLE     | 0.33  |
| DAYS_LAST_PHONE_CHANGE       | 0.00  |
| FLAG_DOCUMENT_2              | 0.00  |
| FLAG_DOCUMENT_3              | 0.00  |
| FLAG_DOCUMENT_4              | 0.00  |
| FLAG_DOCUMENT_5              | 0.00  |
| FLAG_DOCUMENT_6              | 0.00  |
| FLAG_DOCUMENT_7              | 0.00  |
| FLAG_DOCUMENT_8              | 0.00  |
| FLAG_DOCUMENT_9              | 0.00  |
| FLAG_DOCUMENT_10             | 0.00  |
| FLAG_DOCUMENT_11             | 0.00  |
| FLAG_DOCUMENT_12             | 0.00  |
| FLAG_DOCUMENT_13             | 0.00  |



|                            |       |
|----------------------------|-------|
| FLAG_DOCUMENT_14           | 0.00  |
| FLAG_DOCUMENT_15           | 0.00  |
| FLAG_DOCUMENT_16           | 0.00  |
| FLAG_DOCUMENT_17           | 0.00  |
| FLAG_DOCUMENT_18           | 0.00  |
| FLAG_DOCUMENT_19           | 0.00  |
| FLAG_DOCUMENT_20           | 0.00  |
| FLAG_DOCUMENT_21           | 0.00  |
| AMT_REQ_CREDIT_BUREAU_HOUR | 13.50 |
| AMT_REQ_CREDIT_BUREAU_DAY  | 13.50 |
| AMT_REQ_CREDIT_BUREAU_WEEK | 13.50 |
| AMT_REQ_CREDIT_BUREAU_MON  | 13.50 |
| AMT_REQ_CREDIT_BUREAU_QRT  | 13.50 |
| AMT_REQ_CREDIT_BUREAU_YEAR | 13.50 |
| AMT_INCOME_RANGE           | 0.08  |
| AMT_CREDIT_RANGE           | 0.00  |
| AGE                        | 0.00  |
| AGE_GROUP                  | 0.00  |
| YEARS_EMPLOYED             | 0.00  |
| EMPLOYMENT_YEAR            | 27.08 |

dtype: float64

```
In [64]: applicationDF['NAME_TYPE_SUITE'].describe()
```

```
Out[64]: count          306219
unique           7
top      Unaccompanied
freq          248526
Name: NAME_TYPE_SUITE, dtype: object
```

```
In [65]: applicationDF['NAME_TYPE_SUITE'].fillna((applicationDF['NAME_TYPE_SUITE'].mode[0]))
```

```
In [66]: applicationDF['OCCUPATION_TYPE'] = applicationDF['OCCUPATION_TYPE'].cat.add_categories('Unknown')
applicationDF['OCCUPATION_TYPE'].fillna('Unknown', inplace = True)
```

```
In [67]: applicationDF[['AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
                        'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
                        'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR']].describe()
```

Out[67]:

|       | AMT_REQ_CREDIT_BUREAU_HOUR | AMT_REQ_CREDIT_BUREAU_DAY | AMT_REQ_CREDIT_BUREAU_YEAR |
|-------|----------------------------|---------------------------|----------------------------|
| count | 265992.000000              | 265992.000000             | 265992.000000              |
| mean  | 0.006402                   | 0.007000                  | 0.007000                   |
| std   | 0.083849                   | 0.110757                  | 0.110757                   |
| min   | 0.000000                   | 0.000000                  | 0.000000                   |
| 25%   | 0.000000                   | 0.000000                  | 0.000000                   |
| 50%   | 0.000000                   | 0.000000                  | 0.000000                   |
| 75%   | 0.000000                   | 0.000000                  | 0.000000                   |
| max   | 4.000000                   | 9.000000                  | 9.000000                   |

|              |               |               |
|--------------|---------------|---------------|
| <b>count</b> | 265992.000000 | 265992.000000 |
| <b>mean</b>  | 0.006402      | 0.007000      |
| <b>std</b>   | 0.083849      | 0.110757      |
| <b>min</b>   | 0.000000      | 0.000000      |
| <b>25%</b>   | 0.000000      | 0.000000      |
| <b>50%</b>   | 0.000000      | 0.000000      |
| <b>75%</b>   | 0.000000      | 0.000000      |
| <b>max</b>   | 4.000000      | 9.000000      |

```
In [68]: amount = ['AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY', 'AMT_REQ_CREDIT_BUREAU_YEAR',  
                  'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR']  
  
for col in amount:  
    applicationDF[col].fillna(applicationDF[col].median(), inplace = True)
```

```
In [69]: # checking the null value % of each column in previousDF dataframe  
round(applicationDF.isnull().sum() / previousDF.shape[0] * 100.00, 2)
```

```

Out[69]: SK_ID_CURR      0.00
          TARGET          0.00
          NAME_CONTRACT_TYPE 0.00
          CODE_GENDER      0.00
          FLAG_OWN_CAR      0.00
          FLAG_OWN_REALTY   0.00
          CNT_CHILDREN      0.00
          AMT_INCOME_TOTAL  0.00
          AMT_CREDIT        0.00
          AMT_ANNUITY       0.00
          AMT_GOODS_PRICE   0.02
          NAME_TYPE_SUITE   0.00
          NAME_INCOME_TYPE  0.00
          NAME_EDUCATION_TYPE 0.00
          NAME_FAMILY_STATUS 0.00
          NAME_HOUSING_TYPE  0.00
          REGION_POPULATION_RELATIVE 0.00
          DAYS_BIRTH        0.00
          DAYS_EMPLOYED     0.00
          DAYS_REGISTRATION  0.00
          DAYS_ID_PUBLISH   0.00
          OWN_CAR_AGE       12.15
          FLAG_MOBIL        0.00
          FLAG_EMP_PHONE    0.00
          FLAG_WORK_PHONE   0.00
          FLAG_CONT_MOBILE  0.00
          FLAG_PHONE        0.00
          FLAG_EMAIL        0.00
          OCCUPATION_TYPE   0.00
          CNT_FAM_MEMBERS   0.00
          REGION_RATING_CLIENT 0.00
          REGION_RATING_CLIENT_W_CITY 0.00
          WEEKDAY_APPR_PROCESS_START 0.00
          HOUR_APPR_PROCESS_START 0.00
          REG_REGION_NOT_LIVE_REGION 0.00
          REG_REGION_NOT_WORK_REGION 0.00
          LIVE_REGION_NOT_WORK_REGION 0.00
          REG_CITY_NOT_LIVE_CITY 0.00
          REG_CITY_NOT_WORK_CITY 0.00
          LIVE_CITY_NOT_WORK_CITY 0.00
          ORGANIZATION_TYPE 0.00
          EXT_SOURCE_1      10.38
          EXT_SOURCE_2      0.04
          EXT_SOURCE_3      3.65
          APARTMENTS_AVG    9.34
          BASEMENTAREA_AVG  10.77
          YEARS_BEGINEXPLUATATION_AVG 8.98
          YEARS_BUILD_AVG   12.24
          COMMONAREA_AVG    12.86
          ELEVATORS_AVG     9.81
          ENTRANCES_AVG     9.27
          FLOORSMAX_AVG     9.16
          FLOORSMIN_AVG     12.49
          LANDAREA_AVG      10.93

```

|                              |       |
|------------------------------|-------|
| LIVINGAPARTMENTS_AVG         | 12.59 |
| LIVINGAREA_AVG               | 9.24  |
| NONLIVINGAPARTMENTS_AVG      | 12.78 |
| NONLIVINGAREA_AVG            | 10.16 |
| APARTMENTS_MODE              | 9.34  |
| BASEMENTAREA_MODE            | 10.77 |
| YEARS_BEGINEXPLUATATION_MODE | 8.98  |
| YEARS_BUILD_MODE             | 12.24 |
| COMMONAREA_MODE              | 12.86 |
| ELEVATORS_MODE               | 9.81  |
| ENTRANCES_MODE               | 9.27  |
| FLOORSMAX_MODE               | 9.16  |
| FLOORSMIN_MODE               | 12.49 |
| LANDAREA_MODE                | 10.93 |
| LIVINGAPARTMENTS_MODE        | 12.59 |
| LIVINGAREA_MODE              | 9.24  |
| NONLIVINGAPARTMENTS_MODE     | 12.78 |
| NONLIVINGAREA_MODE           | 10.16 |
| APARTMENTS_MEDI              | 9.34  |
| BASEMENTAREA_MEDI            | 10.77 |
| YEARS_BEGINEXPLUATATION_MEDI | 8.98  |
| YEARS_BUILD_MEDI             | 12.24 |
| COMMONAREA_MEDI              | 12.86 |
| ELEVATORS_MEDI               | 9.81  |
| ENTRANCES_MEDI               | 9.27  |
| FLOORSMAX_MEDI               | 9.16  |
| FLOORSMIN_MEDI               | 12.49 |
| LANDAREA_MEDI                | 10.93 |
| LIVINGAPARTMENTS_MEDI        | 12.59 |
| LIVINGAREA_MEDI              | 9.24  |
| NONLIVINGAPARTMENTS_MEDI     | 12.78 |
| NONLIVINGAREA_MEDI           | 10.16 |
| FONDKAPREMONT_MODE           | 12.59 |
| HOUSETYPE_MODE               | 9.24  |
| TOTALAREA_MODE               | 8.89  |
| WALLSMATERIAL_MODE           | 9.36  |
| EMERGENCYSTATE_MODE          | 8.73  |
| OBS_30_CNT_SOCIAL_CIRCLE     | 0.06  |
| DEF_30_CNT_SOCIAL_CIRCLE     | 0.06  |
| OBS_60_CNT_SOCIAL_CIRCLE     | 0.06  |
| DEF_60_CNT_SOCIAL_CIRCLE     | 0.06  |
| DAYS_LAST_PHONE_CHANGE       | 0.00  |
| FLAG_DOCUMENT_2              | 0.00  |
| FLAG_DOCUMENT_3              | 0.00  |
| FLAG_DOCUMENT_4              | 0.00  |
| FLAG_DOCUMENT_5              | 0.00  |
| FLAG_DOCUMENT_6              | 0.00  |
| FLAG_DOCUMENT_7              | 0.00  |
| FLAG_DOCUMENT_8              | 0.00  |
| FLAG_DOCUMENT_9              | 0.00  |
| FLAG_DOCUMENT_10             | 0.00  |
| FLAG_DOCUMENT_11             | 0.00  |
| FLAG_DOCUMENT_12             | 0.00  |
| FLAG_DOCUMENT_13             | 0.00  |

|                            |      |
|----------------------------|------|
| FLAG_DOCUMENT_14           | 0.00 |
| FLAG_DOCUMENT_15           | 0.00 |
| FLAG_DOCUMENT_16           | 0.00 |
| FLAG_DOCUMENT_17           | 0.00 |
| FLAG_DOCUMENT_18           | 0.00 |
| FLAG_DOCUMENT_19           | 0.00 |
| FLAG_DOCUMENT_20           | 0.00 |
| FLAG_DOCUMENT_21           | 0.00 |
| AMT_REQ_CREDIT_BUREAU_HOUR | 0.00 |
| AMT_REQ_CREDIT_BUREAU_DAY  | 0.00 |
| AMT_REQ_CREDIT_BUREAU_WEEK | 0.00 |
| AMT_REQ_CREDIT_BUREAU_MON  | 0.00 |
| AMT_REQ_CREDIT_BUREAU_QRT  | 0.00 |
| AMT_REQ_CREDIT_BUREAU_YEAR | 0.00 |
| AMT_INCOME_RANGE           | 0.01 |
| AMT_CREDIT_RANGE           | 0.00 |
| AGE                        | 0.00 |
| AGE_GROUP                  | 0.00 |
| YEARS_EMPLOYED             | 0.00 |
| EMPLOYMENT_YEAR            | 4.99 |

dtype: float64

```
In [70]: # checking the null value % of each column in previousDF dataframe
round(previousDF.isnull().sum() / previousDF.shape[0] * 100.00,2)
```

```

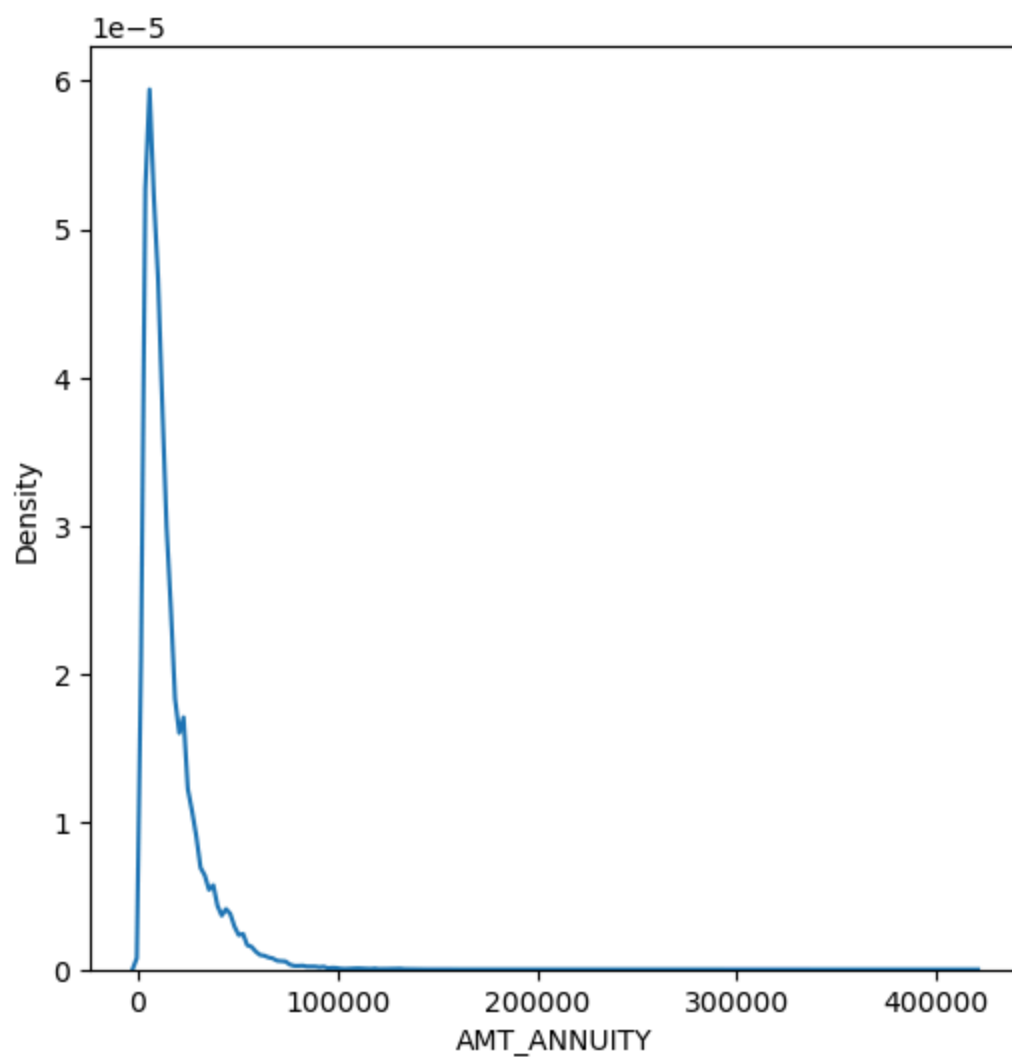
Out[70]: SK_ID_PREV          0.00
         SK_ID_CURR          0.00
         NAME_CONTRACT_TYPE    0.00
         AMT_ANNUITY          22.29
         AMT_APPLICATION      0.00
         AMT_CREDIT           0.00
         AMT_DOWN_PAYMENT     53.64
         AMT_GOODS_PRICE      23.08
         WEEKDAY_APPR_PROCESS_START 0.00
         HOUR_APPR_PROCESS_START 0.00
         FLAG_LAST_APPL_PER_CONTRACT 0.00
         NFLAG_LAST_APPL_IN_DAY 0.00
         RATE_DOWN_PAYMENT    53.64
         RATE_INTEREST_PRIMARY 99.64
         RATE_INTEREST_PRIVILEGED 99.64
         NAME_CASH_LOAN_PURPOSE 0.00
         NAME_CONTRACT_STATUS  0.00
         DAYS_DECISION         0.00
         NAME_PAYMENT_TYPE     0.00
         CODE_REJECT_REASON    0.00
         NAME_TYPE_SUITE       49.12
         NAME_CLIENT_TYPE      0.00
         NAME_GOODS_CATEGORY    0.00
         NAME_PORTFOLIO         0.00
         NAME_PRODUCT_TYPE      0.00
         CHANNEL_TYPE           0.00
         SELLERPLACE_AREA       0.00
         NAME_SELLER_INDUSTRY   0.00
         CNT_PAYMENT            22.29
         NAME_YIELD_GROUP       0.00
         PRODUCT_COMBINATION    0.02
         DAYS_FIRST_DRAWING     40.30
         DAYS_FIRST_DUE         40.30
         DAYS_LAST_DUE_1ST_VERSION 40.30
         DAYS_LAST_DUE         40.30
         DAYS_TERMINATION       40.30
         NFLAG_INSURED_ON_APPROVAL 40.30
         DAYS_DECISION_GROUP    0.00
         dtype: float64

```

```

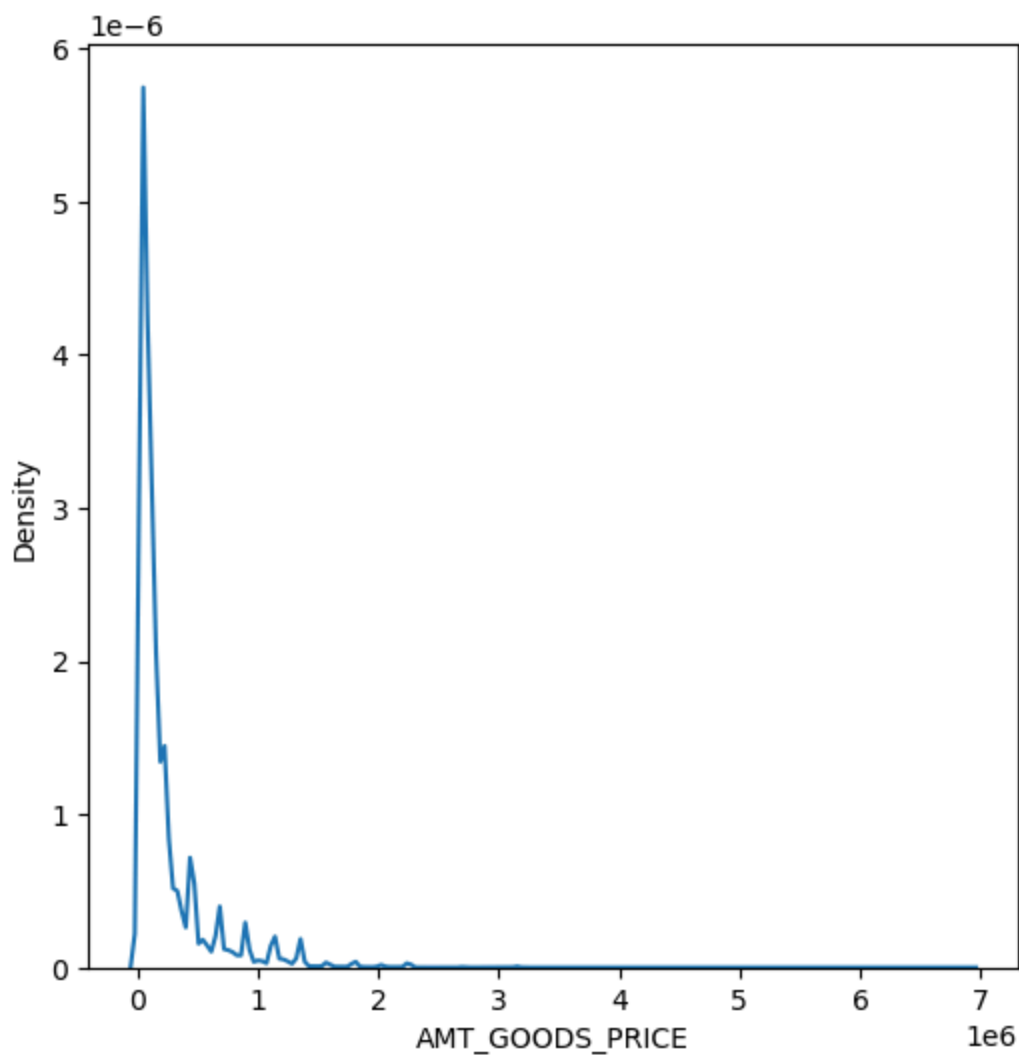
In [71]: plt.figure(figsize=(6,6))
         sns.kdeplot(previousDF['AMT_ANNUITY'])
         plt.show()

```



```
In [72]: previousDF['AMT_ANNUITY'].fillna(previousDF['AMT_ANNUITY'].median(),inplace =
```

```
In [73]: plt.figure(figsize=(6,6))  
sns.kdeplot(previousDF['AMT_GOODS_PRICE'][pd.notnull(previousDF['AMT_GOODS_PRI  
plt.show()
```



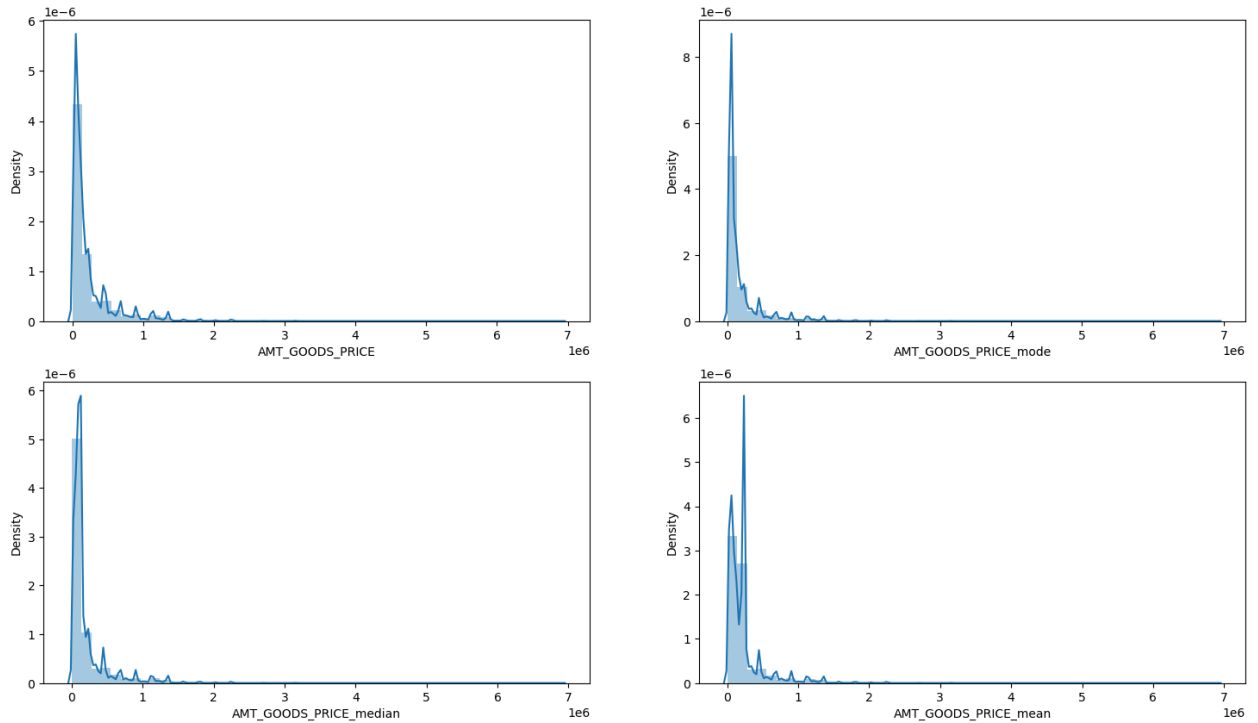
```
In [74]: statsDF = pd.DataFrame() # new dataframe with columns imputed with mode, median, mean
statsDF['AMT_GOODS_PRICE_mode'] = previousDF['AMT_GOODS_PRICE'].fillna(previousDF['AMT_GOODS_PRICE_mode'])
statsDF['AMT_GOODS_PRICE_median'] = previousDF['AMT_GOODS_PRICE'].fillna(previousDF['AMT_GOODS_PRICE_median'])
statsDF['AMT_GOODS_PRICE_mean'] = previousDF['AMT_GOODS_PRICE'].fillna(previousDF['AMT_GOODS_PRICE_mean'])

cols = ['AMT_GOODS_PRICE_mode', 'AMT_GOODS_PRICE_median', 'AMT_GOODS_PRICE_mean']

plt.figure(figsize=(18,10))
plt.suptitle('Distribution of Original data vs imputed data')
plt.subplot(221)
sns.distplot(previousDF['AMT_GOODS_PRICE'][pd.notnull(previousDF['AMT_GOODS_PRICE'])])
for i in enumerate(cols):
    plt.subplot(2,2,i[0]+2)
    sns.distplot(statsDF[i[1]])
```



Distribution of Original data vs imputed data



```
In [75]: previousDF['AMT_GOODS_PRICE'].fillna(previousDF['AMT_GOODS_PRICE'].mode()[0],
```

```
In [76]: previousDF.loc[previousDF['CNT_PAYMENT'].isnull(), 'NAME_CONTRACT_STATUS'].valu
```

```
Out[76]: NAME_CONTRACT_STATUS
Canceled      305805
Refused       40897
Unused offer   25524
Approved         4
Name: count, dtype: int64
```

```
In [77]: previousDF['CNT_PAYMENT'].fillna(0, inplace = True)
```

```
In [78]: # checking the null value % of each column in previousDF dataframe
round(previousDF.isnull().sum() / previousDF.shape[0] * 100.00, 2)
```

```

Out[78]: SK_ID_PREV                0.00
          SK_ID_CURR                0.00
          NAME_CONTRACT_TYPE        0.00
          AMT_ANNUITY                0.00
          AMT_APPLICATION            0.00
          AMT_CREDIT                 0.00
          AMT_DOWN_PAYMENT           53.64
          AMT_GOODS_PRICE            0.00
          WEEKDAY_APPR_PROCESS_START 0.00
          HOUR_APPR_PROCESS_START    0.00
          FLAG_LAST_APPL_PER_CONTRACT 0.00
          NFLAG_LAST_APPL_IN_DAY     0.00
          RATE_DOWN_PAYMENT           53.64
          RATE_INTEREST_PRIMARY       99.64
          RATE_INTEREST_PRIVILEGED    99.64
          NAME_CASH_LOAN_PURPOSE      0.00
          NAME_CONTRACT_STATUS        0.00
          DAYS_DECISION               0.00
          NAME_PAYMENT_TYPE           0.00
          CODE_REJECT_REASON          0.00
          NAME_TYPE_SUITE             49.12
          NAME_CLIENT_TYPE            0.00
          NAME_GOODS_CATEGORY         0.00
          NAME_PORTFOLIO              0.00
          NAME_PRODUCT_TYPE           0.00
          CHANNEL_TYPE                0.00
          SELLERPLACE_AREA            0.00
          NAME_SELLER_INDUSTRY        0.00
          CNT_PAYMENT                0.00
          NAME_YIELD_GROUP            0.00
          PRODUCT_COMBINATION         0.02
          DAYS_FIRST_DRAWING          40.30
          DAYS_FIRST_DUE              40.30
          DAYS_LAST_DUE_1ST_VERSION   40.30
          DAYS_LAST_DUE              40.30
          DAYS_TERMINATION            40.30
          NFLAG_INSURED_ON_APPROVAL   40.30
          DAYS_DECISION_GROUP         0.00
          dtype: float64

```

```

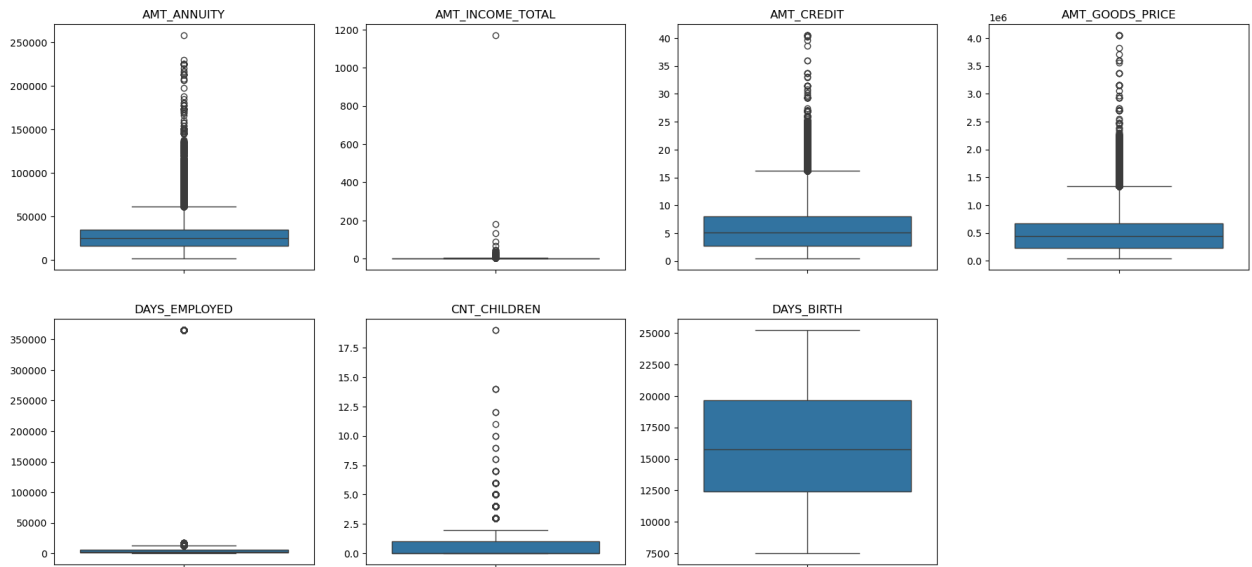
In [79]: plt.figure(figsize=(22,10))

app_outlier_col_1 = ['AMT_ANNUITY', 'AMT_INCOME_TOTAL', 'AMT_CREDIT', 'AMT_GOODS_PRICE']
app_outlier_col_2 = ['CNT_CHILDREN', 'DAYS_BIRTH']
for i in enumerate(app_outlier_col_1):
    plt.subplot(2,4,i[0]+1)
    sns.boxplot(y=applicationDF[i[1]])
    plt.title(i[1])
    plt.ylabel("")

for i in enumerate(app_outlier_col_2):
    plt.subplot(2,4,i[0]+6)
    sns.boxplot(y=applicationDF[i[1]])

```

```
plt.title(i[1])
plt.ylabel("")
```



```
In [80]: applicationDF[['AMT_ANNUITY', 'AMT_INCOME_TOTAL', 'AMT_CREDIT', 'AMT_GOODS_PRICE', 'DAYS_EMPLOYED', 'CNT_CHILDREN', 'DAYS_BIRTH', 'SK_ID_CURR', 'SK_ID_NEXT', 'TARGET']]
```

```
Out[80]:
```

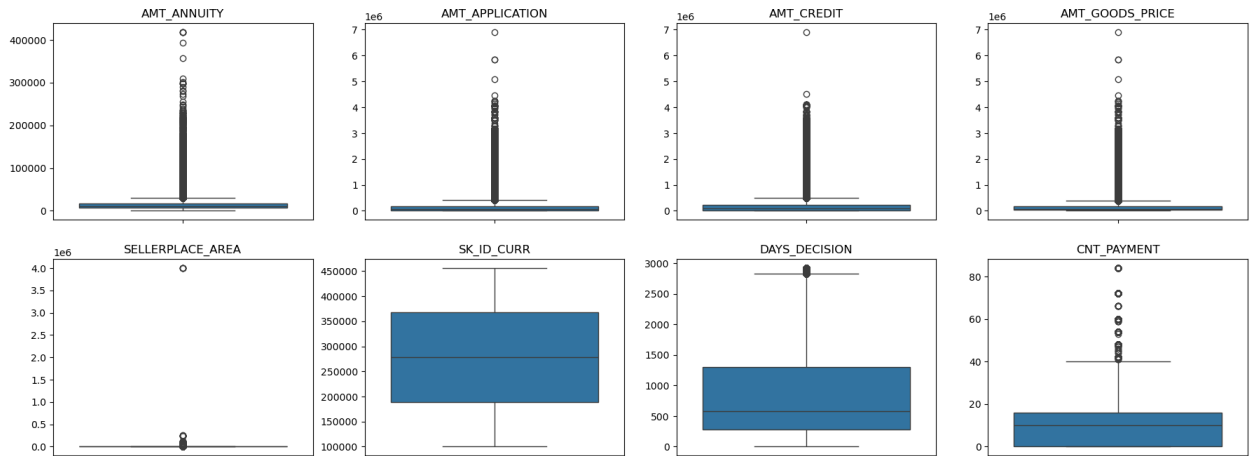
|              | AMT_ANNUITY   | AMT_INCOME_TOTAL | AMT_CREDIT    | AMT_GOODS_PRICE |   |
|--------------|---------------|------------------|---------------|-----------------|---|
| <b>count</b> | 307499.000000 | 307511.000000    | 307511.000000 | 3.072330e+05    | 3 |
| <b>mean</b>  | 27108.573909  | 1.687979         | 5.990260      | 5.383962e+05    |   |
| <b>std</b>   | 14493.737315  | 2.371231         | 4.024908      | 3.694465e+05    |   |
| <b>min</b>   | 1615.500000   | 0.256500         | 0.450000      | 4.050000e+04    |   |
| <b>25%</b>   | 16524.000000  | 1.125000         | 2.700000      | 2.385000e+05    |   |
| <b>50%</b>   | 24903.000000  | 1.471500         | 5.135310      | 4.500000e+05    |   |
| <b>75%</b>   | 34596.000000  | 2.025000         | 8.086500      | 6.795000e+05    |   |
| <b>max</b>   | 258025.500000 | 1170.000000      | 40.500000     | 4.050000e+06    |   |

```
In [81]: plt.figure(figsize=(22,8))

prev_outlier_col_1 = ['AMT_ANNUITY', 'AMT_APPLICATION', 'AMT_CREDIT', 'AMT_GOODS_PRICE', 'DAYS_EMPLOYED', 'CNT_CHILDREN', 'DAYS_BIRTH', 'SK_ID_CURR', 'SK_ID_NEXT', 'TARGET']
prev_outlier_col_2 = ['SK_ID_CURR', 'DAYS_DECISION', 'CNT_PAYMENT']
for i in enumerate(prev_outlier_col_1):
    plt.subplot(2,4,i[0]+1)
    sns.boxplot(y=previousDF[i[1]])
    plt.title(i[1])
    plt.ylabel("")

for i in enumerate(prev_outlier_col_2):
    plt.subplot(2,4,i[0]+6)
    sns.boxplot(y=previousDF[i[1]])
    plt.title(i[1])
```

```
plt.ylabel("")
```



```
In [82]: previousDF[['AMT_ANNUITY', 'AMT_APPLICATION', 'AMT_CREDIT', 'AMT_GOODS_PRICE',
```

```
Out[82]:
```

|              | AMT_ANNUITY  | AMT_APPLICATION | AMT_CREDIT   | AMT_GOODS_PRICE | SEL |
|--------------|--------------|-----------------|--------------|-----------------|-----|
| <b>count</b> | 1.670214e+06 | 1.670214e+06    | 1.670213e+06 | 1.670214e+06    |     |
| <b>mean</b>  | 1.490651e+04 | 1.752339e+05    | 1.961140e+05 | 1.856429e+05    |     |
| <b>std</b>   | 1.317751e+04 | 2.927798e+05    | 3.185746e+05 | 2.871413e+05    |     |
| <b>min</b>   | 0.000000e+00 | 0.000000e+00    | 0.000000e+00 | 0.000000e+00    |     |
| <b>25%</b>   | 7.547096e+03 | 1.872000e+04    | 2.416050e+04 | 4.500000e+04    |     |
| <b>50%</b>   | 1.125000e+04 | 7.104600e+04    | 8.054100e+04 | 7.105050e+04    |     |
| <b>75%</b>   | 1.682403e+04 | 1.803600e+05    | 2.164185e+05 | 1.804050e+05    |     |
| <b>max</b>   | 4.180581e+05 | 6.905160e+06    | 6.905160e+06 | 6.905160e+06    |     |

```
In [83]: import seaborn as sns
```

```
In [84]: def univariate_categorical(feature, ylog=False, label_rotation=False, horizontal_
temp = applicationDF[feature].value_counts()
df1 = pd.DataFrame({feature: temp.index, 'Number of contracts': temp.values

# Calculate the percentage of target=1 per category value
cat_perc = applicationDF[[feature, 'TARGET']].groupby([feature], as_index=False)
cat_perc["TARGET"] = cat_perc["TARGET"]*100
cat_perc.sort_values(by='TARGET', ascending=False, inplace=True)

if(horizontal_layout):
    fig, (ax1, ax2) = plt.subplots(ncols=2, figsize=(12,6))
else:
    fig, (ax1, ax2) = plt.subplots(nrows=2, figsize=(20,24))

# 1. Subplot 1: Count plot of categorical column
# sns.set_palette("Set2")
s = sns.countplot(ax=ax1,
```

```

        x = feature,
        data=applicationDF,
        hue ="TARGET",
        order=cat_perc[feature],
        palette=['g','r'])

    # Define common styling
    ax1.set_title(feature, fontdict={'fontsize' : 10, 'fontweight' : 3, 'color' : 'Blue'})
    ax1.legend(['Repayer', 'Defaulter'])

    # If the plot is not readable, use the log scale.
    if ylog:
        ax1.set_yscale('log')
        ax1.set_ylabel("Count (log)", fontdict={'fontsize' : 10, 'fontweight' : 3, 'color' : 'Blue'})

    if (label_rotation):
        s.set_xticklabels(s.get_xticklabels(), rotation=90)

    # 2. Subplot 2: Percentage of defaulters within the categorical column
    s = sns.barplot(ax=ax2,
                    x = feature,
                    y='TARGET',
                    order=cat_perc[feature],
                    data=cat_perc,
                    palette='Set2')

    if (label_rotation):
        s.set_xticklabels(s.get_xticklabels(), rotation=90)
    plt.ylabel('Percent of Defaulters [%]', fontsize=10)
    plt.tick_params(axis='both', which='major', labelsize=10)
    ax2.set_title(feature + " Defaulter %", fontdict={'fontsize' : 15, 'fontweight' : 5, 'color' : 'Blue'})

    plt.show();

```

```

In [85]: def bivariate_bar(x,y,df,hue,figsize):

    plt.figure(figsize=figsize)
    sns.barplot(x=x,
                y=y,
                data=df,
                hue=hue,
                palette=['g','r'])

    # Defining aesthetics of Labels and Title of the plot using style dictionary
    plt.xlabel(x, fontdict={'fontsize' : 10, 'fontweight' : 3, 'color' : 'Blue'})
    plt.ylabel(y, fontdict={'fontsize' : 10, 'fontweight' : 3, 'color' : 'Blue'})
    plt.title(col, fontdict={'fontsize' : 15, 'fontweight' : 5, 'color' : 'Blue'})
    plt.xticks(rotation=90, ha='right')
    plt.legend(labels = ['Repayer', 'Defaulter'])
    plt.show()

```

```

In [86]: def bivariate_rel(x,y,data, hue, kind, palette, legend,figsize):

```

```
plt.figure(figsize=figsize)
sns.relplot(x=x,
            y=y,
            data=applicationDF,
            hue="TARGET",
            kind=kind,
            palette = ['g','r'],
            legend = False)
plt.legend(['Repayer', 'Defaulter'])
plt.xticks(rotation=90, ha='right')
plt.show()
```

```
In [87]: def univariate_merged(col,df,hue,palette,ylog,figsize):
plt.figure(figsize=figsize)
ax=sns.countplot(x=col,
                data=df,
                hue= hue,
                palette= palette,
                order=df[col].value_counts().index)

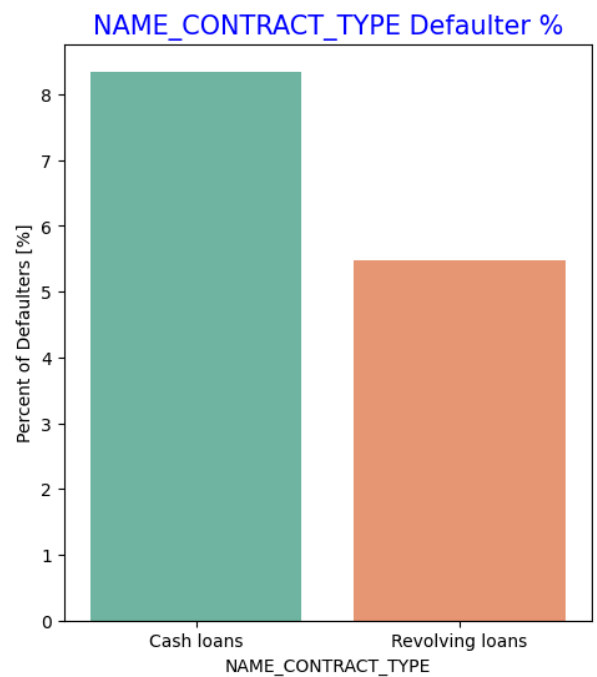
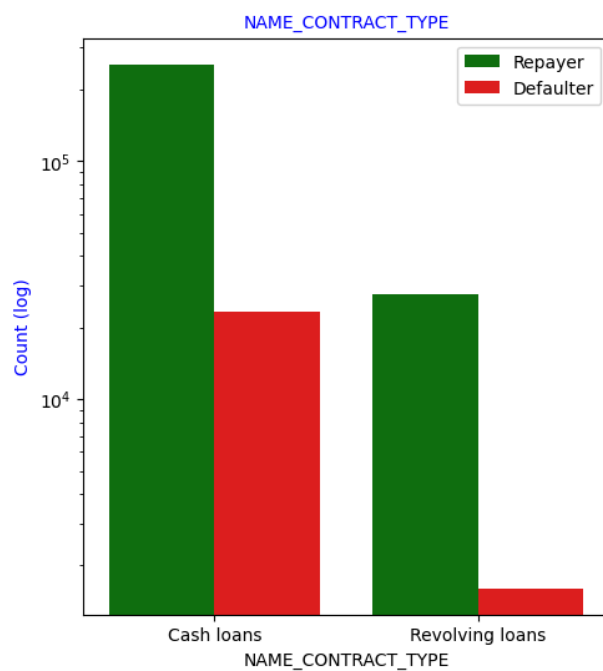
if ylog:
    plt.yscale('log')
    plt.ylabel("Count (log)",fontdict={'fontsize' : 10, 'fontweight' : 3,
else:
    plt.ylabel("Count",fontdict={'fontsize' : 10, 'fontweight' : 3, 'color'

plt.title(col , fontdict={'fontsize' : 15, 'fontweight' : 5, 'color' : 'Bl
plt.legend(loc = "upper right")
plt.xticks(rotation=90, ha='right')

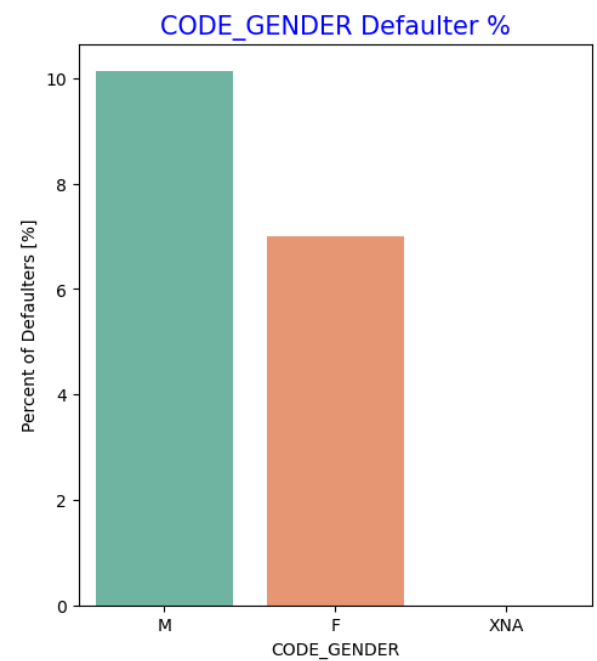
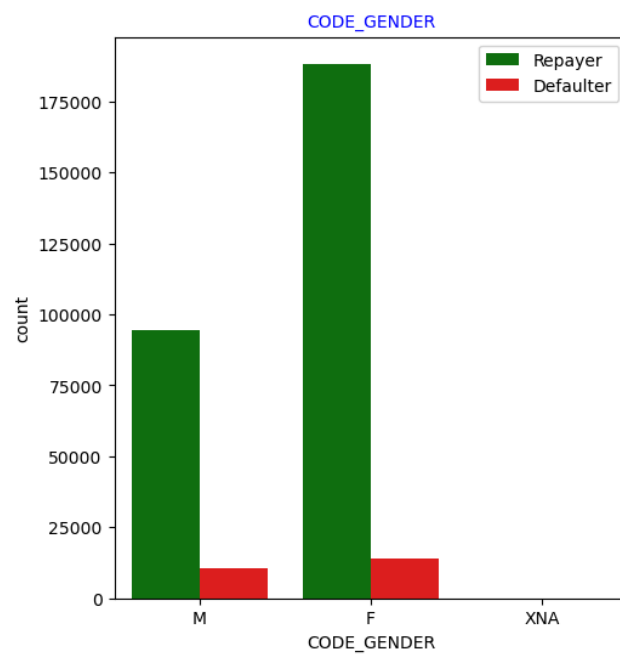
plt.show()
```

```
In [88]: def merged_pointplot(x,y):
plt.figure(figsize=(8,4))
sns.pointplot(x=x,
              y=y,
              hue="TARGET",
              data=loan_process_df,
              palette =['g','r'])
# plt.legend(['Repayer', 'Defaulter'])
```

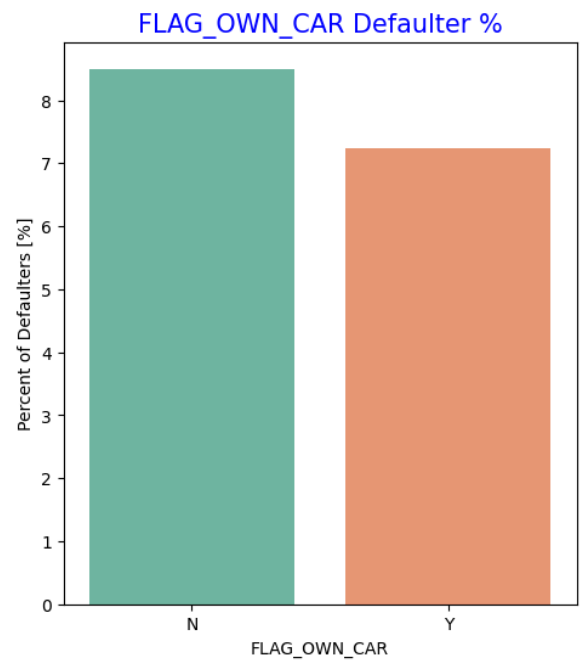
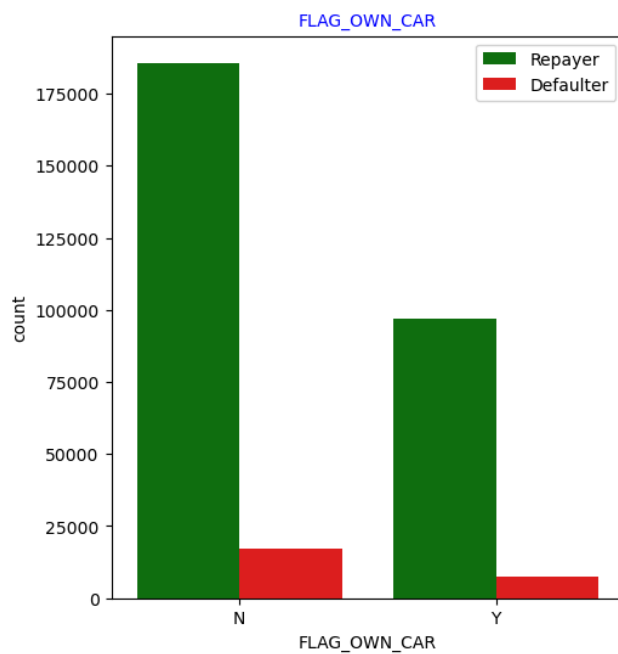
```
In [89]: # Checking the contract type based on loan repayment status
univariate_categorical('NAME_CONTRACT_TYPE',True)
```



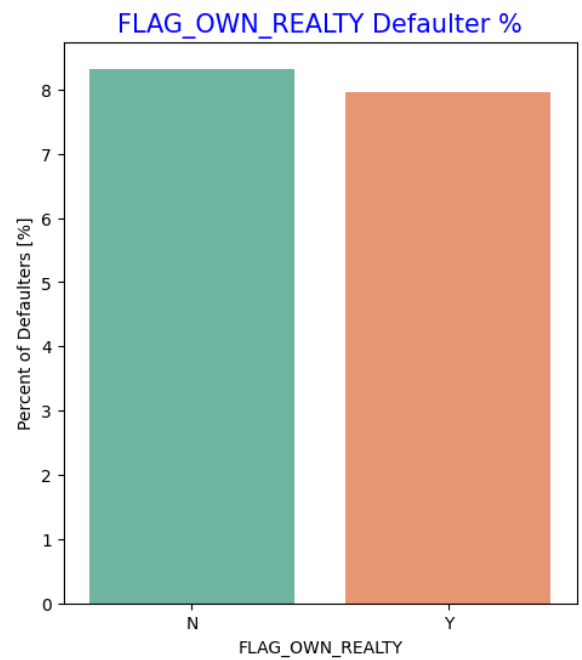
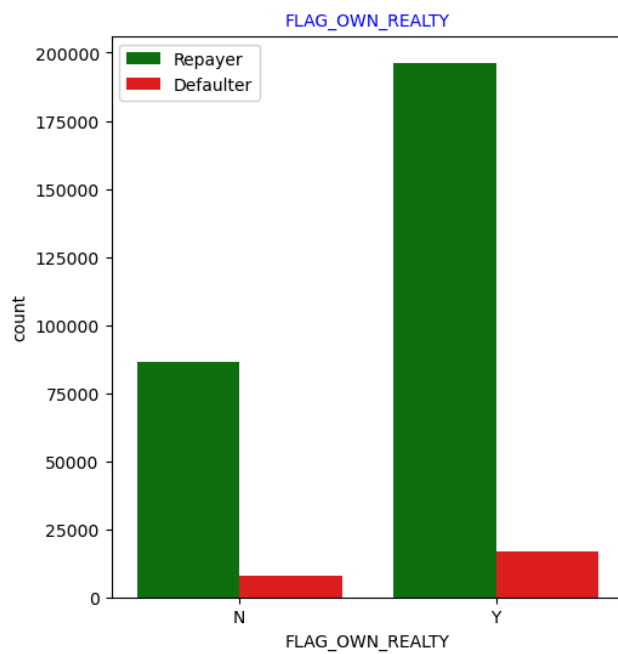
```
In [90]: # Checking the type of Gender on loan repayment status
univariate_categorical('CODE_GENDER')
```



```
In [91]: # Checking if owning a car is related to loan repayment status
univariate_categorical('FLAG_OWN_CAR')
```

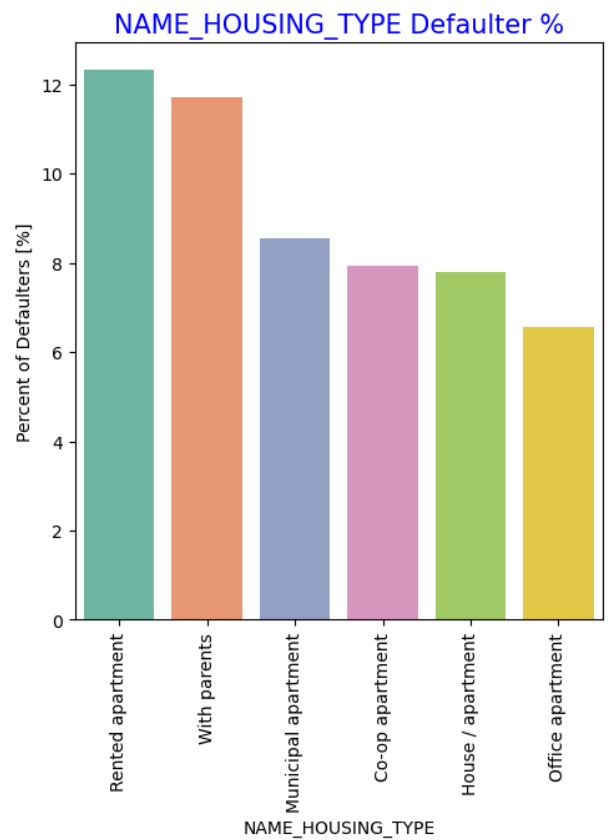
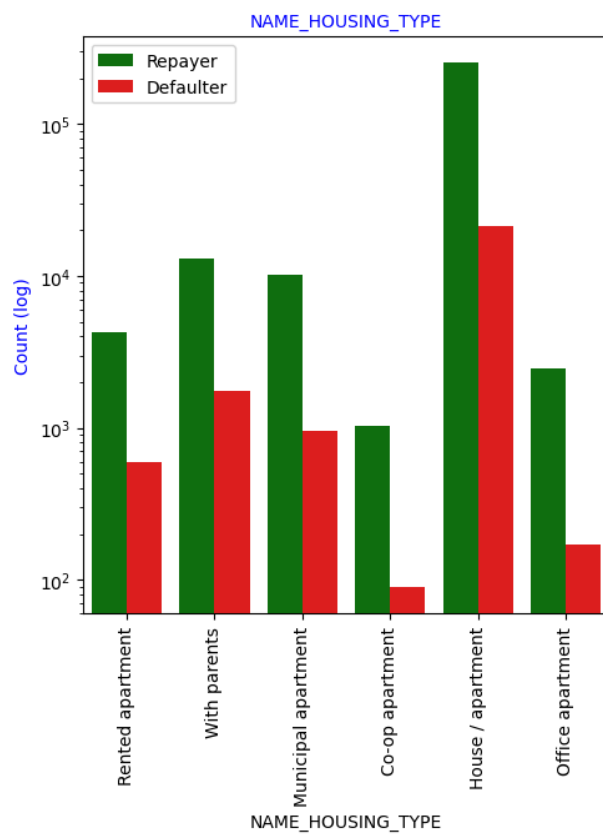


In [92]: *# Checking if owning a realty is related to loan repayment status*  
 univariate\_categorical('FLAG\_OWN\_REALTY')

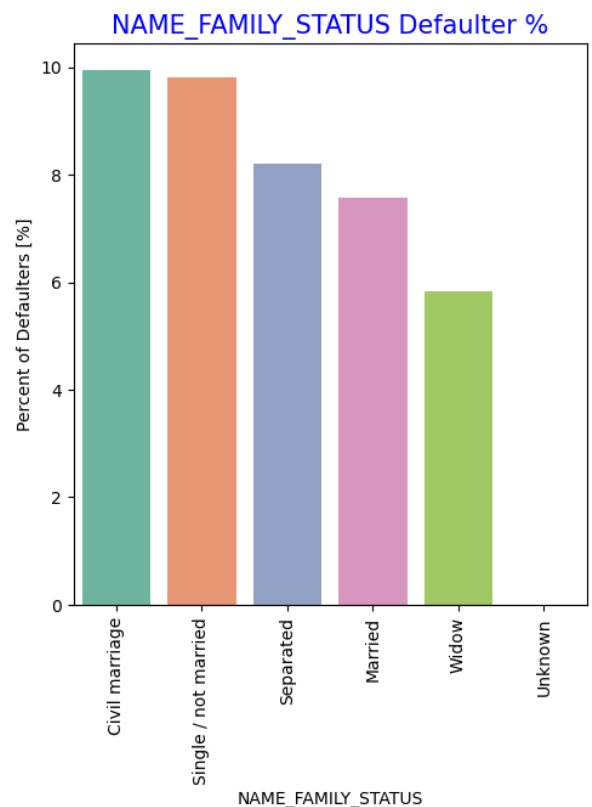
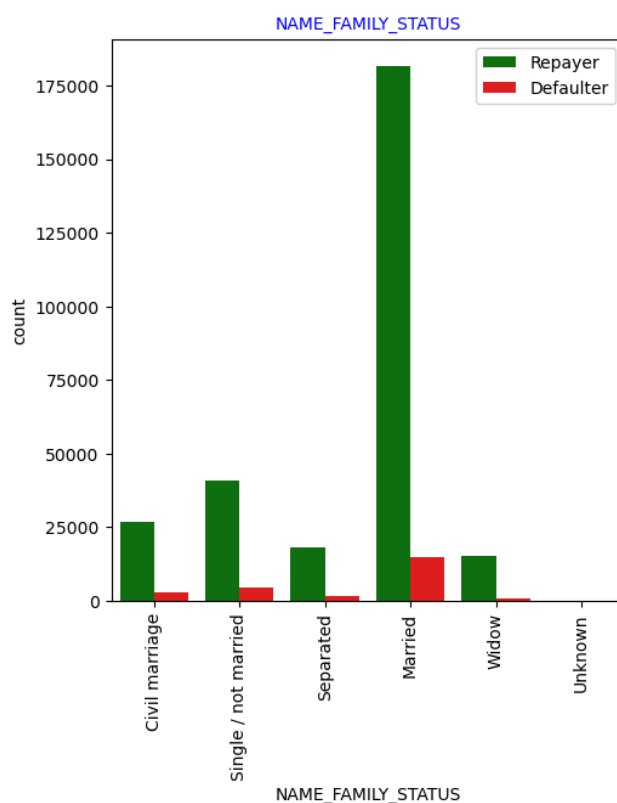


In [93]: *# Analyzing Housing Type based on loan repayment status*  
 univariate\_categorical("NAME\_HOUSING\_TYPE", True, True, True)

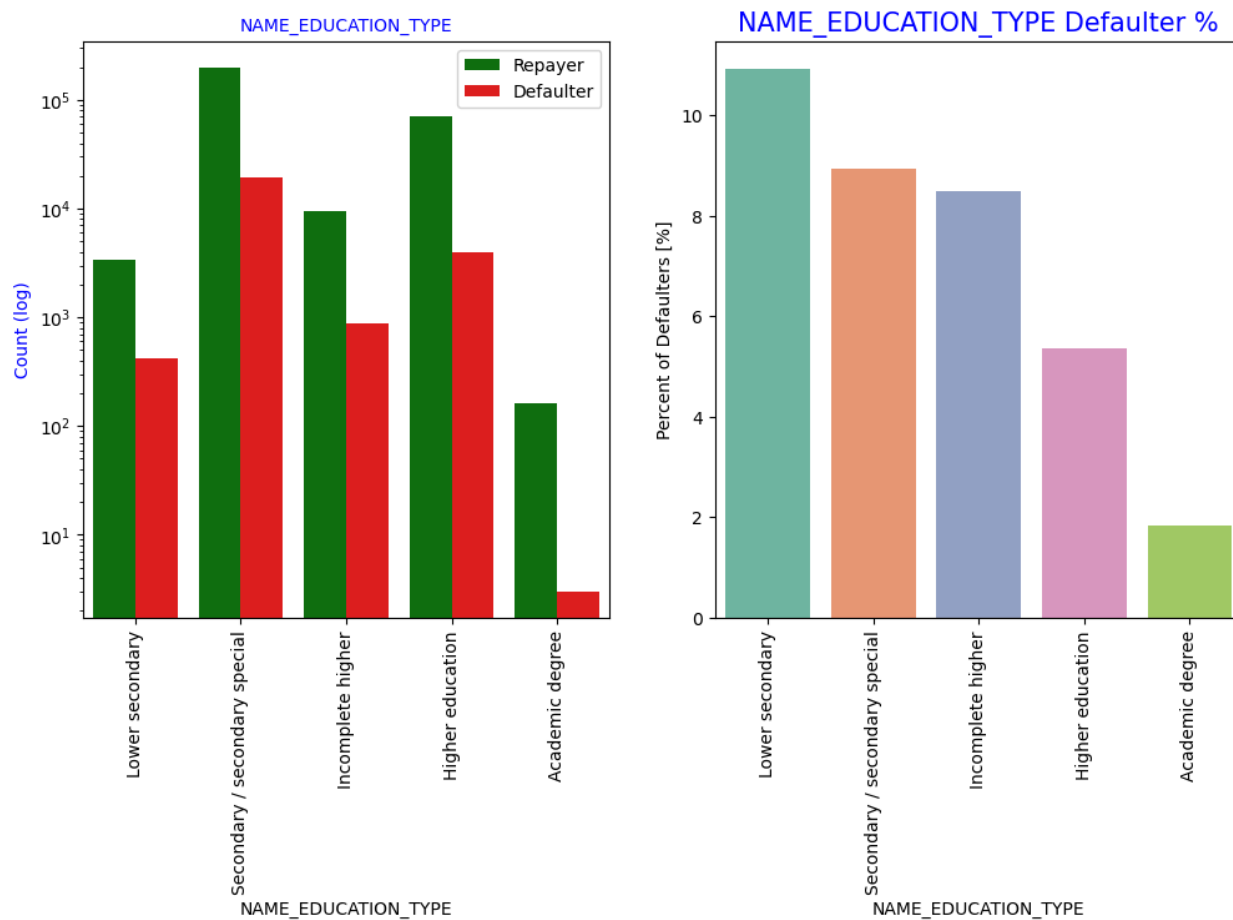




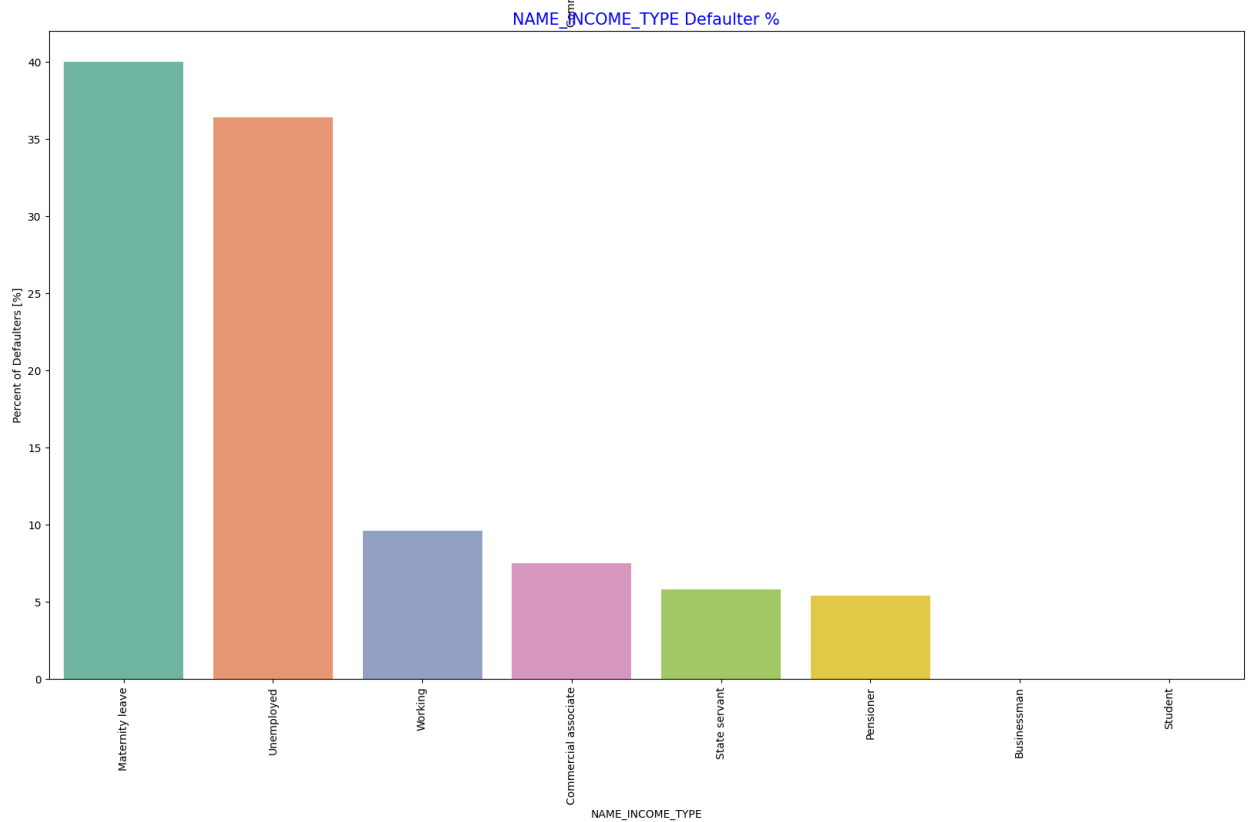
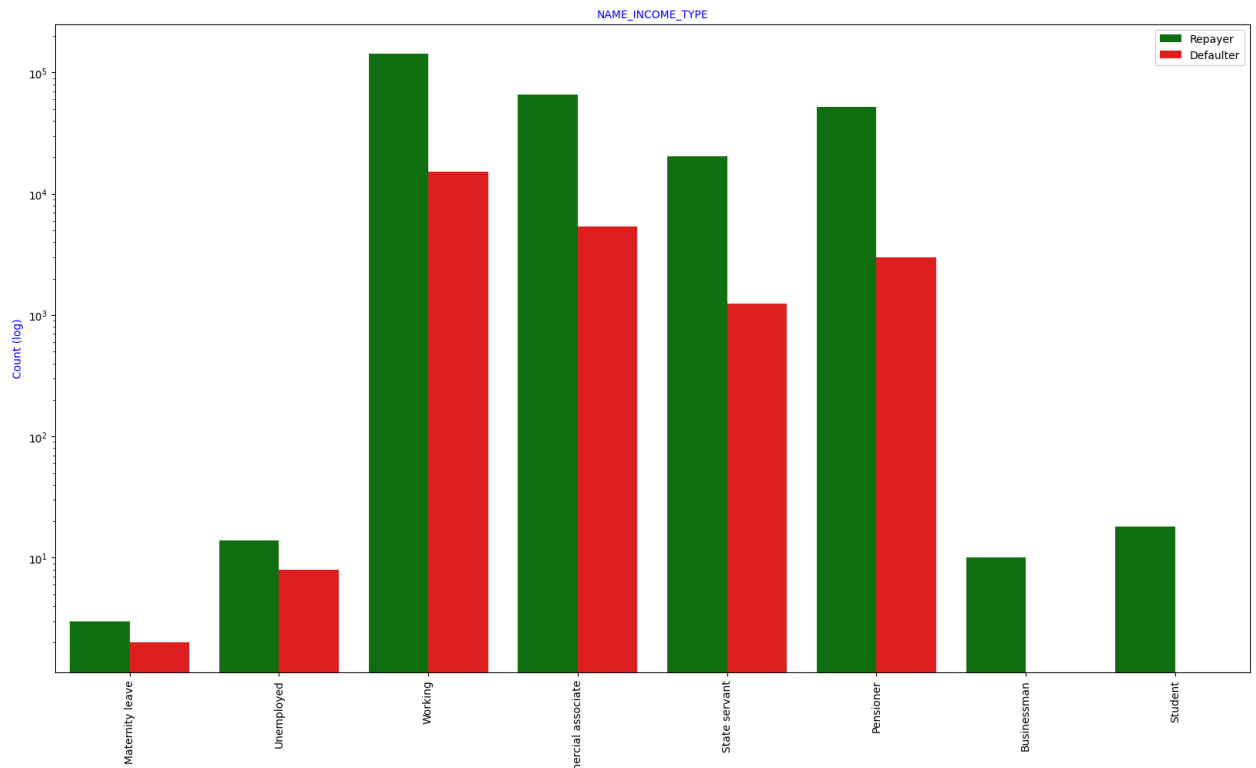
```
In [94]: # Analyzing Family status based on loan repayment status
univariate_categorical("NAME_FAMILY_STATUS", False, True, True)
```



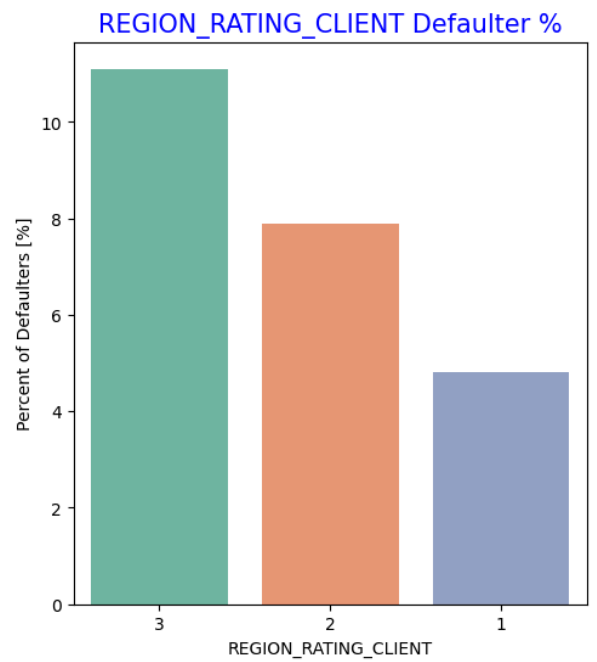
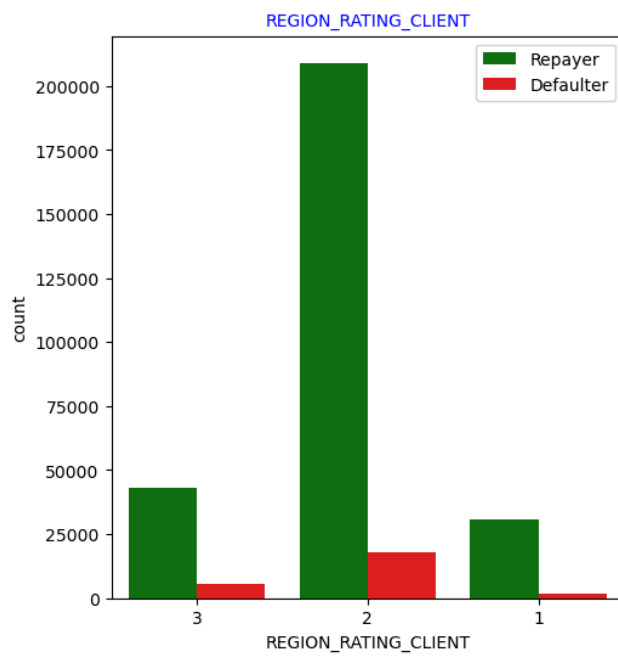
```
In [95]: # Analyzing Education Type based on loan repayment status
univariate_categorical("NAME_EDUCATION_TYPE", True, True, True)
```



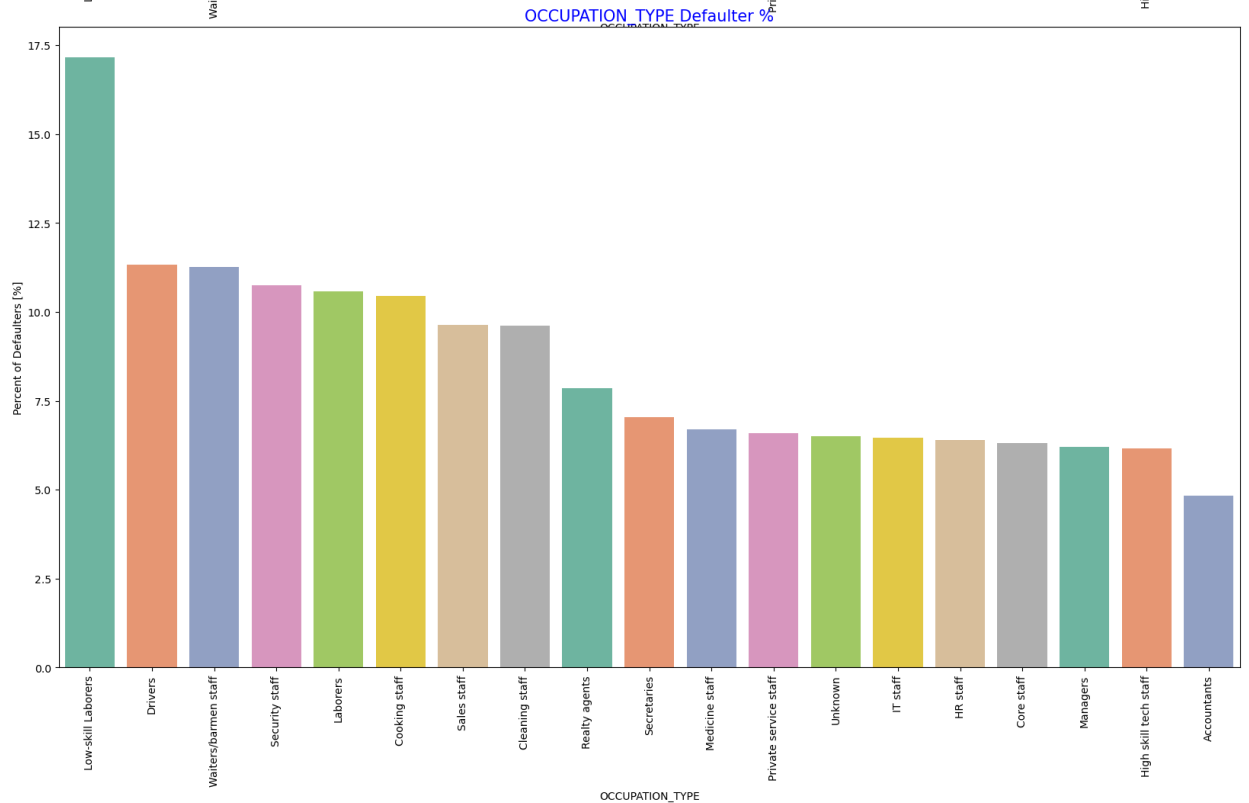
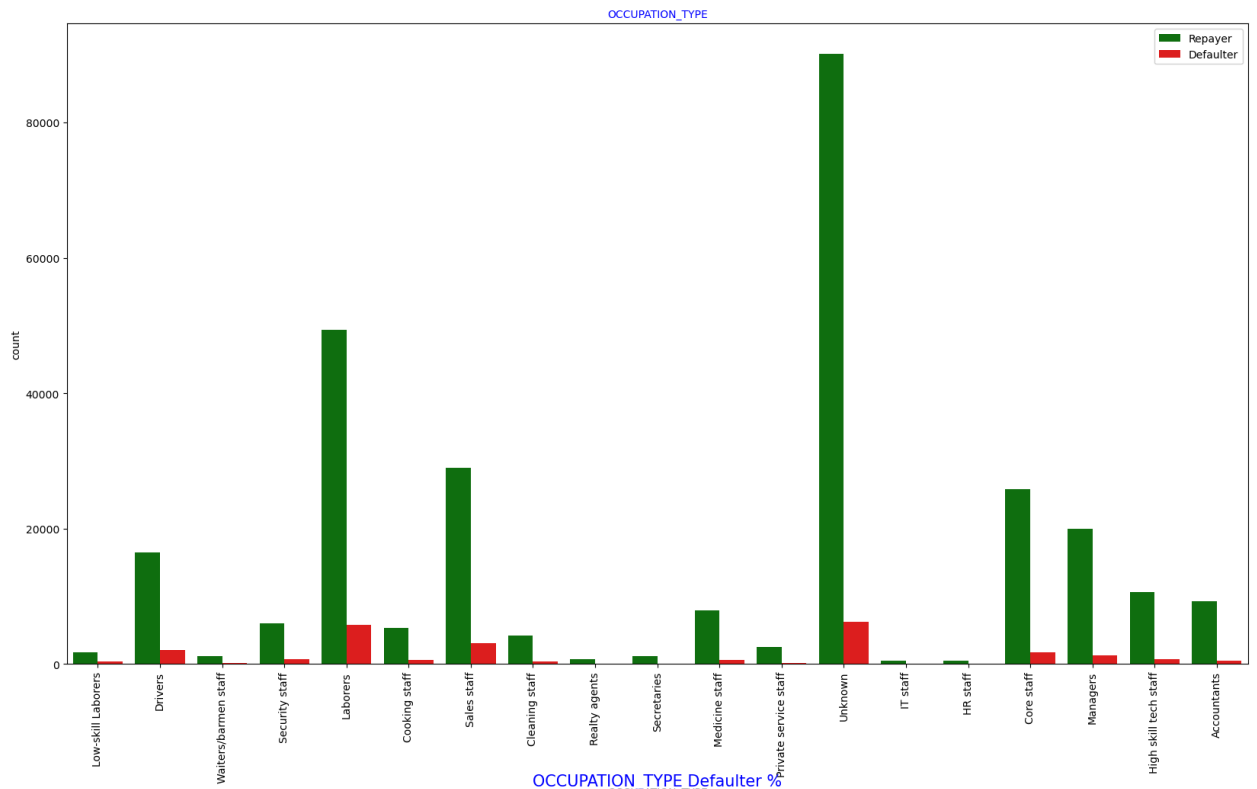
```
In [96]: # Analyzing Income Type based on loan repayment status
univariate_categorical("NAME_INCOME_TYPE", True, True, False)
```



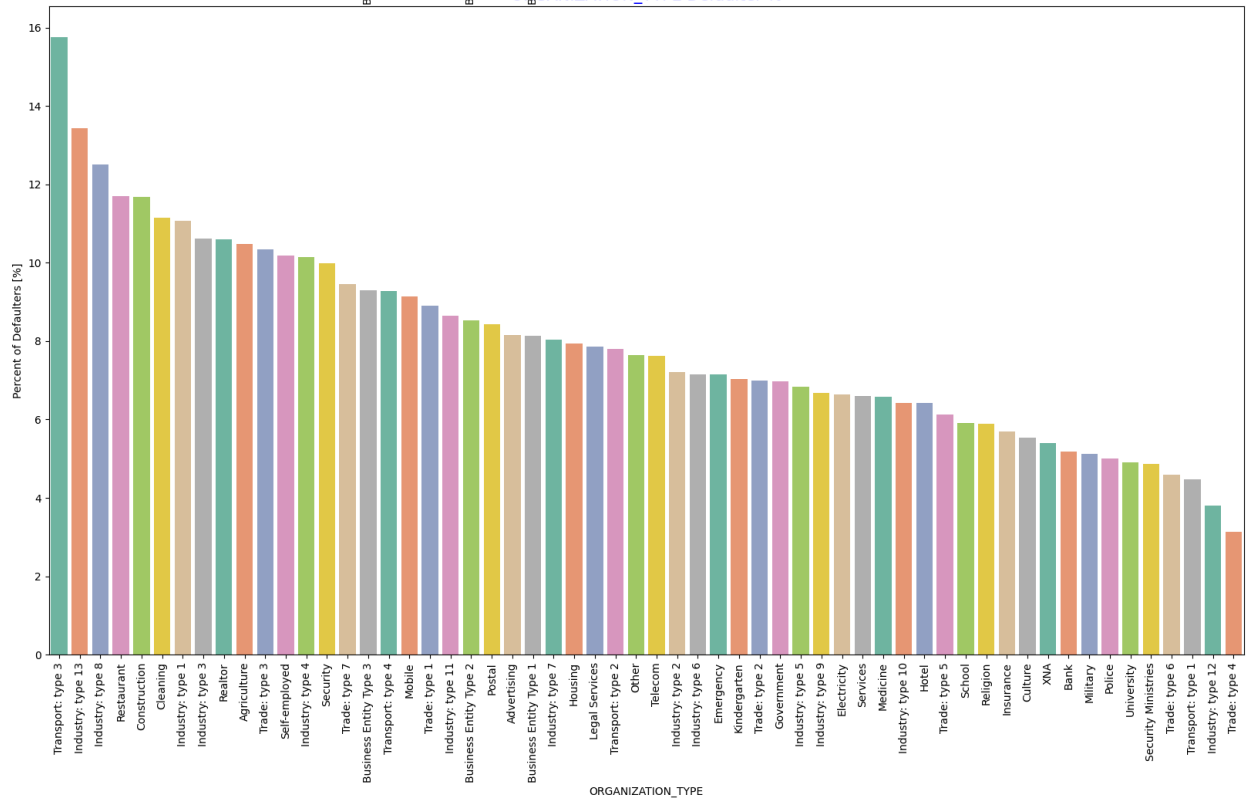
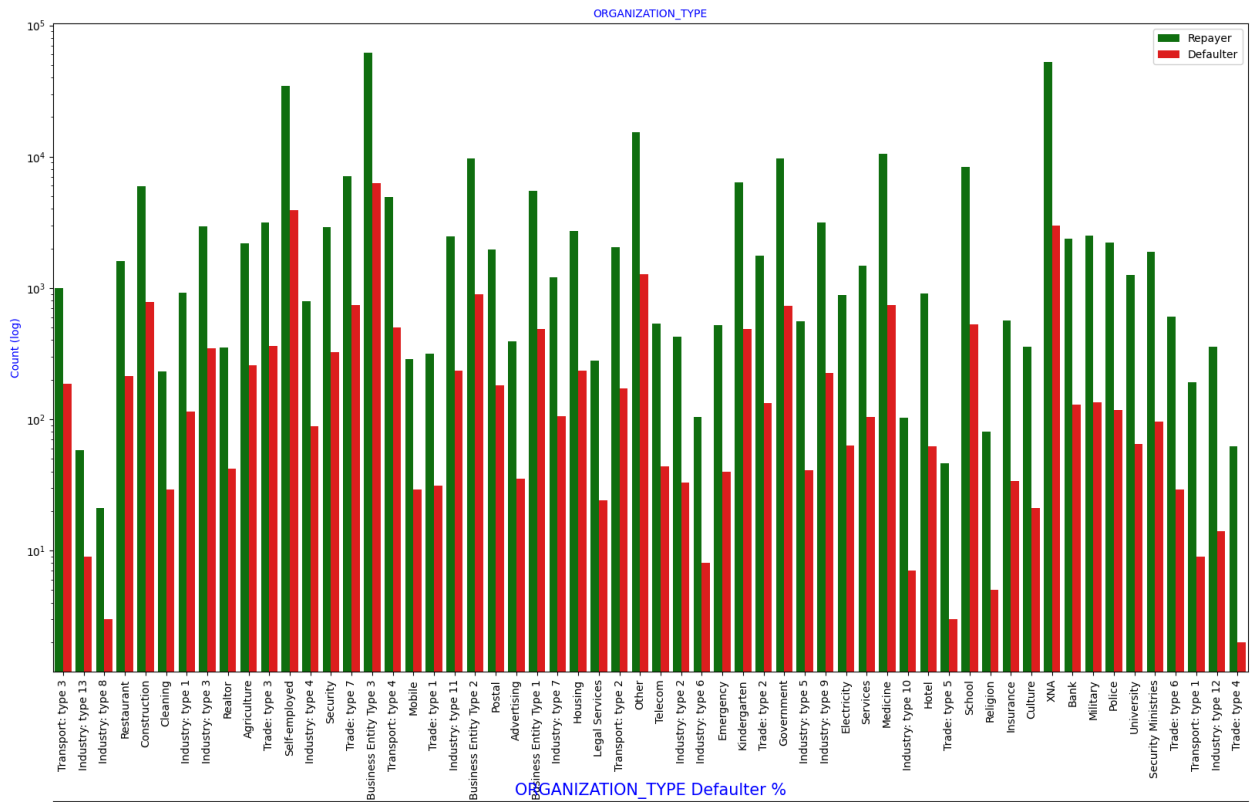
```
In [97]: # Analyzing Region rating where applicant lives based on loan repayment status
univariate_categorical("REGION_RATING_CLIENT", False, False, True)
```



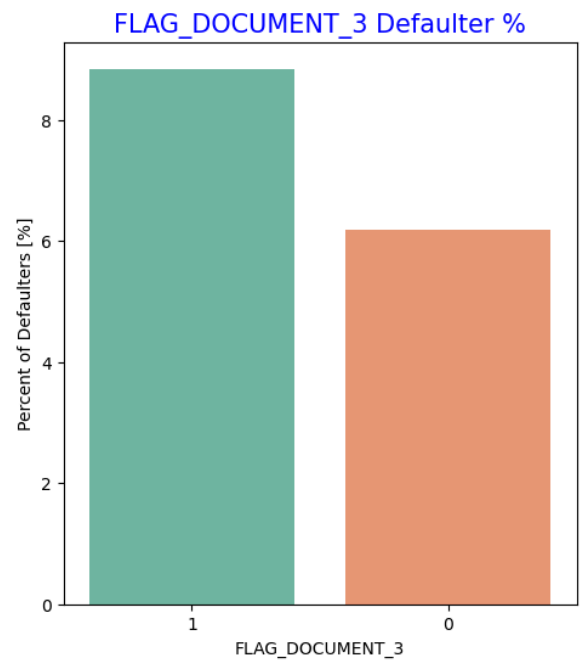
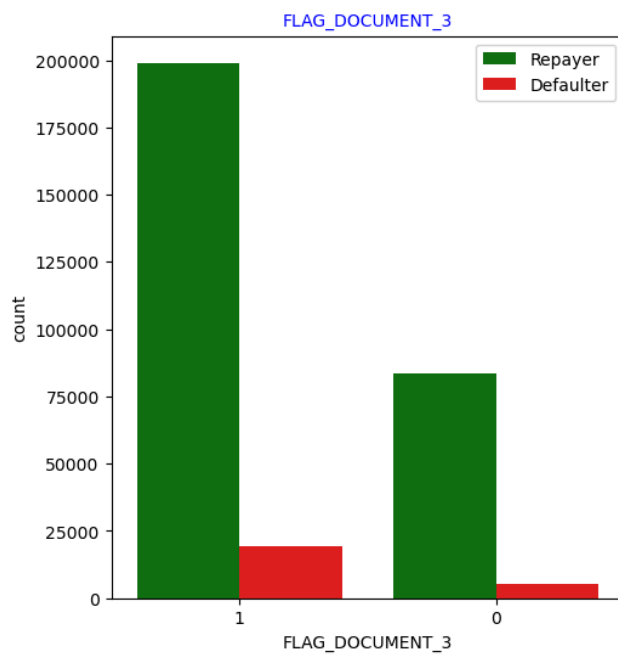
```
In [98]: # Analyzing Occupation Type where applicant lives based on loan repayment status
univariate_categorical("OCCUPATION_TYPE", False, True, False)
```



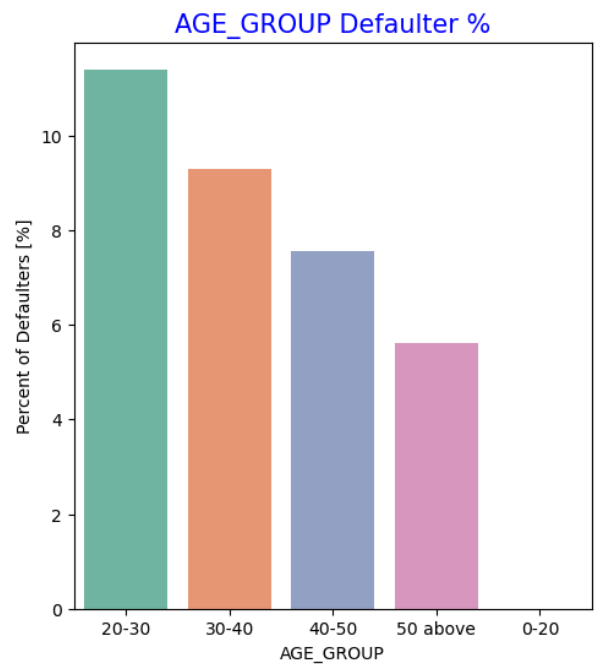
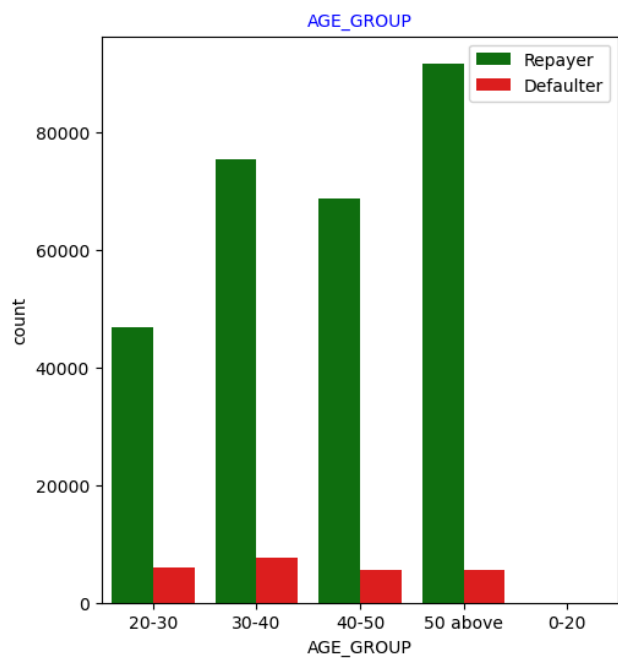
```
In [99]: # Checking Loan repayment status based on Organization type
univariate_categorical("ORGANIZATION_TYPE", True, True, False)
```



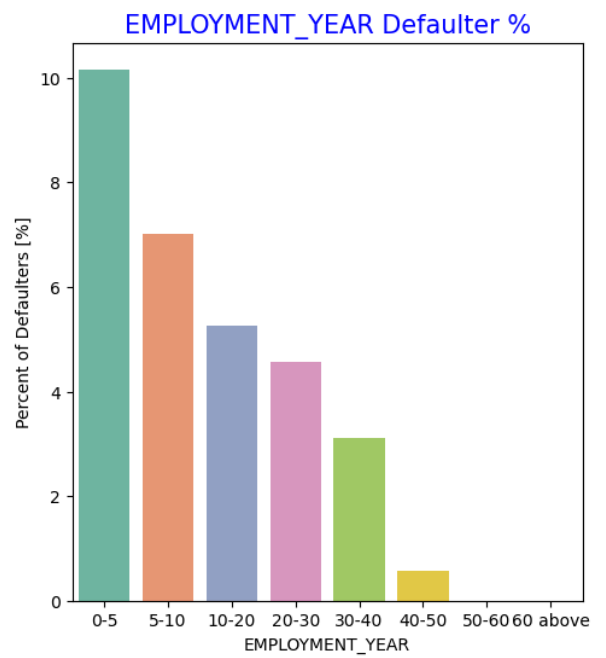
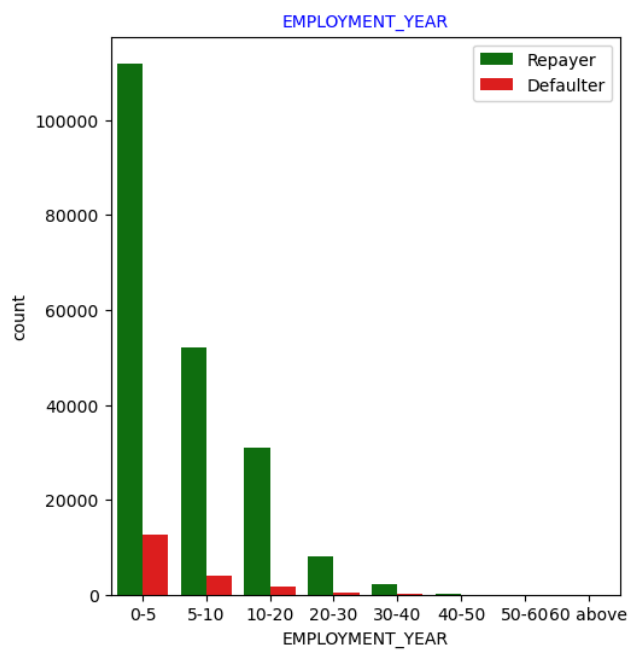
```
In [100... # Analyzing Flag_Doc_3 submission status based on loan repayment status
univariate_categorical("FLAG_DOCUMENT_3",False,False,True)
```



In [101... *# Analyzing Age Group based on loan repayment status*  
`univariate_categorical("AGE_GROUP", False, False, True)`

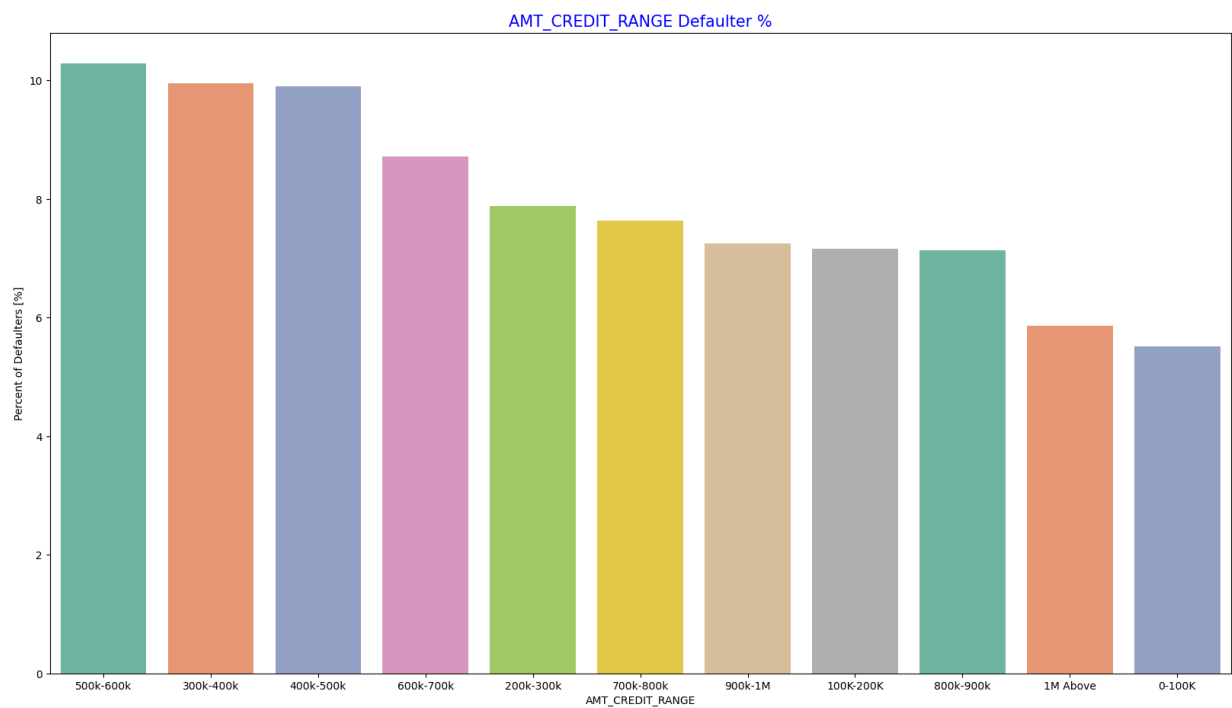
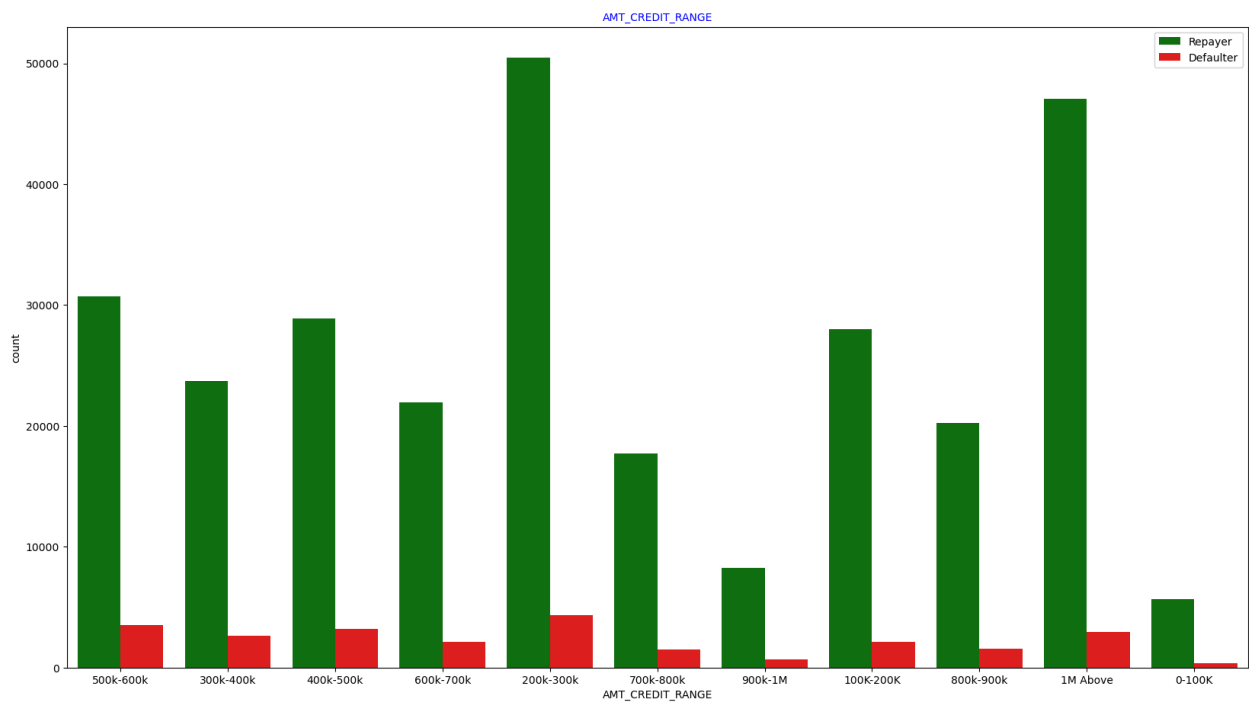


In [102... *# Analyzing Employment\_Year based on loan repayment status*  
`univariate_categorical("EMPLOYMENT_YEAR", False, False, True)`

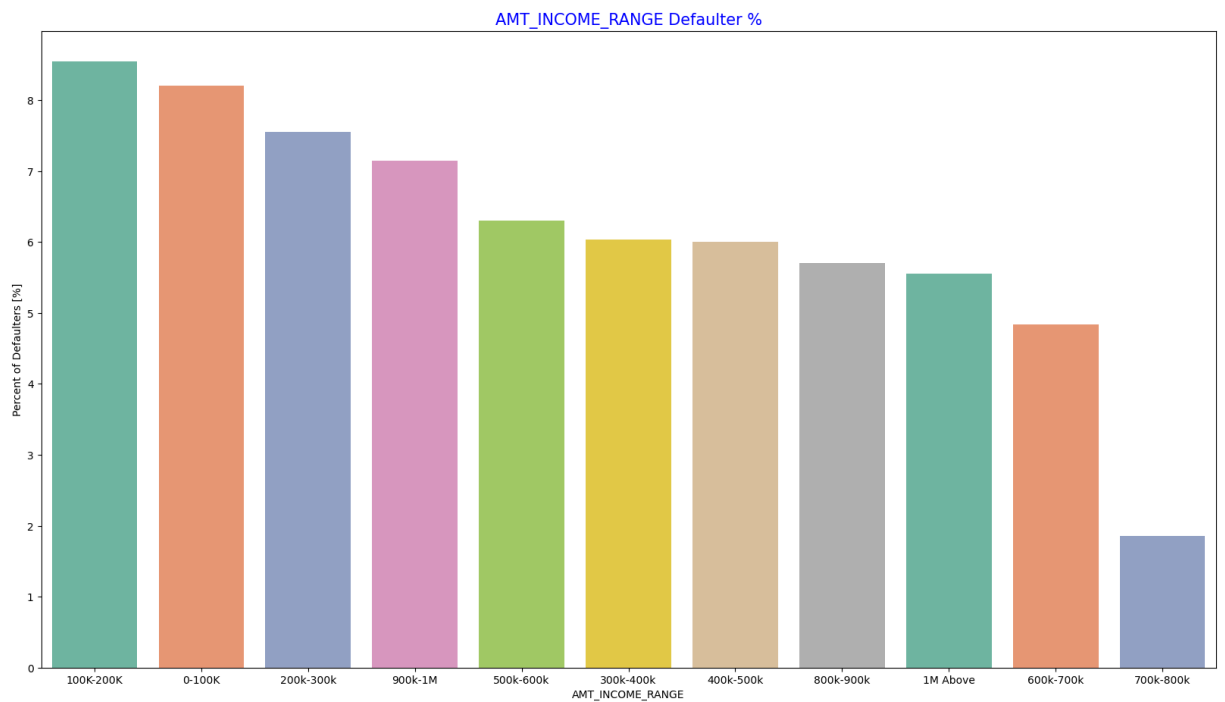
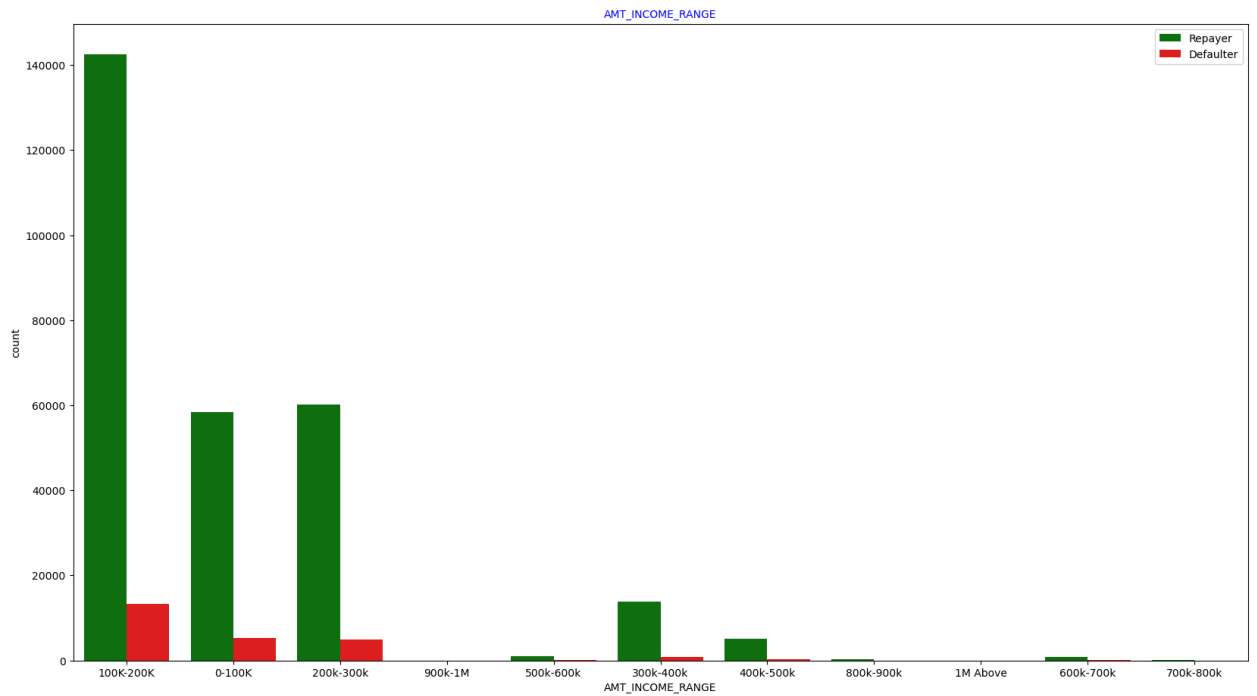


```
In [103... # Analyzing Amount_Credit based on loan repayment status
univariate_categorical("AMT_CREDIT_RANGE",False,False,False)
```

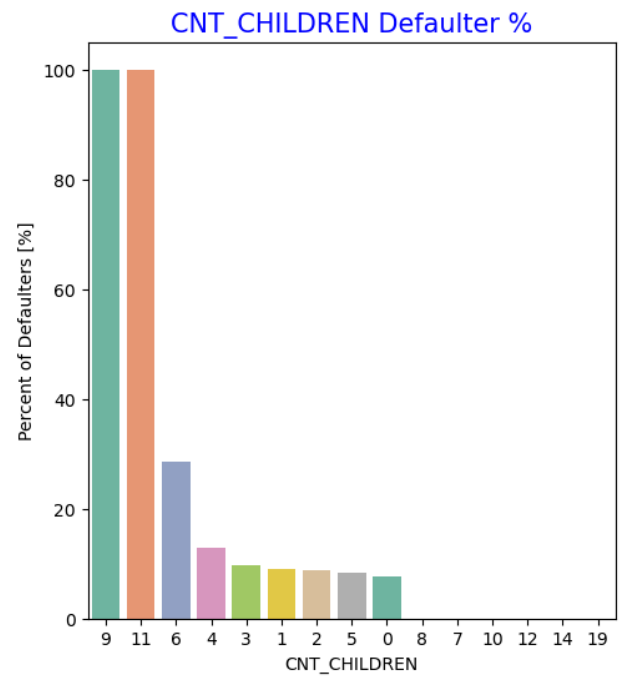
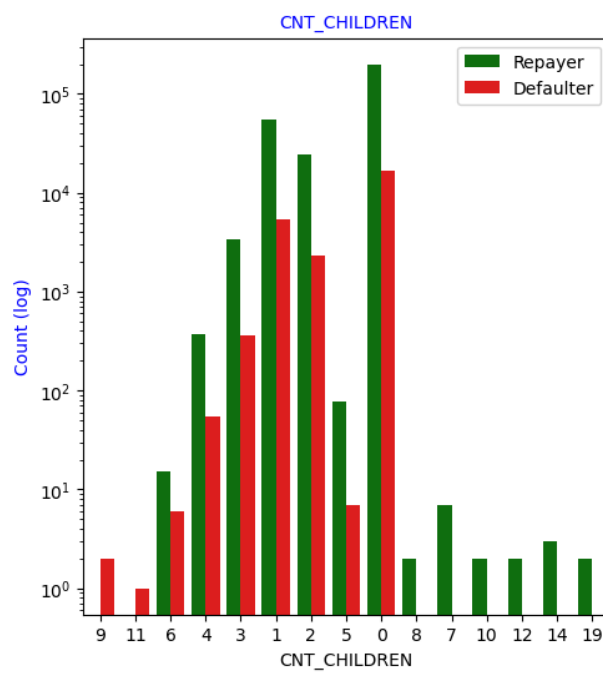




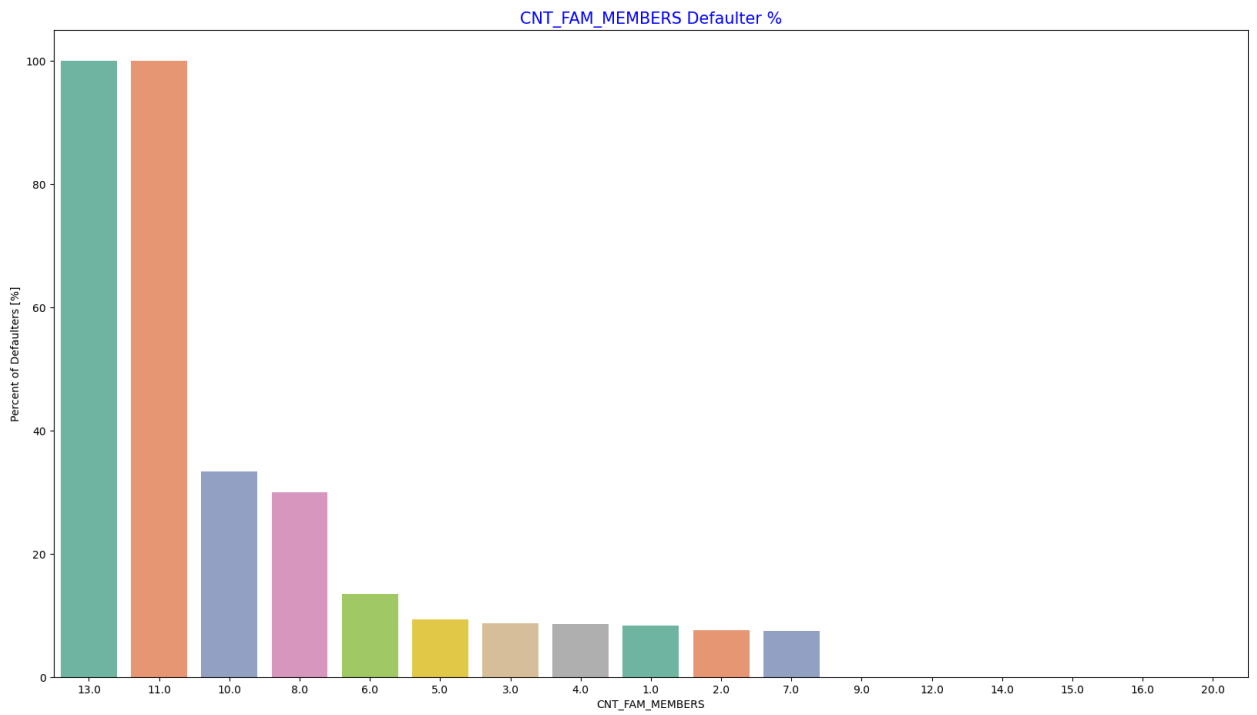
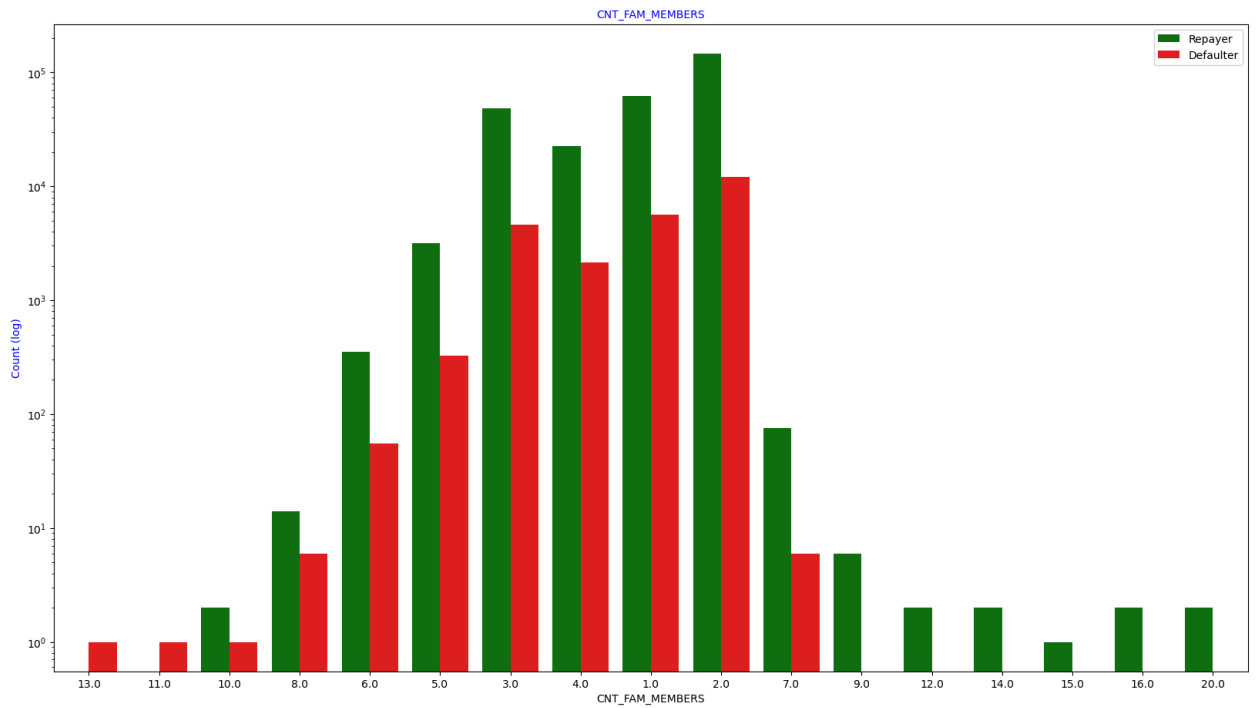
```
In [104... # Analyzing Amount_Income Range based on loan repayment status
univariate_categorical("AMT_INCOME_RANGE",False,False,False)
```



In [105... *# Analyzing Number of children based on loan repayment status*  
 univariate\_categorical("CNT\_CHILDREN", True)



In [106... *# Analyzing Number of family members based on loan repayment status*  
 univariate\_categorical("CNT\_FAM\_MEMBERS", True, False, False)



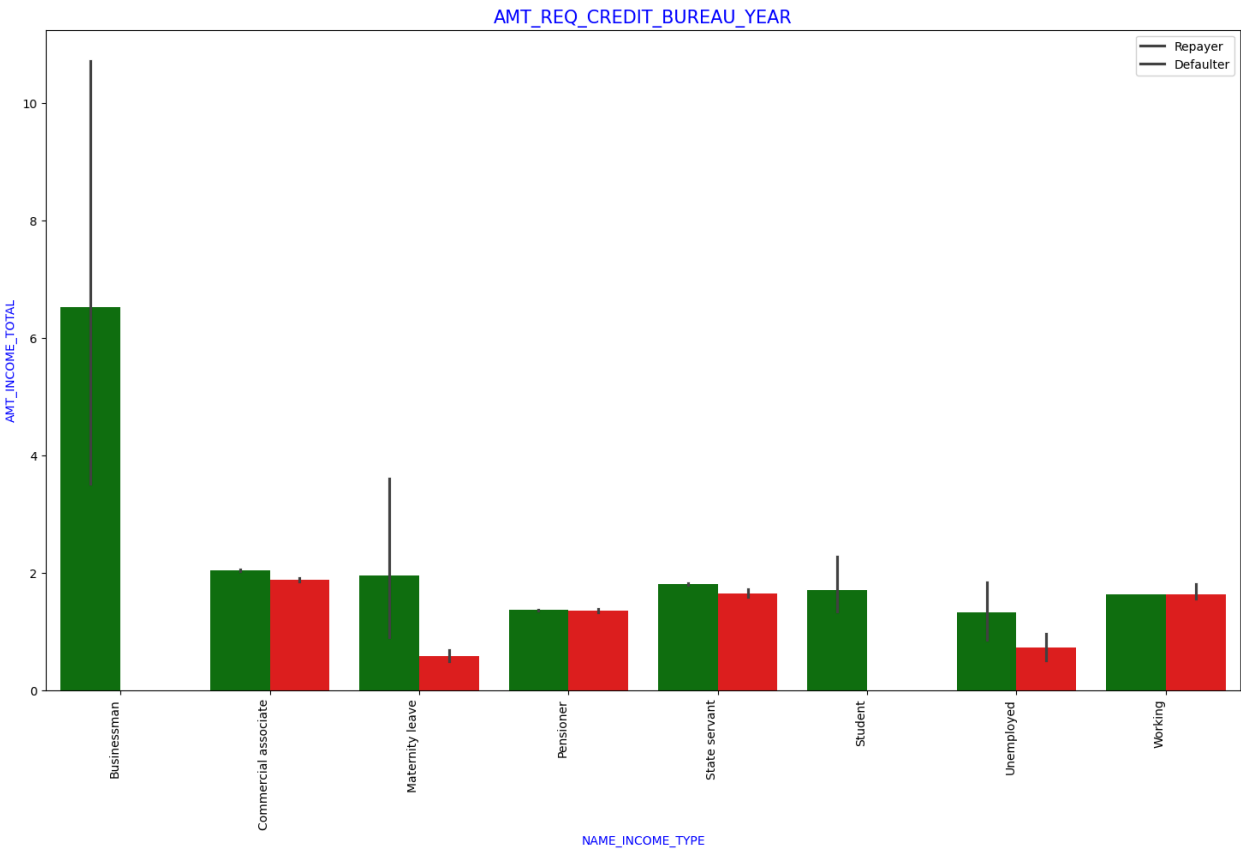
In [107... applicationDF.groupby('NAME\_INCOME\_TYPE')['AMT\_INCOME\_TOTAL'].describe()

Out[107...

|                      | count    | mean     | std      | min    | 25%   | 50%    | 75%     |
|----------------------|----------|----------|----------|--------|-------|--------|---------|
| NAME_INCOME_TYPE     |          |          |          |        |       |        |         |
| Businessman          | 10.0     | 6.525000 | 6.272260 | 1.8000 | 2.250 | 4.9500 | 8.43750 |
| Commercial associate | 71617.0  | 2.029553 | 1.479742 | 0.2655 | 1.350 | 1.8000 | 2.25000 |
| Maternity leave      | 5.0      | 1.404000 | 1.268569 | 0.4950 | 0.675 | 0.9000 | 1.35000 |
| Pensioner            | 55362.0  | 1.364013 | 0.766503 | 0.2565 | 0.900 | 1.1700 | 1.66500 |
| State servant        | 21703.0  | 1.797380 | 1.008806 | 0.2700 | 1.125 | 1.5750 | 2.25000 |
| Student              | 18.0     | 1.705000 | 1.066447 | 0.8100 | 1.125 | 1.5750 | 1.78875 |
| Unemployed           | 22.0     | 1.105364 | 0.880551 | 0.2655 | 0.540 | 0.7875 | 1.35000 |
| Working              | 158774.0 | 1.631699 | 3.075777 | 0.2565 | 1.125 | 1.3500 | 2.02500 |

In [108...

```
# Income type vs Income Amount Range
bivariate_bar("NAME_INCOME_TYPE", "AMT_INCOME_TOTAL", applicationDF, "TARGET", (18
```



In [109...

```
applicationDF.columns
```

```
Out[109... Index(['SK_ID_CURR', 'TARGET', 'NAME_CONTRACT_TYPE', 'CODE_GENDER', 'FLAG_OW
N_CAR', 'FLAG_OW_N_REALTY', 'CNT_CHILDREN', 'AMT_INCOME_TOTAL', 'AMT_CREDIT',
'AMT_ANNUITY',
...
'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON', 'AMT_REQ_CR
EDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR', 'AMT_INCOME_RANGE', 'AMT_CRED
IT_RANGE', 'AGE', 'AGE_GROUP', 'YEARS_EMPLOYED', 'EMPLOYMENT_YEAR'], dtype='o
bject', length=128)
```

```
In [110... # Bifurcating the applicationDF dataframe based on Target value 0 and 1 for co
cols_for_correlation = ['NAME_CONTRACT_TYPE', 'CODE_GENDER', 'FLAG_OW_N_CAR', '
'CNT_CHILDREN', 'AMT_INCOME_TOTAL', 'AMT_CREDIT', 'AMT
'NAME_TYPE_SUITE', 'NAME_INCOME_TYPE', 'NAME_EDUCATION
'NAME_HOUSING_TYPE', 'REGION_POPULATION_RELATIVE', 'DA
'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH', 'OCCUPATION_TY
'REGION_RATING_CLIENT_W_CITY', 'WEEKDAY_APPR_PROCESS_S
'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REG
'REG_CITY_NOT_LIVE_CITY', 'REG_CITY_NOT_WORK_CITY', 'L
'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE'
'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_D
'AMT_REQ_CREDIT_BUREAU_MON', 'AMT_REQ_CREDIT_BUREAU_QR
Repayer_df = applicationDF.loc[applicationDF['TARGET']==0, cols_for_correlatio
Defaulter_df = applicationDF.loc[applicationDF['TARGET']==1, cols_for_correlat
```

```
In [111... Repayer_df.dtypes
```

```

Out[111... NAME_CONTRACT_TYPE      category
          CODE_GENDER        category
          FLAG_OWN_CAR        category
          FLAG_OWN_REALTY     category
          CNT_CHILDREN        int64
          AMT_INCOME_TOTAL    float64
          AMT_CREDIT          float64
          AMT_ANNUITY         float64
          AMT_GOODS_PRICE     float64
          NAME_TYPE_SUITE     category
          NAME_INCOME_TYPE    category
          NAME_EDUCATION_TYPE category
          NAME_FAMILY_STATUS  category
          NAME_HOUSING_TYPE   category
          REGION_POPULATION_RELATIVE float64
          DAYS_BIRTH          int64
          DAYS_EMPLOYED       int64
          DAYS_REGISTRATION   float64
          DAYS_ID_PUBLISH     int64
          OCCUPATION_TYPE     category
          CNT_FAM_MEMBERS     float64
          REGION_RATING_CLIENT category
          REGION_RATING_CLIENT_W_CITY category
          WEEKDAY_APPR_PROCESS_START category
          HOUR_APPR_PROCESS_START int64
          REG_REGION_NOT_LIVE_REGION int64
          REG_REGION_NOT_WORK_REGION category
          LIVE_REGION_NOT_WORK_REGION category
          REG_CITY_NOT_LIVE_CITY category
          REG_CITY_NOT_WORK_CITY category
          LIVE_CITY_NOT_WORK_CITY category
          ORGANIZATION_TYPE   category
          OBS_60_CNT_SOCIAL_CIRCLE float64
          DEF_60_CNT_SOCIAL_CIRCLE float64
          DAYS_LAST_PHONE_CHANGE float64
          FLAG_DOCUMENT_3     int64
          AMT_REQ_CREDIT_BUREAU_HOUR float64
          AMT_REQ_CREDIT_BUREAU_DAY float64
          AMT_REQ_CREDIT_BUREAU_WEEK float64
          AMT_REQ_CREDIT_BUREAU_MON float64
          AMT_REQ_CREDIT_BUREAU_QRT float64
          AMT_REQ_CREDIT_BUREAU_YEAR float64
          dtype: object

```

```

In [112... repayer_numeric = Repayer_df.select_dtypes(include='number')
          corr_repayer = repayer_numeric.corr()

```

```

In [113... df_encoded = pd.get_dummies(Repayer_df, drop_first=True)
          corr_repayer = df_encoded.corr()

```

```

In [114... Repayer_df.dtypes

```

```

Out[114... NAME_CONTRACT_TYPE      category
          CODE_GENDER        category
          FLAG_OWN_CAR        category
          FLAG_OWN_REALTY     category
          CNT_CHILDREN         int64
          AMT_INCOME_TOTAL    float64
          AMT_CREDIT          float64
          AMT_ANNUITY         float64
          AMT_GOODS_PRICE     float64
          NAME_TYPE_SUITE     category
          NAME_INCOME_TYPE    category
          NAME_EDUCATION_TYPE category
          NAME_FAMILY_STATUS   category
          NAME_HOUSING_TYPE    category
          REGION_POPULATION_RELATIVE float64
          DAYS_BIRTH          int64
          DAYS_EMPLOYED       int64
          DAYS_REGISTRATION   float64
          DAYS_ID_PUBLISH     int64
          OCCUPATION_TYPE     category
          CNT_FAM_MEMBERS     float64
          REGION_RATING_CLIENT category
          REGION_RATING_CLIENT_W_CITY category
          WEEKDAY_APPR_PROCESS_START category
          HOUR_APPR_PROCESS_START int64
          REG_REGION_NOT_LIVE_REGION int64
          REG_REGION_NOT_WORK_REGION category
          LIVE_REGION_NOT_WORK_REGION category
          REG_CITY_NOT_LIVE_CITY category
          REG_CITY_NOT_WORK_CITY category
          LIVE_CITY_NOT_WORK_CITY category
          ORGANIZATION_TYPE    category
          OBS_60_CNT_SOCIAL_CIRCLE float64
          DEF_60_CNT_SOCIAL_CIRCLE float64
          DAYS_LAST_PHONE_CHANGE float64
          FLAG_DOCUMENT_3      int64
          AMT_REQ_CREDIT_BUREAU_HOUR float64
          AMT_REQ_CREDIT_BUREAU_DAY float64
          AMT_REQ_CREDIT_BUREAU_WEEK float64
          AMT_REQ_CREDIT_BUREAU_MON float64
          AMT_REQ_CREDIT_BUREAU_QRT float64
          AMT_REQ_CREDIT_BUREAU_YEAR float64
          dtype: object

```

```

In [115... df_encoded = pd.get_dummies(Repayer_df, drop_first=True)

corr = df_encoded.corr()

plt.figure(figsize=(12,12))
sns.heatmap(corr, cmap="RdYlGn")

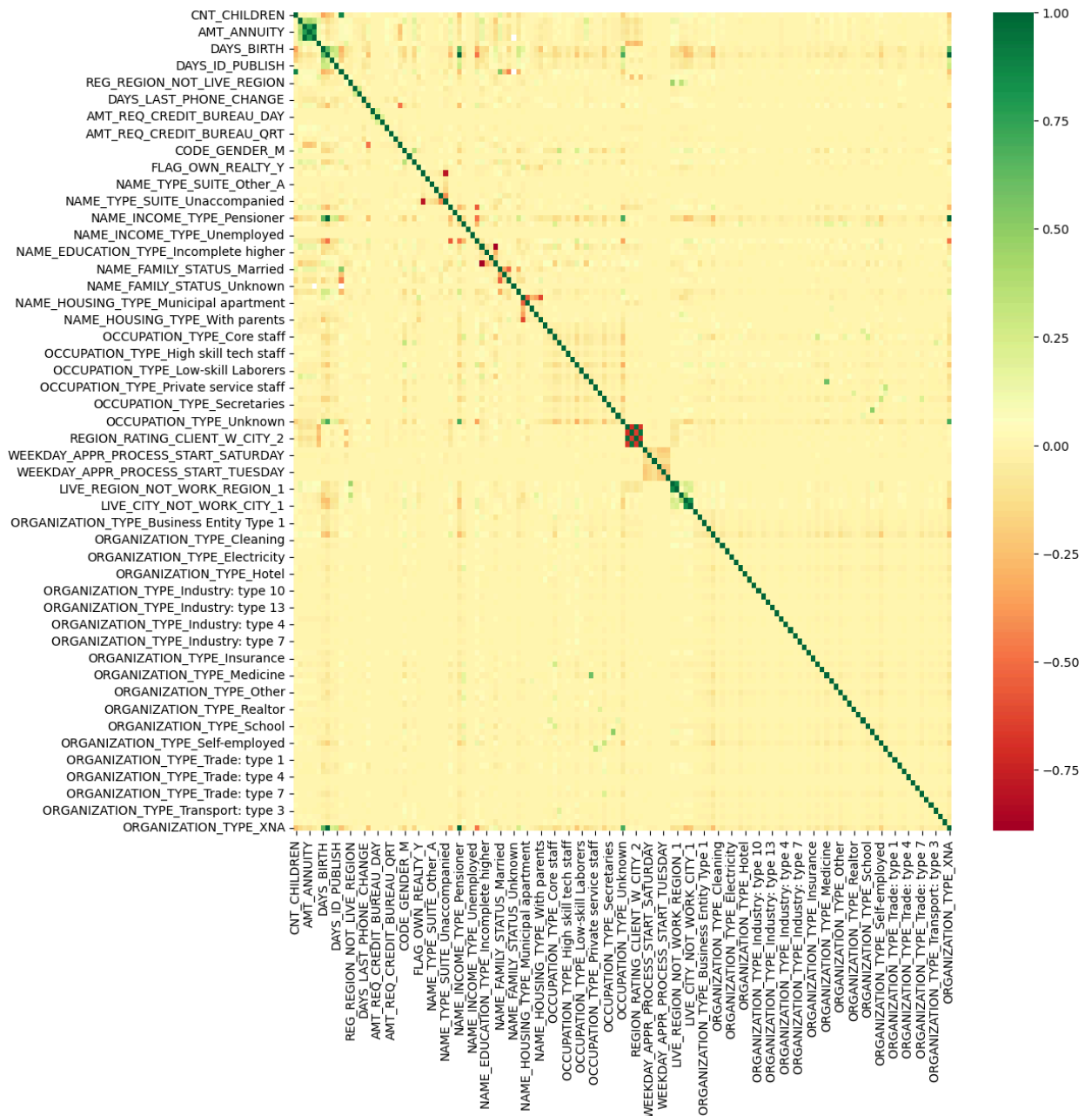
```

```

Out[115... <Axes: >

```





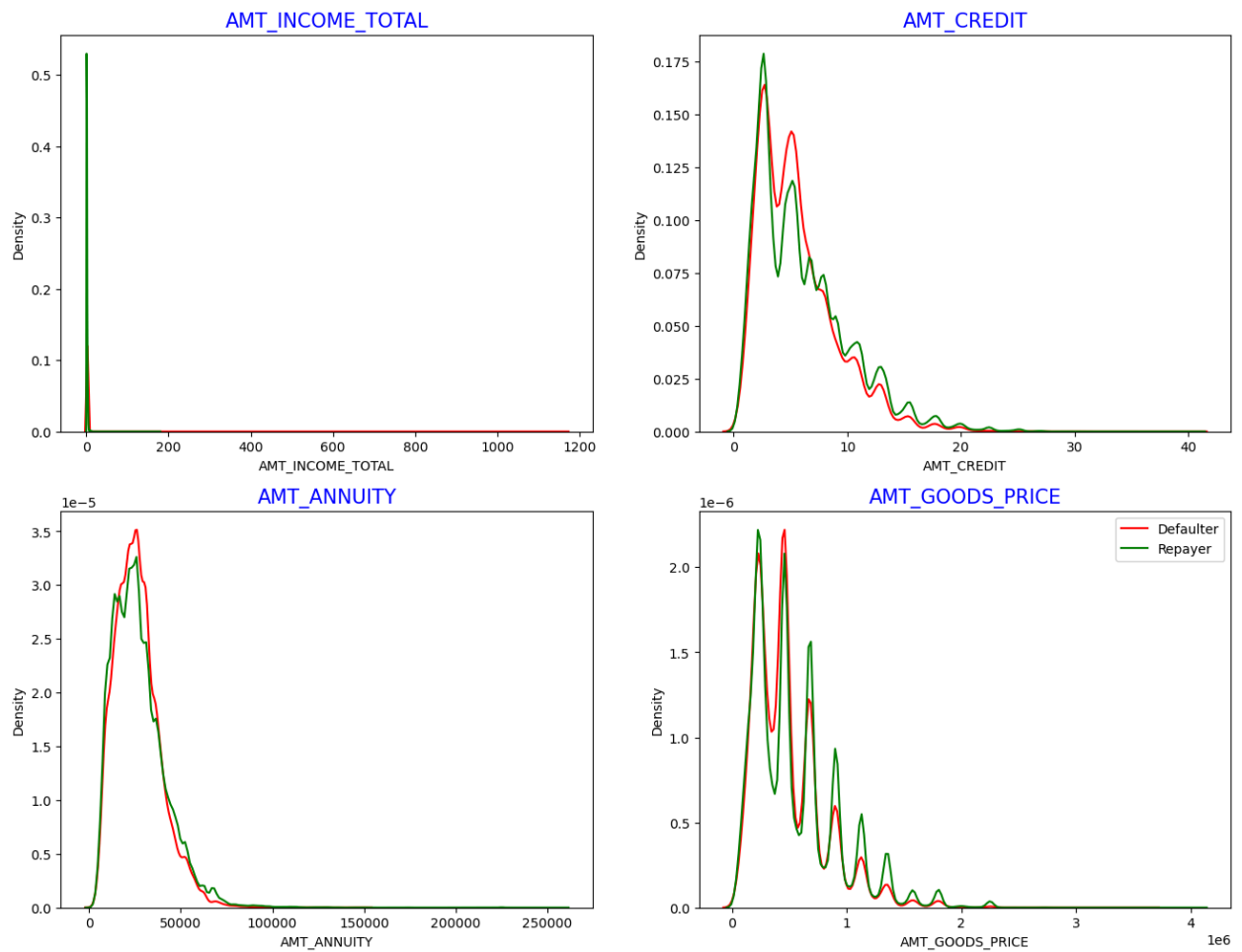
```
In [116... # Plotting the numerical columns related to amount as distribution plot to see
amount = applicationDF[['AMT_INCOME_TOTAL', 'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_C

fig = plt.figure(figsize=(16,12))

for i in enumerate(amount):
    plt.subplot(2,2,i[0]+1)
    sns.distplot(Defaulters_df[i[1]], hist=False, color='r', label="Defaulter")
    sns.distplot(Repayers_df[i[1]], hist=False, color='g', label="Repayer")
    plt.title(i[1], fontdict={'fontsize' : 15, 'fontweight' : 5, 'color' : 'B

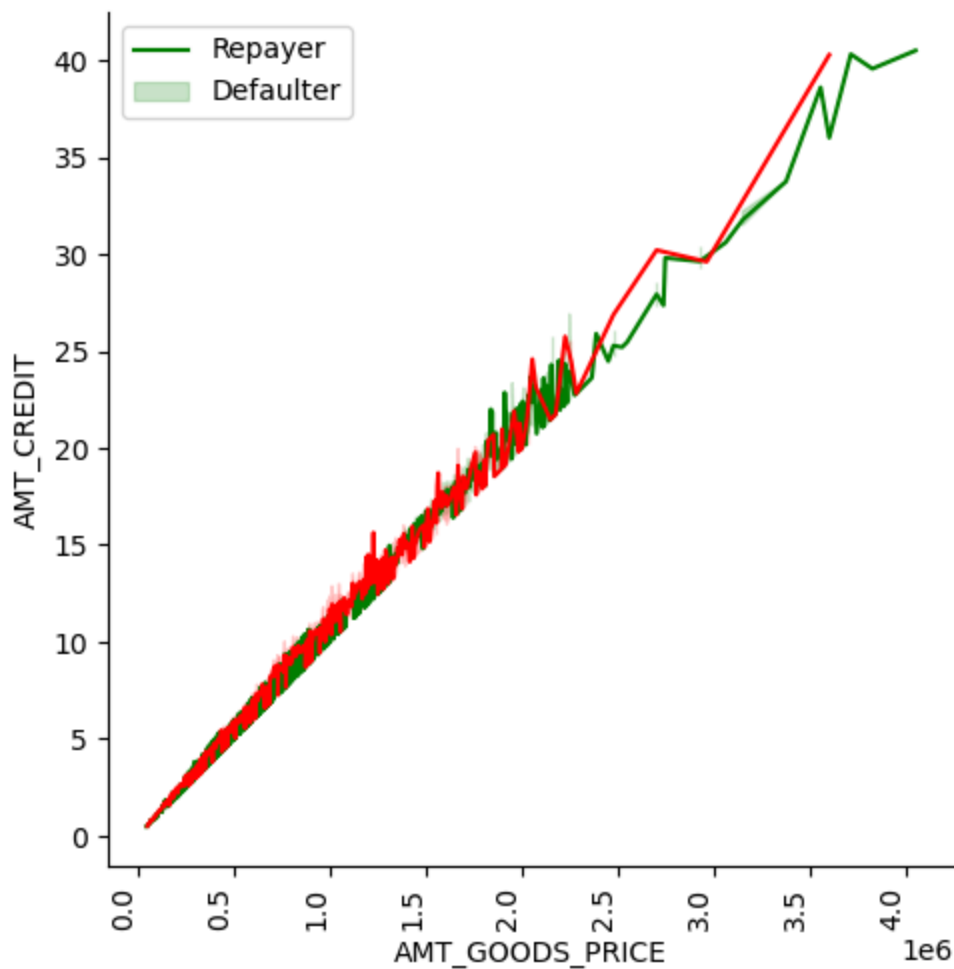
plt.legend()

plt.show()
```



```
In [117... # Checking the relationship between Goods price and credit and comparing with
bivariate_rel('AMT_GOODS_PRICE','AMT_CREDIT',applicationDF,"TARGET", "line", [
```

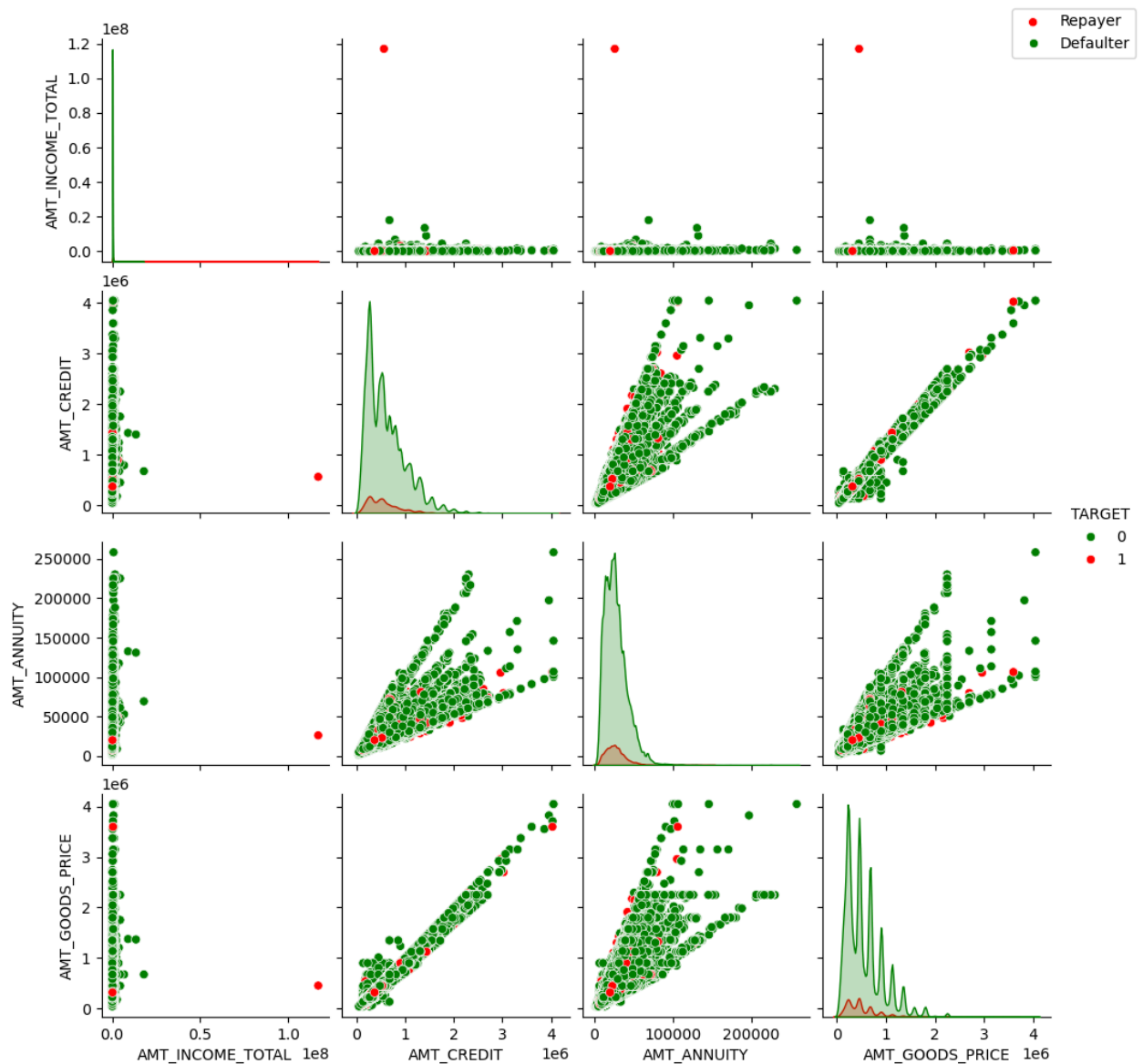
<Figure size 1500x600 with 0 Axes>



```
In [118... import pandas as pd
applicationDF = pd.read_csv(r'C:\Users\HP\Downloads\application_data.csv')
```

```
In [119... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [120... # Plotting pairplot between amount variable to draw reference against loan rep
amount = applicationDF[['AMT_INCOME_TOTAL', 'AMT_CREDIT',
                        'AMT_ANNUITY', 'AMT_GOODS_PRICE', 'TARGET']]
amount = amount[(amount["AMT_GOODS_PRICE"].notnull()) & (amount["AMT_ANNUITY"]
ax= sns.pairplot(amount, hue="TARGET", palette=["g", "r"])
ax.fig.legend(labels=['Repayer', 'Defaulter'])
plt.show()
```



```
In [121...] #merge both the dataframe on SK_ID_CURR with Inner Joins
loan_process_df = pd.merge(applicationDF, previousDF, how='inner', on='SK_ID_C
loan_process_df.head()
```

```
Out[121...]
```

|   | SK_ID_CURR | TARGET | NAME_CONTRACT_TYPE_x | CODE_GENDER | FLAG_OWN_C |
|---|------------|--------|----------------------|-------------|------------|
| 0 | 100002     | 1      | Cash loans           | M           |            |
| 1 | 100003     | 0      | Cash loans           | F           |            |
| 2 | 100003     | 0      | Cash loans           | F           |            |
| 3 | 100003     | 0      | Cash loans           | F           |            |
| 4 | 100004     | 0      | Revolving loans      | M           |            |

```
In [123... #Checking the details of the merged dataframe
loan_process_df.shape
```

```
Out[123... (1413701, 159)
```

```
In [124... # Checking the element count of the dataframe
loan_process_df.size
```

```
Out[124... 224778459
```

```
In [125... # checking the columns and column types of the dataframe
loan_process_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1413701 entries, 0 to 1413700
Columns: 159 entries, SK_ID_CURR to DAYS_DECISION_GROUP
dtypes: category(14), float64(80), int64(46), object(19)
memory usage: 1.5+ GB
```

```
In [126... # Checking merged dataframe numerical columns statistics
loan_process_df.describe()
```

```
Out[126...
```

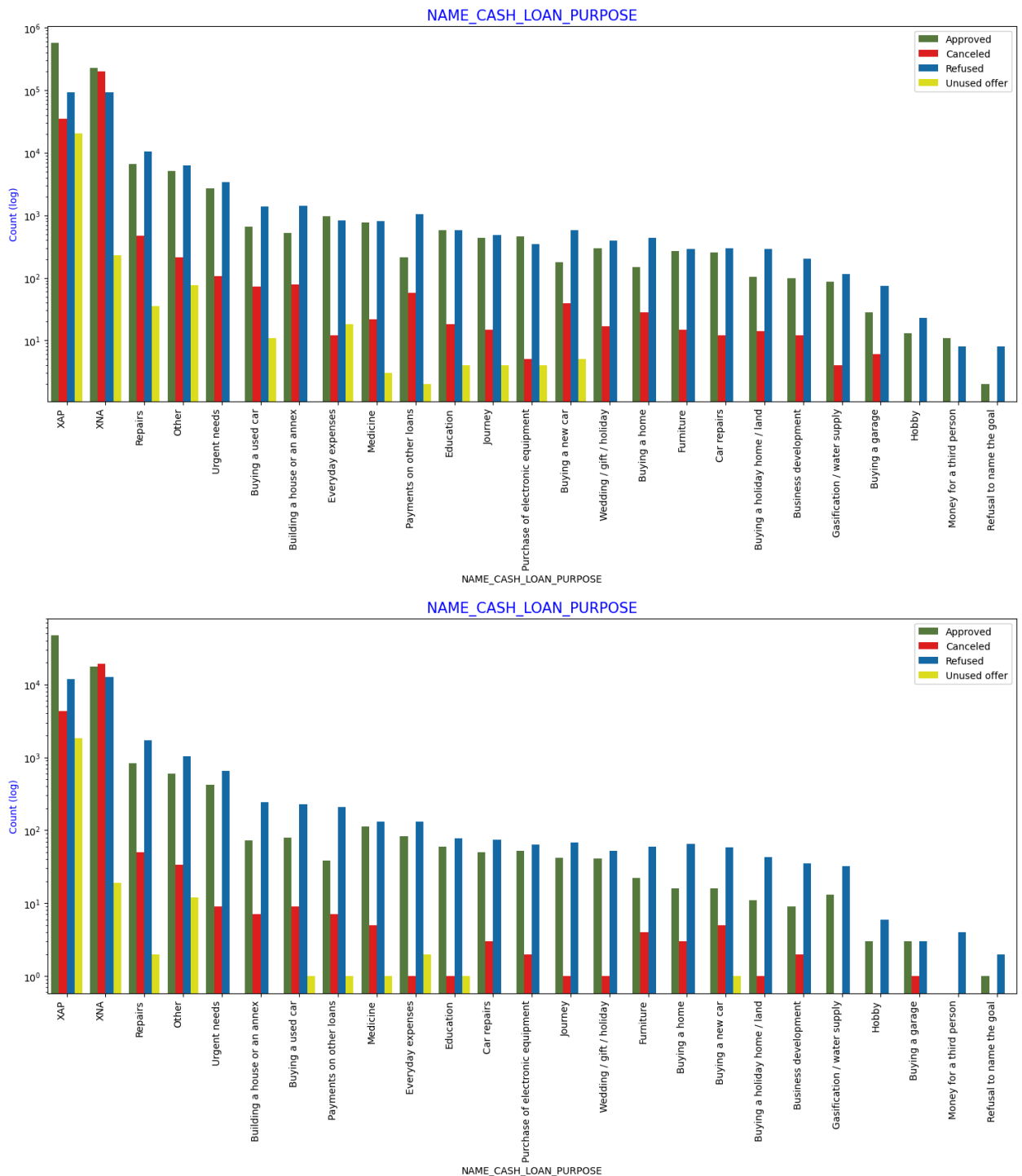
|              | SK_ID_CURR   | TARGET       | CNT_CHILDREN | AMT_INCOME_TOTAL | AMT_C |
|--------------|--------------|--------------|--------------|------------------|-------|
| <b>count</b> | 1.413701e+06 | 1.413701e+06 | 1.413701e+06 | 1.413701e+06     | 1.413 |
| <b>mean</b>  | 2.784813e+05 | 8.655296e-02 | 4.048933e-01 | 1.733160e+05     | 5.875 |
| <b>std</b>   | 1.028118e+05 | 2.811789e-01 | 7.173454e-01 | 1.985734e+05     | 3.849 |
| <b>min</b>   | 1.000020e+05 | 0.000000e+00 | 0.000000e+00 | 2.565000e+04     | 4.500 |
| <b>25%</b>   | 1.893640e+05 | 0.000000e+00 | 0.000000e+00 | 1.125000e+05     | 2.700 |
| <b>50%</b>   | 2.789920e+05 | 0.000000e+00 | 0.000000e+00 | 1.575000e+05     | 5.084 |
| <b>75%</b>   | 3.675560e+05 | 0.000000e+00 | 1.000000e+00 | 2.070000e+05     | 8.079 |
| <b>max</b>   | 4.562550e+05 | 1.000000e+00 | 1.900000e+01 | 1.170000e+08     | 4.050 |

```
In [127... # Bifurcating the applicationDF dataframe based on Target value 0 and 1 for co

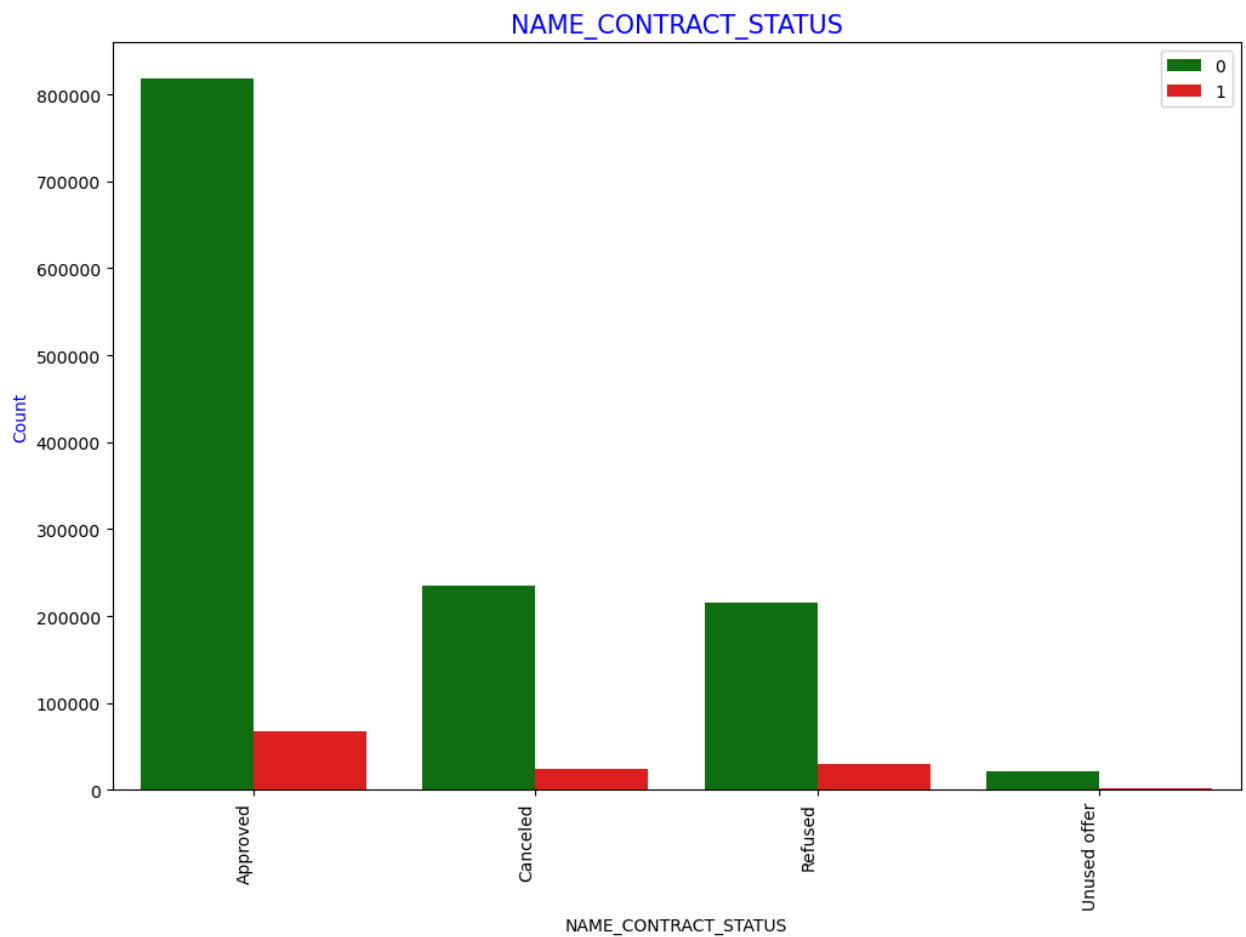
L0 = loan_process_df[loan_process_df['TARGET']==0] # Repayers
L1 = loan_process_df[loan_process_df['TARGET']==1] # Defaulters
```

```
In [128... univariate_merged("NAME_CASH_LOAN_PURPOSE",L0,"NAME_CONTRACT_STATUS",["#548235

univariate_merged("NAME_CASH_LOAN_PURPOSE",L1,"NAME_CONTRACT_STATUS",["#548235
```

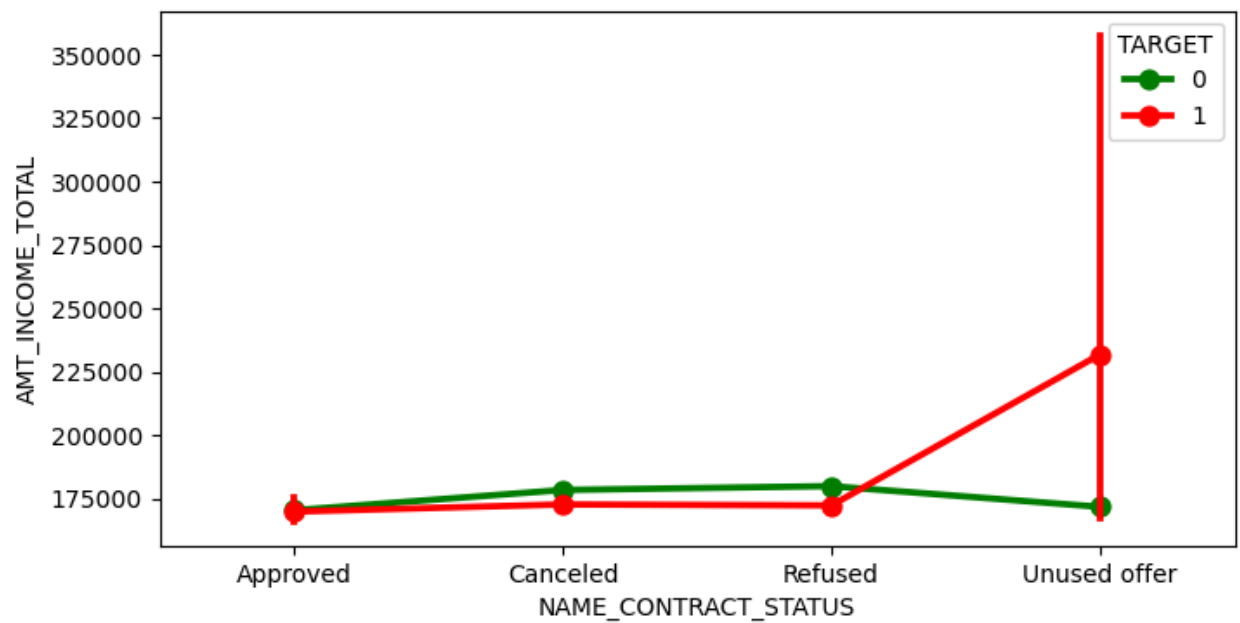


```
In [129... # Checking the Contract Status based on loan repayment status and whether ther
univariate_merged("NAME_CONTRACT_STATUS", loan_process_df, "TARGET", ['g', 'r'], Fa
g = loan_process_df.groupby("NAME_CONTRACT_STATUS")["TARGET"]
df1 = pd.concat([g.value_counts(), round(g.value_counts(normalize=True).mul(100
df1['Percentage'] = df1['Percentage'].astype(str) + "%" # adding percentage sym
print (df1)
```



|                      |        | Counts Percentage |        |
|----------------------|--------|-------------------|--------|
| NAME_CONTRACT_STATUS | TARGET |                   |        |
| Approved             | 0      | 818856            | 92.41% |
|                      | 1      | 67243             | 7.59%  |
| Canceled             | 0      | 235641            | 90.83% |
|                      | 1      | 23800             | 9.17%  |
| Refused              | 0      | 215952            | 88.0%  |
|                      | 1      | 29438             | 12.0%  |
| Unused offer         | 0      | 20892             | 91.75% |
|                      | 1      | 1879              | 8.25%  |

```
In [130... # plotting the relationship between income total and contact status
merged_pointplot("NAME_CONTRACT_STATUS", 'AMT_INCOME_TOTAL')
```



In [131]: `# plotting the relationship between people who defaulted in last 60 days being merged_pointplot("NAME_CONTRACT_STATUS", 'DEF_60_CNT_SOCIAL_CIRCLE')`

